

# CHAPTER 8

## *Determinants*

### 8.1 Introduction

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Each  $n$ -square matrix  $A = [a_{ij}]$  is assigned a special scalar called the *determinant* of  $A$ , denoted by  $\det(A)$  or  $|A|$  or

$$\begin{vmatrix} a_{11} & a_{12} & \dots & a_{1n} \\ a_{21} & a_{22} & \dots & a_{2n} \\ \dots & \dots & \dots & \dots \\ a_{n1} & a_{n2} & \dots & a_{nn} \end{vmatrix}$$

We emphasize that an  $n \times n$  array of scalars enclosed by straight lines, called a *determinant of order  $n$* , is not a matrix but denotes the determinant of the enclosed array of scalars (i.e., the enclosed matrix).

The determinant function was first discovered during the investigation of systems of linear equations. We shall see that the determinant is an indispensable tool in investigating and obtaining properties of square matrices.

The definition of the determinant and most of its properties also apply in the case where the entries of a matrix come from a commutative ring.

We begin with a special case of determinants of orders 1, 2, and 3. Then we define a determinant of arbitrary order. This general definition is preceded by a discussion of permutations, which is necessary for our general definition of the determinant.

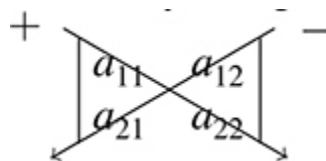
## 8.2 Determinants of Orders 1 and 2

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Determinants of orders 1 and 2 are defined as follows:

$$|a_{11}| = a_{11} \quad \text{and} \quad \begin{vmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{vmatrix} = a_{11}a_{22} - a_{12}a_{21}$$

Thus, the determinant of a  $1 \times 1$  matrix  $A = [a_{11}]$  is the scalar  $a_{11}$ ; that is,  $\det(A) = |a_{11}| = a_{11}$ . The determinant of order two may easily be remembered by using the following diagram:



That, is, the determinant is equal to the product of the elements along the plus-labeled arrow minus the product of the elements along the minus-labeled arrow. (There is an analogous diagram for determinants of order 3, but not for higher-order determinants.)

### EXAMPLE 8.1

- (a) Because the determinant of order 1 is the scalar itself, we have:

$$\det(27) = 27, \quad \det(-7) = -7, \quad \det(t - 3) = t - 3$$

(b)

$$\begin{vmatrix} 5 & 3 \\ 4 & 6 \end{vmatrix} = 5(6) - 3(4) = 30 - 12 = 18, \quad \begin{vmatrix} 3 & 2 \\ -5 & 7 \end{vmatrix} = 21 + 10 = 31$$

## Application to Linear Equations

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Consider two linear equations in two unknowns, say

$$a_1x + b_1y = c_1$$

$$a_2x + b_2y = c_2$$

Let  $D = a_1 b_2 - a_2 b_1$ , the determinant of the matrix of coefficients. Then the system has a unique solution if and only if  $D \neq 0$ . In such a case, the unique solution may be expressed completely in terms of determinants as follows:

$$x = \frac{N_x}{D} = \frac{b_2 c_1 - b_1 c_2}{a_1 b_2 - a_2 b_1} = \frac{\begin{vmatrix} c_1 & b_1 \\ c_2 & b_2 \end{vmatrix}}{\begin{vmatrix} a_1 & b_1 \\ a_2 & b_2 \end{vmatrix}}, \quad y = \frac{N_y}{D} = \frac{a_1 c_2 - a_2 c_1}{a_1 b_2 - a_2 b_1} = \frac{\begin{vmatrix} a_1 & c_1 \\ a_2 & c_2 \end{vmatrix}}{\begin{vmatrix} a_1 & b_1 \\ a_2 & b_2 \end{vmatrix}}$$

Here  $D$  appears in the denominator of both quotients. The numerators  $N_x$  and  $N_y$  of the quotients for  $x$  and  $y$ , respectively, can be obtained by substituting the column of constant terms in place of the column of coefficients of the given unknown in the matrix of coefficients. On the other hand, if  $D = 0$ , then the system may have no solution or more than one solution.

**EXAMPLE 8.2** Solve by determinants the system

$$\begin{cases} 4x - 3y = 15 \\ 2x + 5y = 1 \end{cases}$$

First find the determinant  $D$  of the matrix of coefficients:

$$D = \begin{vmatrix} 4 & -3 \\ 2 & 5 \end{vmatrix} = 4(5) - (-3)(2) = 20 + 6 = 26$$

Because  $D \neq 0$ , the system has a unique solution. To obtain the numerators  $N_x$  and  $N_y$ , simply replace, in the matrix of coefficients, the coefficients of  $x$  and  $y$ , respectively, by the constant terms, and then take their determinants:

$$N_x = \begin{vmatrix} 15 & -3 \\ 1 & 5 \end{vmatrix} = 75 + 3 = 78 \quad N_y = \begin{vmatrix} 4 & 15 \\ 2 & 1 \end{vmatrix} = 4 - 30 = -26$$

Then the unique solution of the system is

$$x = \frac{N_x}{D} = \frac{78}{26} = 3, \quad y = \frac{N_y}{D} = \frac{-26}{26} = -1$$

## 8.3 Determinants of Order 3

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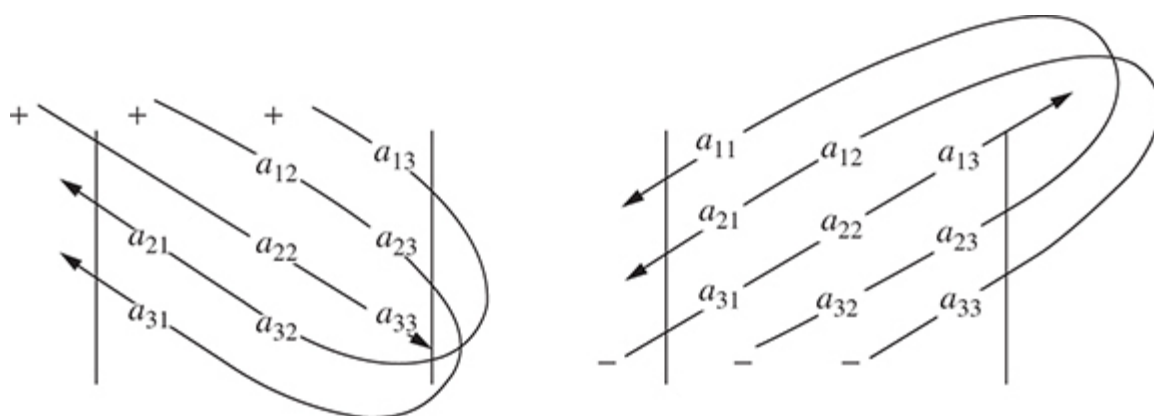
Consider an arbitrary  $3 \times 3$  matrix  $A = [a_{ij}]$ . The determinant of  $A$  is defined as follows:

$$\det(A) = \begin{vmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{vmatrix} = a_{11}a_{22}a_{33} + a_{12}a_{23}a_{31} + a_{13}a_{21}a_{32} - a_{13}a_{22}a_{31} - a_{12}a_{21}a_{33} - a_{11}a_{23}a_{32}$$

Observe that there are six products, each product consisting of three elements of the original matrix. Three of the products are plus-labeled (keep their sign) and three of the products are minus-labeled (change their sign).

The diagrams in [Fig. 8-1](#) may help us to remember the above six products in  $\det(A)$ . That is, the determinant is equal to the sum of the products of the elements along the three plus-labeled arrows in [Fig. 8-1](#) plus the sum of the negatives of the products of the elements along the three minus-labeled arrows. We emphasize that there are no such

diagrammatic devices with which to remember determinants of higher order.



**Figure 8-1**

### EXAMPLE 8.3

Let  $A = \begin{bmatrix} 2 & 1 & 1 \\ 0 & 5 & -2 \\ 1 & -3 & 4 \end{bmatrix}$  and  $B = \begin{bmatrix} 3 & 2 & 1 \\ -4 & 5 & -1 \\ 2 & -3 & 4 \end{bmatrix}$ . Find  $\det(A)$  and  $\det(B)$ .

Use the diagrams in [Fig. 8-1](#):

$$\begin{aligned} \det(A) &= 2(5)(4) + 1(-2)(1) + 1(-3)(0) - 1(5)(1) - (-3)(-2)(2) - 4(1)(0) \\ &= 40 - 2 + 0 - 5 - 12 - 0 = 21 \end{aligned}$$

$$\det(B) = 60 - 4 + 12 - 10 - 9 + 32 = 81$$

### Alternative Form for a Determinant of Order 3

The determinant of the  $3 \times 3$  matrix  $A = [a_{ij}]$  may be rewritten as follows:

$$\begin{aligned} \det(A) &= a_{11}(a_{22}a_{33} - a_{23}a_{32}) - a_{12}(a_{21}a_{33} - a_{23}a_{31}) + a_{13}(a_{21}a_{32} - a_{22}a_{31}) \\ &= a_{11} \begin{vmatrix} a_{22} & a_{23} \\ a_{32} & a_{33} \end{vmatrix} - a_{12} \begin{vmatrix} a_{21} & a_{23} \\ a_{31} & a_{33} \end{vmatrix} + a_{13} \begin{vmatrix} a_{21} & a_{22} \\ a_{31} & a_{32} \end{vmatrix} \end{aligned}$$

which is a linear combination of three determinants of order 2 whose coefficients (with alternating signs) form the first

row of the given matrix. This linear combination may be indicated in the form

$$a_{11} \begin{vmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{vmatrix} - a_{12} \begin{vmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{vmatrix} + a_{13} \begin{vmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{vmatrix}$$

Note that each  $2 \times 2$  matrix can be obtained by deleting, in the original matrix, the row and column containing its coefficient.

#### EXAMPLE 8.4

$$\begin{vmatrix} 1 & 2 & 3 \\ 4 & -2 & 3 \\ 0 & 5 & -1 \end{vmatrix} = 1 \begin{vmatrix} 1 & 2 & 3 \\ 4 & -2 & 3 \\ 0 & 5 & -1 \end{vmatrix} - 2 \begin{vmatrix} 1 & 2 & 3 \\ 4 & -2 & 3 \\ 0 & 5 & -1 \end{vmatrix} + 3 \begin{vmatrix} 1 & 2 & 3 \\ 4 & -2 & 3 \\ 0 & 5 & -1 \end{vmatrix}$$

$$= 1 \begin{vmatrix} -2 & 3 \\ 5 & -1 \end{vmatrix} - 2 \begin{vmatrix} 4 & 3 \\ 0 & -1 \end{vmatrix} + 3 \begin{vmatrix} 4 & -2 \\ 0 & 5 \end{vmatrix}$$

$$= 1(2 - 15) - 2(-4 + 0) + 3(20 + 0) = -13 + 8 + 60 = 55$$

## 8.4 Permutations

A permutation  $\sigma$  of the set  $\{1, 2, \dots, n\}$  is a one-to-one mapping of the set onto itself or, equivalently, a rearrangement of the numbers  $1, 2, \dots, n$ . Such a permutation  $\sigma$  is denoted by

$$\sigma = \begin{pmatrix} 1 & 2 & \dots & n \\ j_1 & j_2 & \dots & j_n \end{pmatrix} \quad \text{or} \quad \sigma = j_1 j_2 \dots j_n, \quad \text{where } j_i = \sigma(i)$$

The set of all such permutations is denoted by  $S_n$ , and the number of such permutations is  $n!$ . If  $\sigma \in S_n$ , then the inverse mapping  $\sigma^{-1} \in S_n$ ; and if  $\sigma, \tau \in S_n$ , then the composition mapping  $\sigma \circ \tau \in S_n$ . Also, the identity mapping  $\varepsilon = \sigma \circ \sigma^{-1} \in S_n$ . (In fact,  $\varepsilon = 123 \dots n$ .)

### EXAMPLE 8.5

- (a) There are  $2! = 2 \cdot 1 = 2$  permutations in  $S_2$ ; they are 12 and 21.
- (b) There are  $3! = 3 \cdot 2 \cdot 1 = 6$  permutations in  $S_3$ ; they are 123, 132, 213, 231, 312, 321.

### Sign (Parity) of a Permutation

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Consider an arbitrary permutation  $\sigma$  in  $S_n$ , say  $\sigma = j_1 j_2 \dots j_n$ . We say  $\sigma$  is an even or odd permutation according to whether there is an even or odd number of inversions in  $\sigma$ . By an *inversion* in  $\sigma$  we mean a pair of integers  $(i, k)$  such that  $i > k$ , but  $i$  precedes  $k$  in  $\sigma$ . We then define the sign or parity of  $\sigma$ , written  $\text{sgn } \sigma$ , by

$$\text{sgn } \sigma = \begin{cases} 1 & \text{if } \sigma \text{ is even} \\ -1 & \text{if } \sigma \text{ is odd} \end{cases}$$

### EXAMPLE 8.6

- (a) Find the sign of  $\sigma = 35142$  in  $S_5$ .

For each element  $k$ , we count the number of elements  $i$  such that  $i > k$  and  $i$  precedes  $k$  in  $\sigma$ . There are

- 2 numbers (3 and 5) greater than and preceding 1,
- 3 numbers (3; 5; and 4) greater than and preceding 2,
- 1 number (5) greater than and preceding 4

(There are no numbers greater than and preceding either 3 or 5.) Because there are, in all, six inversions,  $\sigma$  is even and  $\text{sgn } \sigma = 1$ .

(b) The identity permutation  $\varepsilon = 123\dots n$  is even because there are no inversions in  $\varepsilon$ .

(c) In  $S_2$ , the permutation 12 is even and 21 is odd. In  $S_3$ , the permutations 123, 231, 312 are even and the permutations 132, 213, 321 are odd.

(d) Let  $\tau$  be the permutation that interchanges two numbers  $i$  and  $j$  and leaves the other numbers fixed. That is,

$$\tau(i) = j, \quad \tau(j) = i, \quad \tau(k) = k, \quad \text{where } k \neq i, j$$

We call  $\tau$  a *transposition*. If  $i < j$ , then there are  $2(j - i) - 1$  inversions in  $\tau$ , and hence, the transposition  $\tau$  is odd.

**Remark:** One can show that, for any  $n$ , half of the permutations in  $S_n$  are even and half of them are odd. For example, 3 of the 6 permutations in  $S_3$  are even, and 3 are odd.

## 8.5 Determinants of Arbitrary Order

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Let  $A = [a_{ij}]$  be a square matrix of order  $n$  over a field  $K$ .

Consider a product of  $n$  elements of  $A$  such that one and only one element comes from each row and one and only one element comes from each column. Such a product can be written in the form

$$a_{1j_1} a_{2j_2} \cdots a_{nj_n}$$

that is, where the factors come from successive rows, and so the first subscripts are in the natural order  $1, 2, \dots, n$ . Now

because the factors come from different columns, the sequence of second subscripts forms a permutation  $\sigma = j_1 j_2 \dots j_n$  in  $S_n$ . Conversely, each permutation in  $S_n$  determines a product of the above form. Thus, the matrix  $A$  contains  $n!$  such products.

**DEFINITION 8.1:** The determinant of  $A = [a_{ij}]$ , denoted by  $\det(A)$  or  $|A|$ , is the sum of all the above  $n!$  products, where each such product is multiplied by  $\operatorname{sgn} \sigma$ . That is,

$$|A| = \sum_{\sigma} (\operatorname{sgn} \sigma) a_{1j_1} a_{2j_2} \cdots a_{nj_n}$$

or

$$|A| = \sum_{\sigma \in S_n} (\operatorname{sgn} \sigma) a_{1\sigma(1)} a_{2\sigma(2)} \cdots a_{n\sigma(n)}$$

The determinant of the  $n$ -square matrix  $A$  is said to be of order  $n$ .

The next example shows that the above definition agrees with the previous definition of determinants of orders 1, 2, and 3.

### EXAMPLE 8.7

- (a) Let  $A = [a_{11}]$  be a  $1 \times 1$  matrix. Because  $S_1$  has only one permutation, which is even,  $\det(A) = a_{11}$ , the number itself.
- (b) Let  $A = [a_{ij}]$  be a  $2 \times 2$  matrix. In  $S_2$ , the permutation 12 is even and the permutation 21 is odd. Hence,

$$\det(A) = \begin{vmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{vmatrix} = a_{11}a_{22} - a_{12}a_{21}$$

(c) Let  $A = [a_{ij}]$  be a  $3 \times 3$  matrix. In  $S_3$ , the permutations 123, 231, 312 are even, and the permutations 321, 213, 132 are odd. Hence,

$$\det(A) = \begin{vmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{vmatrix} = a_{11}a_{22}a_{33} + a_{12}a_{23}a_{31} + a_{13}a_{21}a_{32} - a_{13}a_{22}a_{31} - a_{12}a_{21}a_{33} - a_{11}a_{23}a_{32}$$

**Remark:** As  $n$  increases, the number of terms in the determinant becomes astronomical. Accordingly, we use indirect methods to evaluate determinants rather than the definition of the determinant. In fact, we prove a number of properties about determinants that will permit us to shorten the computation considerably. In particular, we show that a determinant of order  $n$  is equal to a linear combination of determinants of order  $n - 1$ , as in the case  $n = 3$  above.

## 8.6 Properties of Determinants

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We now list basic properties of the determinant.

**THEOREM 8.1:** The determinant of a matrix  $A$  and its transpose  $A^T$  are equal; that is,  $|A| = |A^T|$ .

By this theorem (proved in [Problem 8.22](#)), any theorem about the determinant of a matrix  $A$  that concerns the rows of  $A$  will have an analogous theorem concerning the columns of  $A$ .

The next theorem (proved in [Problem 8.24](#)) gives certain cases for which the determinant can be obtained immediately.

**THEOREM 8.2:** Let  $A$  be a square matrix.

- (i) If  $A$  has a row (column) of zeros, then  $|A| = 0$ .

- (ii) If  $A$  has two identical rows (columns), then  $|A| = 0$ .
- (iii) If  $A$  is triangular (i.e.,  $A$  has zeros above or below the diagonal), then  $|A| =$  product of diagonal elements. Thus, in particular,  $|I| = 1$ , where  $I$  is the identity matrix.

The next theorem (proved in Problems 8.23 and 8.25) shows how the determinant of a matrix is affected by the elementary row and column operations.

**THEOREM 8.3:** Suppose  $B$  is obtained from  $A$  by an elementary row (column) operation.

- (i) If two rows (columns) of  $A$  were interchanged, then  $|B| = -|A|$ .
- (ii) If a row (column) of  $A$  were multiplied by a scalar  $k$ , then  $|B| = k|A|$ .
- (iii) If a multiple of a row (column) of  $A$  were added to another row (column) of  $A$ , then  $|B| = |A|$ .

## **Major Properties of Determinants**

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We now state two of the most important and useful theorems on determinants.

**THEOREM 8.4:** The determinant of a product of two matrices  $A$  and  $B$  is the product of their determinants; that is,

$$\det(AB) = \det(A) \det(B)$$

The above theorem says that the determinant is a multiplicative function.

**THEOREM 8.5:** Let  $A$  be a square matrix. Then the following are equivalent:

- (i)  $A$  is invertible; that is,  $A$  has an inverse  $A^{-1}$ .
- (ii)  $AX = 0$  has only the zero solution.
- (iii) The determinant of  $A$  is not zero; that is,  $\det(A) \neq 0$ .

**Remark:** Depending on the author and the text, a nonsingular matrix  $A$  is defined to be an invertible matrix  $A$ , or a matrix  $A$  for which  $|A| \neq 0$ , or a matrix  $A$  for which  $AX = 0$  has only the zero solution. The above theorem shows that all such definitions are equivalent.

We will prove Theorems 8.4 and 8.5 (in Problems 8.29 and 8.28, respectively) using the theory of elementary matrices and the following lemma (proved in [Problem 8.26](#)), which is a special case of [Theorem 8.4](#).

**LEMMA 8.6:** Let  $E$  be an elementary matrix. Then, for any matrix  $A$ ,  $|EA| = |E||A|$ .

Recall that matrices  $A$  and  $B$  are similar if there exists a nonsingular matrix  $P$  such that  $B = P^{-1}AP$ . Using the multiplicative property of the determinant ([Theorem 8.4](#)), one can easily prove ([Problem 8.31](#)) the following theorem.

**THEOREM 8.7:** Suppose  $A$  and  $B$  are similar matrices. Then  $|A| = |B|$ .

## 8.7 Minors and Cofactors

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Consider an  $n$ -square matrix  $A = [a_{ij}]$ . Let  $M_{ij}$  denote the  $(n - 1)$ -square submatrix of  $A$  obtained by deleting its  $i$ th row and  $j$ th column. The determinant  $|M_{ij}|$  is called the *minor* of

the element  $a_{ij}$  of  $A$ , and we define the *cofactor* of  $a_{ij}$ , denoted by  $A_{ij}$ , to be the “signed” minor:

$$A_{ij} = (-1)^{i+j} |M_{ij}|$$

Note that the “signs”  $(-1)^{i+j}$  accompanying the minors form a chessboard pattern with +’s on the main diagonal:

$$\begin{bmatrix} + & - & + & - & \dots \\ - & + & - & + & \dots \\ + & - & + & - & \dots \\ \dots & \dots & \dots & \dots & \dots \end{bmatrix}$$

We emphasize that  $M_{ij}$  denotes a matrix, whereas  $A_{ij}$  denotes a scalar.

**Remark:** The sign  $(-1)^{i+j}$  of the cofactor  $A_{ij}$  is frequently obtained using the checkerboard pattern. Specifically, beginning with + and alternating signs:

$$+, -, +, -, \dots,$$

count from the main diagonal to the appropriate square.

**EXAMPLE 8.8** Let  $A = \begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \\ 7 & 8 & 9 \end{bmatrix}$ . Find the following minors and cofactors: (a)  $|M_{23}|$  and  $A_{23}$ , (b)  $|M_{31}|$  and  $A_{31}$ .

$$(a) \quad |M_{23}| = \begin{vmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \\ 7 & 8 & 9 \end{vmatrix} = \begin{vmatrix} 1 & 2 \\ 7 & 8 \end{vmatrix} = 8 - 14 = -6, \text{ and so } A_{23} = (-1)^{2+3} |M_{23}| = -(-6) = 6$$

$$(b) \quad |M_{31}| = \begin{vmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \\ 7 & 8 & 9 \end{vmatrix} = \begin{vmatrix} 2 & 3 \\ 5 & 6 \end{vmatrix} = 12 - 15 = -3, \text{ and so } A_{31} = (-1)^{1+3}|M_{31}| = +(-3) = -3$$

## Laplace Expansion

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The following theorem (proved in Problem 8.32) holds.

**THEOREM 8.8:** (Laplace) The determinant of a square matrix  $A = [a_{ij}]$  is equal to the sum of the products obtained by multiplying the elements of any row (column) by their respective cofactors:

$$|A| = a_{i1}A_{i1} + a_{i2}A_{i2} + \cdots + a_{in}A_{in} = \sum_{j=1}^n a_{ij}A_{ij}$$

$$|A| = a_{1j}A_{1j} + a_{2j}A_{2j} + \cdots + a_{nj}A_{nj} = \sum_{i=1}^n a_{ij}A_{ij}$$

The above formulas for  $|A|$  are called the *Laplace expansions* of the determinant of  $A$  by the  $i$ th row and the  $j$ th column. Together with the elementary row (column) operations, they offer a method of simplifying the computation of  $|A|$ , as described below.

## 8.8 Evaluation of Determinants

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The following algorithm reduces the evaluation of a determinant of order  $n$  to the evaluation of a determinant of order  $n - 1$ .

**ALGORITHM 8.1:** (Reduction of the order of a determinant) The input is a nonzero  $n$ -square matrix  $A = [a_{ij}]$  with  $n > 1$ .

**Step 1.** Choose an element  $a_{ij} = 1$  or, if lacking,  $a_{ij} \neq 0$ .

**Step 2.** Using  $a_{ij}$  as a pivot, apply elementary row (column) operations to put 0's in all the other positions in the column (row) containing  $a_{ij}$ .

**Step 3.** Expand the determinant by the column (row) containing  $a_{ij}$ .

The following remarks are in order.

**Remark 1:** Algorithm 8.1 is usually used for determinants of order 4 or more. With determinants of order less than 4, one uses the specific formulas for the determinant.

**Remark 2:** Gaussian elimination or, equivalently, repeated use of Algorithm 8.1 together with row interchanges can be used to transform a matrix  $A$  into an upper triangular matrix whose determinant is the product of its diagonal entries. However, one must keep track of the number of row interchanges, because each row interchange changes the sign of the determinant.

**EXAMPLE 8.9** Use Algorithm 8.1 to find the determinant of

$$A = \begin{bmatrix} 5 & 4 & 2 & 1 \\ 2 & 3 & 1 & -2 \\ -5 & -7 & -3 & 9 \\ 1 & -2 & -1 & 4 \end{bmatrix}.$$

Use  $a_{23} = 1$  as a pivot to put 0's in the other positions of the third column; that is, apply the row operations "Replace  $R_1$  by  $-2R_2 + R_1$ ," "Replace  $R_3$  by  $3R_2 + R_3$ ," and "Replace  $R_4$  by  $R_2 + R_4$ ." By [Theorem 8.3](#) (iii), the value of the determinant does not change under these operations. Thus,

$$|A| = \begin{vmatrix} 5 & 4 & 2 & 1 \\ 2 & 3 & 1 & -2 \\ -5 & -7 & -3 & 9 \\ 1 & -2 & -1 & 4 \end{vmatrix} = \begin{vmatrix} 1 & -2 & 0 & 5 \\ 2 & 3 & 1 & -2 \\ 1 & 2 & 0 & 3 \\ 3 & 1 & 0 & 2 \end{vmatrix}$$

Now expand by the third column. Specifically, neglect all terms that contain 0 and use the fact that the sign of the minor  $M_{23}$  is  $(-1)^{2+3} = 1$ . Thus,

$$|A| = - \begin{vmatrix} 1 & 2 & 0 & 5 \\ 2 & 3 & 1 & -2 \\ 1 & 2 & 0 & 3 \\ 3 & 1 & 0 & 2 \end{vmatrix} = - \begin{vmatrix} 1 & -2 & 5 \\ 1 & 2 & 3 \\ 3 & 1 & 2 \end{vmatrix} = -(4 - 18 + 5 - 30 - 3 + 4) = -(-38) = 38$$

## 8.9 Classical Adjoint

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Let  $A = [a_{ij}]$  be an  $n \times n$  matrix over a field  $K$  and let  $A_{ij}$  denote the cofactor of  $a_{ij}$ . The *classical adjoint* of  $A$ , denoted by  $\text{adj } A$ , is the transpose of the matrix of cofactors of  $A$ . Namely,

$$\text{adj } A = [A_{ij}]^T$$

We say “classical adjoint” instead of simply “adjoint” because the term “adjoint” is currently used for an entirely different concept.

**EXAMPLE 8.10**  $A = \begin{bmatrix} 2 & 3 & -4 \\ 0 & -4 & 2 \\ 1 & -1 & 5 \end{bmatrix}$  The cofactors of the nine elements of  $A$  follow:

$$\begin{aligned}
A_{11} &= + \begin{vmatrix} -4 & 2 \\ -1 & 5 \end{vmatrix} = -18, & A_{12} &= - \begin{vmatrix} 0 & 2 \\ 1 & 5 \end{vmatrix} = 2, & A_{13} &= + \begin{vmatrix} 0 & -4 \\ 1 & -1 \end{vmatrix} = 4 \\
A_{21} &= - \begin{vmatrix} 3 & -4 \\ -1 & 5 \end{vmatrix} = -11, & A_{22} &= + \begin{vmatrix} 2 & -4 \\ 1 & 5 \end{vmatrix} = 14, & A_{23} &= - \begin{vmatrix} 2 & 3 \\ 1 & -1 \end{vmatrix} = 5 \\
A_{31} &= + \begin{vmatrix} 3 & -4 \\ -4 & 2 \end{vmatrix} = -10, & A_{32} &= - \begin{vmatrix} 2 & -4 \\ 0 & 2 \end{vmatrix} = -4, & A_{33} &= + \begin{vmatrix} 2 & 3 \\ 0 & -4 \end{vmatrix} = -8
\end{aligned}$$

The transpose of the above matrix of cofactors yields the classical adjoint of  $A$ ; that is,

$$\text{adj } A = \begin{bmatrix} -18 & -11 & -10 \\ 2 & 14 & -4 \\ 4 & 5 & -8 \end{bmatrix}$$

The following theorem (proved in [Problem 8.34](#)) holds.

**THEOREM 8.9:** Let  $A$  be any square matrix. Then

$$A(\text{adj } A) = (\text{adj } A)A = |A|I$$

where  $I$  is the identity matrix. Thus, if  $|A| \neq 0$ ,

$$A^{-1} = \frac{1}{|A|} (\text{adj } A)$$

**EXAMPLE 8.11** Let  $A$  be the matrix in [Example 8.10](#). We have

$$\det(A) = -40 + 6 + 0 - 16 + 4 + 0 = -46$$

Thus,  $A$  does have an inverse, and, by [Theorem 8.9](#),

$$A^{-1} = \frac{1}{|A|} (\text{adj } A) = -\frac{1}{46} \begin{bmatrix} -18 & -11 & -10 \\ 2 & 14 & -4 \\ 4 & 5 & -8 \end{bmatrix} = \begin{bmatrix} \frac{9}{23} & \frac{11}{46} & \frac{5}{23} \\ -\frac{1}{23} & -\frac{7}{23} & \frac{2}{23} \\ -\frac{2}{23} & -\frac{5}{46} & \frac{4}{23} \end{bmatrix}$$

## 8.10 Applications to Linear Equations, Cramer's Rule

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Consider a system  $AX = B$  of  $n$  linear equations in  $n$  unknowns. Here  $A = [a_{ij}]$  is the (square) matrix of coefficients and  $B = [b_i]$  is the column vector of constants. Let  $A_i$  be the matrix obtained from  $A$  by replacing the  $i$ th column of  $A$  by the column vector  $B$ . Furthermore, let

$$D = \det(A), \quad N_1 = \det(A_1), \quad N_2 = \det(A_2), \quad \dots, \quad N_n = \det(A_n)$$

The fundamental relationship between determinants and the solution of the system  $AX = B$  follows.

**THEOREM 8.10:** The (square) system  $AX = B$  has a solution if and only if  $D \neq 0$ . In this case, the unique solution is given by

$$x_1 = \frac{N_1}{D}, \quad x_2 = \frac{N_2}{D}, \quad \dots, \quad x_n = \frac{N_n}{D}$$

The above theorem (proved in [Problem 8.10](#)) is known as *Cramer's rule* for solving systems of linear equations. We emphasize that the theorem only refers to a system with the same number of equations as unknowns, and that it only gives the solution when  $D \neq 0$ . In fact, if  $D = 0$ , the theorem does not tell us whether or not the system has a solution. However, in the case of a homogeneous system, we have the following useful result (to be proved in [Problem 8.54](#)).

**THEOREM 8.11:** A square homogeneous system  $AX = 0$  has a nonzero solution if and only if  $D = |A| = 0$ .

**EXAMPLE 8.12** Solve the system using determinants

$$\begin{cases} x + y + z = 5 \\ x - 2y - 3z = -1 \\ 2x + y - z = 3 \end{cases}$$

First compute the determinant  $D$  of the matrix of coefficients:

$$D = \begin{vmatrix} 1 & 1 & 1 \\ 1 & -2 & -3 \\ 2 & 1 & -1 \end{vmatrix} = 2 - 6 + 1 + 4 + 3 + 1 = 5$$

Because  $D \neq 0$ , the system has a unique solution. To compute  $N_x$ ,  $N_y$ ,  $N_z$ , we replace, respectively, the coefficients of  $x$ ,  $y$ ,  $z$  in the matrix of coefficients by the constant terms. This yields

$$N_x = \begin{vmatrix} 5 & 1 & 1 \\ -1 & -2 & -3 \\ 3 & 1 & -1 \end{vmatrix} = 20, \quad N_y = \begin{vmatrix} 1 & 5 & 1 \\ 1 & -1 & -3 \\ 2 & 3 & -1 \end{vmatrix} = -10, \quad N_z = \begin{vmatrix} 1 & 1 & 5 \\ 1 & -2 & -1 \\ 2 & 1 & 3 \end{vmatrix} = 15$$

Thus, the unique solution of the system is  $x = N_x/D = 4$ ,  $y = N_y/D = -2$ ,  $z = N_z/D = 3$ ; that is, the vector  $u = (4, -2, 3)$ .

## 8.11 Submatrices, Minors, Principal Minors

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Let  $A = [a_{ij}]$  be a square matrix of order  $n$ . Consider any  $r$  rows and  $r$  columns of  $A$ . That is, consider any set  $I = (i_1, i_2, \dots, i_r)$  of  $r$  row indices and any set  $J = (j_1, j_2, \dots, j_r)$  of  $r$  column indices. Then  $I$  and  $J$  define an  $r \times r$  submatrix of  $A$ , denoted by  $A(I; J)$ , obtained by deleting the rows and columns of  $A$  whose subscripts do not belong to  $I$  or  $J$ , respectively. That is,

$$A(I; J) = [a_{st} : s \in I, t \in J]$$

The determinant  $|A(I; J)|$  is called a *minor* of  $A$  of order  $r$  and

$$(-1)^{i_1+i_2+\dots+i_r+j_1+j_2+\dots+j_r} |A(I; J)|$$

is the corresponding signed minor. (Note that a minor of order  $n - 1$  is a minor in the sense of Section 8.7, and the corresponding signed minor is a cofactor.) Furthermore, if  $I'$  and  $J'$  denote, respectively, the remaining row and column indices, then

$$|A(I'; J')|$$

denotes the *complementary minor*, and its sign (Problem 8.74) is the same sign as the minor.

**EXAMPLE 8.13** Let  $A = [a_{ij}]$  be a 5-square matrix, and let  $I = \{1, 2, 4\}$  and  $J = \{2, 3, 5\}$ . Then  $I' = \{3, 5\}$  and  $J' = \{1, 4\}$ , and the corresponding minor  $|M|$  and complementary minor  $|M'|$  are as follows:

$$|M| = |A(I; J)| = \begin{vmatrix} a_{12} & a_{13} & a_{15} \\ a_{22} & a_{23} & a_{25} \\ a_{42} & a_{43} & a_{45} \end{vmatrix} \quad \text{and} \quad |M'| = |A(I'; J')| = \begin{vmatrix} a_{31} & a_{34} \\ a_{51} & a_{54} \end{vmatrix}$$

Because  $1 + 2 + 4 + 2 + 3 + 5 = 17$  is odd,  $-|M|$  is the signed minor, and  $-|M'|$  is the signed complementary minor.

## Principal Minors

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A minor is *principal* if the row and column indices are the same, or equivalently, if the diagonal elements of the minor come from the diagonal of the matrix. We note that the sign

of a principal minor is always +1, because the sum of the row and identical column subscripts must always be even.

**EXAMPLE 8.14**  $A = \begin{bmatrix} 1 & 2 & -1 \\ 3 & 5 & 4 \\ -3 & 1 & -2 \end{bmatrix}$ . Find the sums  $C_1$ ,  $C_2$ , and  $C_3$  of the principal minors of  $A$  of orders 1, 2, and 3, respectively.

(a) There are three principal minors of order 1. These are

$$|1| = 1, \quad |5| = 5, \quad |-2| = -2, \quad \text{and so} \quad C_1 = 1 + 5 - 2 = 4$$

Note that  $C_1$  is simply the trace of  $A$ . Namely,  $C_1 = \text{tr}(A)$ .

(b) There are three ways to choose two of the three diagonal elements, and each choice gives a minor of order 2. These are

$$\begin{vmatrix} 1 & 2 \\ 3 & 5 \end{vmatrix} = -1, \quad \begin{vmatrix} 1 & -1 \\ -3 & -2 \end{vmatrix} = 1, \quad \begin{vmatrix} 5 & 4 \\ 1 & -2 \end{vmatrix} = -14$$

(Note that these minors of order 2 are the cofactors  $A_{33}$ ,  $A_{22}$ , and  $A_{11}$  of  $A$ , respectively.) Thus,

$$C_2 = -1 + 1 - 14 = -14$$

(c) There is only one way to choose three of the three diagonal elements. Thus, the only minor of order 3 is the determinant of  $A$  itself. Thus,

$$C_3 = |A| = -10 - 24 - 3 - 15 - 4 + 12 = -44$$

## 8.12 Block Matrices and Determinants

---

The following theorem (proved in [Problem 8.36](#)) is one main result of this section.

**THEOREM 8.12:** Suppose  $M$  is an upper (lower) triangular block matrix with the diagonal blocks  $A_1, A_2, \dots, A_n$ . Then

$$\det(M) = \det(A_1) \det(A_2) \dots \det(A_n)$$

**EXAMPLE 8.15** Find  $|M|$  where  $M = \left[ \begin{array}{cc|cc|cc} 2 & 3 & 4 & 7 & 8 \\ -1 & 5 & 3 & 2 & 1 \\ 0 & 0 & 2 & -1 & 5 \\ 0 & 0 & 3 & -1 & 4 \\ 0 & 0 & 5 & 2 & 6 \end{array} \right]$

Note that  $M$  is an upper triangular block matrix. Evaluate the determinant of each diagonal block:

$$\begin{vmatrix} 2 & 3 \\ -1 & 5 \end{vmatrix} = 10 + 3 = 13, \quad \begin{vmatrix} 2 & 1 & 5 \\ 3 & -1 & 4 \\ 5 & 2 & 6 \end{vmatrix} = -12 + 20 + 30 + 25 - 16 - 18 = 29$$

Then  $|M| = 13 (29) = 377$ .

Another result of this section follows:

**THEOREM 8.13:** Consider the block matrix  $M = \begin{bmatrix} A & B \\ C & D \end{bmatrix}$  where  $A$  is nonsingular,  $A$  is  $r \times r$  and  $D$  is  $s \times s$ . Then  $\det(M) = \det(A) \det(D - CA^{-1}B)$

**Proof:** Follows from the fact that

$$M = \begin{bmatrix} I & 0 \\ CA^{-1} & I \end{bmatrix} \begin{bmatrix} A & B \\ 0 & D - CA^{-1}B \end{bmatrix} \text{ and } \text{Theorem 8.12}.$$

## 8.13 Determinants and Volume

---

Determinants are related to the notions of area and volume as follows. Let  $u_1, u_2, \dots, u_n$  be vectors in  $\mathbf{R}^n$ . Let  $S$  be the (solid) parallelopiped determined by the vectors; that is,

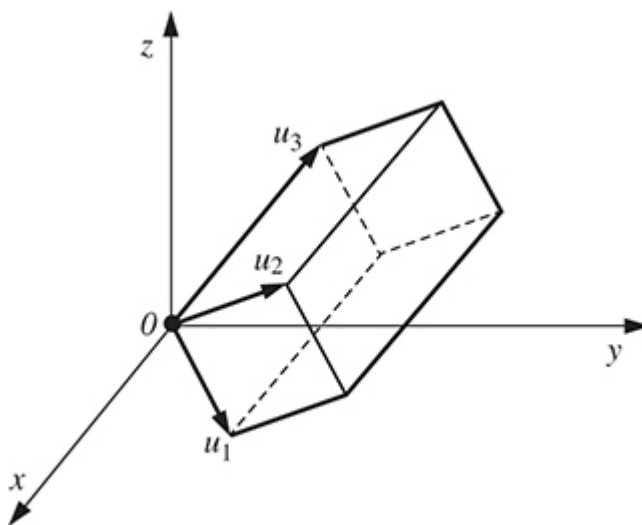
$$S = \{a_1 u_1 + a_2 u_2 + \cdots + a_n u_n : 0 \leq a_i \leq 1 \text{ for } i = 1, \dots, n\}$$

(When  $n = 2$ ,  $S$  is a parallelogram.) Let  $V(S)$  denote the volume of  $S$  (or area of  $S$  when  $n = 2$ ). Then

$$V(S) = \text{absolute value of } \det(A)$$

where  $A$  is the matrix with rows  $u_1, u_2, \dots, u_n$ . In general,  $V(S) = 0$  if and only if the vectors  $u_1, \dots, u_n$  do not form a coordinate system for  $\mathbf{R}^n$  (i.e., if and only if the vectors are linearly dependent).

**EXAMPLE 8.16** Let  $u_1 = (1, 1, 0)$ ,  $u_2 = (1, 1, 1)$ ,  $u_3 = (0, 2, 3)$ . Find the volume  $V(S)$  of the parallelo-piped  $S$  in  $\mathbf{R}^3$  (Fig. 8-2) determined by the three vectors.



## Figure 8-2

Evaluate the determinant of the matrix whose rows are  $u_1, u_2, u_3$ :

$$\begin{vmatrix} 1 & 1 & 0 \\ 1 & 1 & 1 \\ 0 & 2 & 3 \end{vmatrix} = 3 + 0 + 0 - 0 - 2 - 3 = -2$$

Hence,  $V(S) = |-2| = 2$ .

## 8.14 Determinant of a Linear Operator

---

Let  $F$  be a linear operator on a vector space  $V$  with finite dimension. Let  $A$  be the matrix representation of  $F$  relative to some basis  $S$  of  $V$ . Then we define the determinant of  $F$ , written  $\det(F)$ , by

$$\det(F) = |A|$$

If  $B$  were another matrix representation of  $F$  relative to another basis  $S'$  of  $V$ , then  $A$  and  $B$  are similar matrices ([Theorem 6.7](#)) and  $|B| = |A|$  ([Theorem 8.7](#)). In other words, the above definition  $\det(F)$  is independent of the particular basis  $S$  of  $V$ . (We say that the definition is *well defined*.)

The next theorem (to be proved in [Problem 8.62](#)) follows from analogous theorems on matrices.

**THEOREM 8.14:** Let  $F$  and  $G$  be linear operators on a vector space  $V$ . Then

- (i)  $\det(F \circ G) = \det(F) \det(G)$ .
- (ii)  $F$  is invertible if and only if  $\det(F) \neq 0$ .

**EXAMPLE 8.17** Let  $F$  be the following linear operator on  $\mathbf{R}^3$  and let  $A$  be the matrix that represents  $F$  relative to the usual basis of  $\mathbf{R}^3$ :

$$F(x, y, z) = (2x - 4y + z, x - 2y + 3z, 5x + y - z) \quad \text{and} \quad A = \begin{bmatrix} 2 & -4 & 1 \\ 1 & -2 & 3 \\ 5 & 1 & -1 \end{bmatrix}$$

Then

$$\det(F) = |A| = 4 - 60 + 1 + 10 - 6 - 4 = -55$$

## 8.15 Multilinearity and Determinants

---

Let  $V$  be a vector space over a field  $K$ . Let  $A = V^n$ ; that is,  $A$  consists of all the  $n$ -tuples

$$A = (A_1, A_2, \dots, A_n)$$

where the  $A_i$  are vectors in  $V$ . The following definitions apply.

**DEFINITION 8.2:** A function  $D: A \rightarrow K$  is said to be *multilinear* if it is linear in each component:

(i) If  $A_j = B + C$ , then

$$D(A) = D(\dots, B + C, \dots) = D(\dots, B, \dots) + D(\dots, C, \dots)$$

(ii) If  $A_j = kB$ , where  $k \in K$ , then

$$D(A) = D(\dots, kB, \dots) = kD(\dots, B, \dots)$$

We also say *n-linear* for multilinear if there are  $n$  components.

**DEFINITION 8.3:** A function  $D: A \rightarrow K$  is said to be *alternating* if  $D(A) = 0$  whenever  $A$  has two identical elements:

$$D(A_1, A_2, \dots, A_n) = 0 \quad \text{whenever} \quad A_i = A_j, \quad i \neq j$$

Now let  $\mathbf{M}$  denote the set of all  $n$ -square matrices  $A$  over a field  $K$ . We may view  $A$  as an  $n$ -tuple consisting of its row vectors  $A_1, A_2, \dots, A_n$ ; that is, we may view  $A$  in the form  $A = (A_1, A_2, \dots, A_n)$ .

The following theorem (proved in [Problem 8.37](#)) characterizes the determinant function.

**THEOREM 8.15:** There exists a unique function  $D: M \rightarrow K$  such that

- (i)  $D$  is multilinear, (ii)  $D$  is alternating,
- (iii)  $D(I) = 1$ .

This function  $D$  is the determinant function; that is,  $D(A) = |A|$ , for any matrix  $A \in M$ .

## SOLVED PROBLEMS

### Computation of Determinants

**8.1.** Evaluate the determinant of each of the following matrices:

$$(a) \ A = \begin{bmatrix} 6 & 5 \\ 2 & 3 \end{bmatrix}, (b) \ B = \begin{bmatrix} 2 & -3 \\ 4 & 7 \end{bmatrix}, (c) \ C = \begin{bmatrix} 4 & -5 \\ -1 & -2 \end{bmatrix}, (d) \ D = \begin{bmatrix} t-5 & 6 \\ 3 & t+2 \end{bmatrix}$$

Use the formula  $\begin{vmatrix} a & b \\ c & d \end{vmatrix} = ad - bc$ :

$$(a) \ |A| = 6(3) - 5(2) = 18 - 10 = 8$$

$$(b) |B| = 14 + 12 = 26$$

$$(c) |C| = -8 - 5 = 13$$

$$(d) |D| = (t - 5)(t + 2) - 18 = t^2 - 3t - 10 - 18 = t^2 - 10t - 28$$

**8.2.** Evaluate the determinant of each of the following matrices:

$$A = \begin{bmatrix} 2 & 3 & 4 \\ 5 & 4 & 3 \\ 1 & 2 & 1 \end{bmatrix}, (b) \quad B = \begin{bmatrix} 1 & -2 & 3 \\ 2 & 4 & -1 \\ 1 & 5 & -2 \end{bmatrix}, (c) \quad C = \begin{bmatrix} 1 & 3 & -5 \\ 3 & -1 & 2 \\ 1 & -2 & 1 \end{bmatrix}$$

Use the diagram in [Fig. 8-1](#) to obtain the six products:

$$(a) |A| = 2(4)(1) + 3(3)(1) + 4(2)(5) - 1(4)(4) - 2(3)(2) - 1(3)(5) = 8 + 9 + 40 - 16 - 12 - 15 = 14$$

$$(b) |B| = -8 + 2 + 30 - 12 + 5 - 8 = 9$$

$$(c) |C| = -1 + 6 + 30 - 5 + 4 - 9 = 25$$

**8.3.** Compute the determinant of each of the following matrices:

$$(a) \quad A = \begin{bmatrix} 2 & 3 & 4 \\ 5 & 6 & 7 \\ 8 & 9 & 1 \end{bmatrix}, (b) \quad B = \begin{bmatrix} 4 & -6 & 8 & 9 \\ 0 & -2 & 7 & -3 \\ 0 & 0 & 5 & 6 \\ 0 & 0 & 0 & 3 \end{bmatrix}, (c) \quad C = \begin{bmatrix} \frac{1}{2} & -1 & -\frac{1}{3} \\ \frac{3}{4} & \frac{1}{2} & -1 \\ 1 & -4 & 1 \end{bmatrix}.$$

(a) One can simplify the entries by first subtracting twice the first row from the second row—that is, by applying the row operation “Replace  $R_2$  by  $2_1 + R_2$ .” Then

$$|A| = \begin{vmatrix} 2 & 3 & 4 \\ 5 & 6 & 7 \\ 8 & 9 & 1 \end{vmatrix} = \begin{vmatrix} 2 & 3 & 4 \\ 1 & 0 & -1 \\ 8 & 9 & 1 \end{vmatrix} = 0 - 24 + 36 - 0 + 18 - 3 = 27$$

(b)  $B$  is triangular, so  $|B|$  = product of the diagonal entries = 120.

(c) The arithmetic is simpler if fractions are first eliminated. Hence, multiply the first row  $R_1$  by 6 and the second row  $R_2$  by 4. Then

$$|24C| = \begin{vmatrix} 3 & -6 & -2 \\ 3 & 2 & -4 \\ 1 & -4 & 1 \end{vmatrix} = 6 + 24 + 24 + 4 - 48 + 18 = 28, \quad \text{so } |C| = \frac{28}{24} = \frac{7}{6}$$

**8.4.** Compute the determinant of each of the following matrices:

$$(a) \quad A = \begin{bmatrix} 2 & 5 & -3 & -2 \\ -2 & -3 & 2 & -5 \\ 1 & 3 & -2 & 2 \\ -1 & -6 & 4 & 3 \end{bmatrix}, \quad (b) \quad B = \begin{bmatrix} 6 & 2 & 1 & 0 & 5 \\ 2 & 1 & 1 & -2 & 1 \\ 1 & 1 & 2 & -2 & 3 \\ 3 & 0 & 2 & 3 & -1 \\ -1 & -1 & -3 & 4 & 2 \end{bmatrix}$$

(a) Use  $a_{31} = 1$  as a pivot to put 0's in the first column, by applying the row operations "Replace  $R_1$  by  $-2R_3 + R_1$ ," "Replace  $R_2$  by  $2R_3 + R_2$ ," and "Replace  $R_4$  by  $R_3 + R_4$ ." Then

$$\begin{aligned} |A| &= \begin{vmatrix} 2 & 5 & -3 & -2 \\ -2 & -3 & 2 & -5 \\ 1 & 3 & -2 & 2 \\ -1 & -6 & 4 & 3 \end{vmatrix} = \begin{vmatrix} 0 & -1 & 1 & -6 \\ 0 & 3 & -2 & -1 \\ 1 & 3 & -2 & 2 \\ 0 & -3 & 2 & 5 \end{vmatrix} = \begin{vmatrix} -1 & 1 & -6 \\ 3 & -2 & -1 \\ -3 & 2 & 5 \end{vmatrix} \\ &= 10 + 3 - 36 + 36 - 2 - 15 = -4 \end{aligned}$$

- (b) First reduce  $|B|$  to a determinant of order 4, and then to a determinant of order 3, for which we can use Fig. 8-1. First use  $c_{22} = 1$  as a pivot to put 0's in the second column, by applying the row operations "Replace  $R_1$  by  $-2R_2 + R_1$ ," "Replace  $R_3$  by  $-R_2 + R_3$ ," and "Replace  $R_5$  by  $R_2 + R_5$ ." Then

$$\begin{aligned}
 |B| &= \begin{vmatrix} 2 & 0 & -1 & 4 & 3 \\ 2 & 1 & 1 & -2 & 1 \\ -1 & 0 & 1 & 0 & 2 \\ 3 & 0 & 2 & 3 & -1 \\ 1 & 0 & -2 & 2 & 3 \end{vmatrix} = \begin{vmatrix} 2 & -1 & 4 & 3 \\ -1 & 1 & 0 & 2 \\ 3 & 2 & 3 & -1 \\ 1 & -2 & 2 & 3 \end{vmatrix} = \begin{vmatrix} 1 & 1 & 4 & 5 \\ 0 & 1 & 0 & 0 \\ 5 & 2 & 3 & -5 \\ -1 & -2 & 2 & 7 \end{vmatrix} \\
 &= \begin{vmatrix} 1 & 4 & 5 \\ 5 & 3 & -5 \\ -1 & 2 & 7 \end{vmatrix} = 21 + 20 + 50 + 15 + 10 - 140 = -34
 \end{aligned}$$

## Cofactors, Classical Adjoint, Minors, Principal Minors

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8.5. Let  $A = \begin{bmatrix} 2 & 1 & -3 & 4 \\ 5 & -4 & 7 & -2 \\ 4 & 0 & 6 & -3 \\ 3 & -2 & 5 & 2 \end{bmatrix}$ .

- Find  $A_{23}$ , the cofactor (signed minor) of 7 in  $A$ .
- Find the minor and the signed minor of the submatrix  $M = A(2, 4; 2, 3)$ .
- Find the principal minor determined by the first and third diagonal entries—that is, by  $M = A(1, 3; 1, 3)$ .
- Take the determinant of the submatrix of  $A$  obtained by deleting row 2 and column 3 (those which

contain the 7), and multiply the determinant by  $(-1)^{2+3}$ :

$$A_{23} = - \begin{vmatrix} 2 & 1 & 4 \\ 4 & 0 & -3 \\ 3 & -2 & 2 \end{vmatrix} = -(-61) = 61$$

The exponent  $2 + 3$  comes from the subscripts of  $A_{23}$ — that is, from the fact that 7 appears in row 2 and column 3.

- (b) The row subscripts are 2 and 4 and the column subscripts are 2 and 3. Hence, the minor is the determinant

$$|M| = \begin{vmatrix} a_{22} & a_{23} \\ a_{42} & a_{43} \end{vmatrix} = \begin{vmatrix} -4 & 7 \\ -2 & 5 \end{vmatrix} = -20 + 14 = -6$$

and the signed minor is  $(-1)^{2+4+2+3}|M| = -|M| = -(-6) = 6$ .

- (c) The principal minor is the determinant

$$|M| = \begin{vmatrix} a_{11} & a_{13} \\ a_{31} & a_{33} \end{vmatrix} = \begin{vmatrix} 2 & -3 \\ 4 & 6 \end{vmatrix} = 12 + 12 = 24$$

Note that now the diagonal entries of the submatrix are diagonal entries of the original matrix. Also, the sign of the principal minor is positive.

## 8.6.

Let  $B = \begin{bmatrix} 1 & 1 & 1 \\ 2 & 3 & 4 \\ 5 & 8 & 9 \end{bmatrix}$ . Find: (a)  $|B|$ , (b)  $\text{adj } B$ , (c)  $B^{-1}$  using  $\text{adj } B$ .

(a)  $|B| = 27 + 20 + 16 - 15 - 32 - 18 = -2$

- (b) Take the transpose of the matrix of cofactors:

$$\text{adj } B = \begin{bmatrix} \begin{vmatrix} 3 & 4 \\ 8 & 9 \end{vmatrix} & -\begin{vmatrix} 2 & 4 \\ 5 & 9 \end{vmatrix} & \begin{vmatrix} 2 & 3 \\ 5 & 8 \end{vmatrix} \\ -\begin{vmatrix} 1 & 1 \\ 8 & 9 \end{vmatrix} & \begin{vmatrix} 1 & 1 \\ 5 & 9 \end{vmatrix} & -\begin{vmatrix} 1 & 1 \\ 5 & 8 \end{vmatrix} \\ \begin{vmatrix} 1 & 1 \\ 3 & 4 \end{vmatrix} & -\begin{vmatrix} 1 & 1 \\ 2 & 4 \end{vmatrix} & \begin{vmatrix} 1 & 1 \\ 2 & 3 \end{vmatrix} \end{bmatrix}^T = \begin{bmatrix} -5 & 2 & 1 \\ -1 & 4 & -3 \\ 1 & -2 & 1 \end{bmatrix}^T = \begin{bmatrix} -5 & -1 & 1 \\ 2 & 4 & -2 \\ 1 & -3 & 1 \end{bmatrix}$$

(c) Because

$$|B| \neq 0, B^{-1} = \frac{1}{|B|} (\text{adj } B) = \frac{1}{-2} \begin{bmatrix} -5 & -1 & 1 \\ 2 & 4 & -2 \\ 1 & -3 & 1 \end{bmatrix} = \begin{bmatrix} \frac{5}{2} & \frac{1}{2} & -\frac{1}{2} \\ -1 & -2 & 1 \\ -\frac{1}{2} & \frac{3}{2} & -\frac{1}{2} \end{bmatrix}$$

**8.7.** Let  $A = \begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \\ 0 & 7 & 8 \end{bmatrix}$ , and let  $S_k$  denote the sum of its principal minors of order  $k$ . Find  $S_k$  for

(a)  $k = 1$ , (b)  $k = 2$ , (c)  $k = 3$ .

(a) The principal minors of order 1 are the diagonal elements. Thus,  $S_1$  is the trace of  $A$ ; that is,

$$S_1 = \text{tr}(A) = 1 + 5 + 8 = 14$$

(b) The principal minors of order 2 are the cofactors of the diagonal elements. Thus,

$$S_2 = A_{11} + A_{22} + A_{33} = \begin{vmatrix} 5 & 6 \\ 7 & 8 \end{vmatrix} + \begin{vmatrix} 1 & 3 \\ 0 & 8 \end{vmatrix} + \begin{vmatrix} 1 & 2 \\ 4 & 5 \end{vmatrix} = -2 + 8 - 3 = 3$$

(c) There is only one principal minor of order 3, the determinant of  $A$ . Then

$$S_3 = |A| = 40 + 0 + 84 - 0 - 42 - 64 = 18$$

**8.8.** Let  $A = \begin{bmatrix} 1 & 3 & 0 & -1 \\ -4 & 2 & 5 & 1 \\ 1 & 0 & 3 & -2 \\ 3 & -2 & 1 & 4 \end{bmatrix}$ . Find the number  $N_k$  and sum  $S_k$  of principal minors of order:

- (a)  $k = 1$ , (b)  $k = 2$ , (c)  $k = 3$ , (d)  $k = 4$ .

Each (nonempty) subset of the diagonal (or equivalently, each nonempty subset of  $\{1, 2, 3, 4\}$ ) determines a principal minor of  $A$ , and

$$N_k = \binom{n}{k} = \frac{n!}{k!(n-k)!} \text{ of them are of order } k.$$

$$\text{Thus, } N_1 = \binom{4}{1} = 4, \quad N_2 = \binom{4}{2} = 6, \quad N_3 = \binom{4}{3} = 4, \quad N_4 = \binom{4}{4} = 1$$

$$(a) \ S_1 = |1| + |2| + |3| + |4| = 1 + 2 + 3 + 4 = 10$$

$$(b) \ S_2 = \begin{vmatrix} 1 & 3 \\ -4 & 2 \end{vmatrix} + \begin{vmatrix} 1 & 0 \\ 1 & 3 \end{vmatrix} + \begin{vmatrix} 1 & -1 \\ 3 & 4 \end{vmatrix} + \begin{vmatrix} 2 & 5 \\ 0 & 3 \end{vmatrix} + \begin{vmatrix} 2 & 1 \\ -2 & 4 \end{vmatrix} + \begin{vmatrix} 3 & -2 \\ 1 & 4 \end{vmatrix} \\ = 14 + 3 + 7 + 6 + 10 + 14 = 54$$

$$(c) \ S_3 = \begin{vmatrix} 1 & 3 & 0 \\ -4 & 2 & 5 \\ 1 & 0 & 3 \end{vmatrix} + \begin{vmatrix} 1 & 3 & -1 \\ -4 & 2 & 1 \\ 3 & -2 & 4 \end{vmatrix} + \begin{vmatrix} 1 & 0 & -1 \\ 1 & 3 & -2 \\ 3 & 1 & 4 \end{vmatrix} + \begin{vmatrix} 2 & 5 & 1 \\ 0 & 3 & -2 \\ -2 & 1 & 4 \end{vmatrix} \\ = 57 + 65 + 22 + 54 = 198$$

$$(d) \ S_4 = \det(A) = 378$$

## Determinants and Systems of Linear Equations

**8.9.** Use determinants to solve the system

$$\begin{cases} 3y + 2x = z + 1 \\ 3x + 2z = 8 - 5y \\ 3z - 1 = x - 2y \end{cases}$$

First arrange the equation in standard form, then compute the determinant  $D$  of the matrix of coefficients:

$$\begin{array}{rcl} 2x + 3y - z & = & 1 \\ 3x + 5y + 2z & = & 8 \\ x - 2y - 3z & = & -1 \end{array} \quad \text{and} \quad D = \begin{vmatrix} 2 & 3 & -1 \\ 3 & 5 & 2 \\ 1 & -2 & -3 \end{vmatrix} = -30 + 6 + 6 + 5 + 8 + 27 = 22$$

Because  $D \neq 0$ , the system has a unique solution. To compute  $N_x, N_y, N_z$ , we replace, respectively, the coefficients of  $x, y, z$  in the matrix of coefficients by the constant terms. Then

$$N_x = \begin{vmatrix} 1 & 3 & -1 \\ 8 & 5 & 2 \\ -1 & -2 & -1 \end{vmatrix} = 66, \quad N_y = \begin{vmatrix} 2 & 1 & -1 \\ 3 & 8 & 2 \\ 1 & -1 & -3 \end{vmatrix} = -22, \quad N_z = \begin{vmatrix} 2 & 3 & 1 \\ 3 & 5 & 8 \\ 1 & -2 & -1 \end{vmatrix} = 44$$

Thus,

$$x = \frac{N_x}{D} = \frac{66}{22} = 3, \quad y = \frac{N_y}{D} = \frac{-22}{22} = -1, \quad z = \frac{N_z}{D} = \frac{44}{22} = 2$$

**8.10.** Consider the system 
$$\begin{cases} kx + y + z = 1 \\ x + ky + z = 1 \\ x + y + kz = 1 \end{cases}$$

Use determinants to find those values of  $k$  for which the system has

(a) a unique solution, (b) more than one solution, (c) no solution.

(a) The system has a unique solution when  $D \neq 0$ , where  $D$  is the determinant of the matrix of coefficients. Compute

$$D = \begin{vmatrix} k & 1 & 1 \\ 1 & k & 1 \\ 1 & 1 & k \end{vmatrix} = k^3 + 1 + 1 - k - k - k = k^3 - 3k + 2 = (k - 1)^2(k + 2)$$

Thus, the system has a unique solution when

$$(k-1)^2(k+2) \neq 0, \quad \text{when } k \neq 1 \text{ and } k \neq -2$$

(b and c) Gaussian elimination shows that the system has more than one solution when  $k = 1$ , and the system has no solution when  $k = -2$ .

## Miscellaneous Problems

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**8.11.** Find the volume  $V(S)$  of the parallelepiped  $S$  in  $\mathbf{R}^3$  determined by the vectors:

(a)  $u_1 = (1, 1, 1)$   $u_2 = (1, 3, -4)$ ,  $u_3 = (1, 2, -5)$ .

(b)  $u_1 = (1, 2, 4)$   $u_2 = (2, 1, -3)$ ,  $u_3 = (5, 7, -9)$ .

$V(S)$  is the absolute value of the determinant of the matrix  $M$  whose rows are the given vectors. Thus,

(a)  $|M| = \begin{vmatrix} 1 & 1 & 1 \\ 1 & 3 & -4 \\ 1 & 2 & -5 \end{vmatrix} = -15 - 4 + 2 - 3 + 8 + 5 = -7$ . Hence,

$$V(S) = |-7| = 7.$$

(b)  $|M| = \begin{vmatrix} 1 & 2 & 4 \\ 2 & 1 & -3 \\ 5 & 7 & 9 \end{vmatrix} = 9 - 30 + 56 - 20 + 21 - 36 = 0$ . Thus,

$$V(S) = 0, \text{ or, in other words, } u_1, u_2, u_3$$

lie in a plane and are linearly dependent.

**8.12.** Find  $\det(M)$  where

$$M = \begin{bmatrix} 3 & 4 & 0 & 0 & 0 \\ 2 & 5 & 0 & 0 & 0 \\ 0 & 9 & 2 & 0 & 0 \\ 0 & 5 & 0 & 6 & 7 \\ 0 & 0 & 4 & 3 & 4 \end{bmatrix} = \begin{bmatrix} 3 & 4 & 0 & 0 & 0 \\ 2 & 5 & 0 & 0 & 0 \\ 0 & 9 & 2 & 0 & 0 \\ 0 & 5 & 0 & 6 & 7 \\ 0 & 0 & 4 & 3 & 4 \end{bmatrix}$$

$M$  is a (lower) triangular block matrix; hence, evaluate the determinant of each diagonal block:

$$\begin{vmatrix} 3 & 4 \\ 2 & 5 \end{vmatrix} = 15 - 8 = 7, \quad |2| = 2, \quad \begin{vmatrix} 6 & 7 \\ 3 & 4 \end{vmatrix} = 24 - 21 = 3$$

Thus,  $|M| = 7(2)(3) = 42$ .

**8.13.** Find the determinant of  $F: \mathbf{R}^3 \rightarrow \mathbf{R}^3$  defined by

$$F(x, y, z) = (x + 3y - 4z, 2y + 7z, x + 5y - 3z)$$

The determinant of a linear operator  $F$  is equal to the determinant of any matrix that represents  $F$ . Thus first find the matrix  $A$  representing  $F$  in the usual basis (whose rows, respectively, consist of the coefficients of  $x, y, z$ ). Then

$$A = \begin{bmatrix} 1 & 3 & -4 \\ 0 & 2 & 7 \\ 1 & 5 & -3 \end{bmatrix}, \quad \text{and so} \quad \det(F) = |A| = -6 + 21 + 0 + 8 - 35 - 0 = -8$$

**8.14.** Write out  $g = g(x_1, x_2, x_3, x_4)$  explicitly where

$$g(x_1, x_2, \dots, x_n) = \prod_{i < j} (x_i - x_j).$$

The symbol  $\prod$  is used for a product of terms in the same way that the symbol  $\sum$  is used for a sum of terms. That is,  $\prod_{i < j} (x_i - x_j)$  means the product of all terms  $(x_i - x_j)$  for which  $i < j$ . Hence,

$$g = g(x_1, \dots, x_4) = (x_1 - x_2)(x_1 - x_3)(x_1 - x_4)(x_2 - x_3)(x_2 - x_4)(x_3 - x_4)$$

**8.15.** Let  $D$  be a 2-linear, alternating function. Show that  $D(A, B) = -D(B, A)$ .

Because  $D$  is alternating,  $D(A, A) = 0$ ,  $D(B, B) = 0$ .  
Hence,

$$D(A + B, A + B) = D(A, A) + D(A, B) + D(B, A) + D(B, B) = D(A, B) + D(B, A)$$

However,  $D(A + B, A + B) = 0$ . Hence,  $D(A, B) = -D(B, A)$ , as required.

## Permutations

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**8.16.** Determine the parity (sign) of the permutation  $\sigma = 364152$ .

Count the number of inversions. That is, for each element  $k$ , count the number of elements  $i$  in  $\sigma$  such that  $i > k$  and  $i$  precedes  $k$  in  $\sigma$ . Namely,

|          |                        |          |              |
|----------|------------------------|----------|--------------|
| $k = 1:$ | 3 numbers (3, 6, 4)    | $k = 4:$ | 1 number (6) |
| $k = 2:$ | 4 numbers (3, 6, 4, 5) | $k = 5:$ | 1 number (6) |
| $k = 3:$ | 0 numbers              | $k = 6:$ | 0 numbers    |

Because  $3 + 4 + 0 + 1 + 1 + 0 = 9$  is odd,  $\sigma$  is an odd permutation, and  $\text{sgn } \sigma = -1$ .

**8.17.** Let  $\sigma = 24513$  and  $\tau = 41352$  be permutations in  $S_5$ . Find (a)  $\tau \circ \sigma$ , (b)  $\sigma^{-1}$ .

Recall that  $\sigma = 24513$  and  $\tau = 41352$  are short ways of writing

$$\sigma = \begin{pmatrix} 1 & 2 & 3 & 4 & 5 \\ 2 & 4 & 5 & 1 & 3 \end{pmatrix} \quad \text{or} \quad \sigma(1) = 2, \quad \sigma(2) = 4, \quad \sigma(3) = 5, \quad \sigma(4) = 1, \quad \sigma(5) = 3$$

$$\tau = \begin{pmatrix} 1 & 2 & 3 & 4 & 5 \\ 4 & 1 & 3 & 5 & 2 \end{pmatrix} \quad \text{or} \quad \tau(1) = 4, \quad \tau(2) = 1, \quad \tau(3) = 3, \quad \tau(4) = 5, \quad \tau(5) = 2$$

(a) The effects of  $\sigma$  and then  $\tau$  on 1, 2, 3, 4, 5 are as follows:

$$1 \rightarrow 2 \rightarrow 1, \quad 2 \rightarrow 4 \rightarrow 5, \quad 3 \rightarrow 5 \rightarrow 2, \quad 4 \rightarrow 1 \rightarrow 4, \quad 5 \rightarrow 3 \rightarrow 3$$

[That is, for example,  $(\iota \circ \sigma)(1) = \iota(\sigma(1)) = \iota(2) = 1$ .]

Thus,  $\iota \circ \sigma = 15243$ .

(b) By definition,  $\sigma^{-1}(j) = k$  if and only if  $\sigma(k) = j$ .

Hence,

$$\sigma^{-1} = \begin{pmatrix} 2 & 4 & 5 & 1 & 3 \\ 1 & 2 & 3 & 4 & 5 \end{pmatrix} = \begin{pmatrix} 1 & 2 & 3 & 4 & 5 \\ 4 & 1 & 5 & 2 & 3 \end{pmatrix} \quad \text{or} \quad \sigma^{-1} = 41523$$

**8.18.** Let  $\sigma = j_1 j_2 \dots j_n$  be any permutation in  $S_n$ . Show that, for each inversion  $(i, k)$  where  $i > k$  but  $i$  precedes  $k$  in  $\sigma$ , there is a pair  $(i^*, j^*)$  such that

$$i^* < k^* \quad \text{and} \quad \sigma(i^*) > \sigma(j^*) \quad (1)$$

and vice versa. Thus,  $\sigma$  is even or odd according to whether there is an even or an odd number of pairs satisfying (1).

Choose  $i^*$  and  $k^*$  so that  $\sigma(i^*) = i$  and  $\sigma(k^*) = k$ . Then  $i > k$  if and only if  $\sigma(i^*) > \sigma(k^*)$ , and  $i$  precedes  $k$  in  $\sigma$  if and only if  $i^* < k^*$ .

**8.19.** Consider the polynomials  $g = g(x_1, \dots, x_n)$  and  $\sigma(g)$ , defined by

$$g = g(x_1, \dots, x_n) = \prod_{i < j} (x_i - x_j) \quad \text{and} \quad \sigma(g) = \prod_{i < j} (x_{\sigma(i)} - x_{\sigma(j)})$$

(See [Problem 8.14](#).) Show that  $\sigma(g) = g$  when  $\sigma$  is an even permutation, and  $\sigma(g) = -g$  when  $\sigma$  is an odd permutation. That is,  $\sigma(g) = (\text{sgn } \sigma)g$ .

Because  $\sigma$  is one-to-one and onto,

$$\sigma(g) = \prod_{i < j} (x_{\sigma(i)} - x_{\sigma(j)}) = \prod_{i < j \text{ or } i > j} (x_i - x_j)$$

Thus,  $\sigma(g)$  or  $\sigma(g) = -g$  according to whether there is an even or an odd number of terms of the form  $x_i - x_j$ , where  $i > j$ . Note that for each pair  $(i, j)$  for which

$$i < j \quad \text{and} \quad \sigma(i) > \sigma(j)$$

there is a term  $(x_{\sigma(i)} - x_{\sigma(j)})$  in  $\sigma(g)$  for which  $\sigma(i) > \sigma(j)$ . Because  $\sigma$  is even if and only if there is an even number of pairs satisfying (1), we have  $\sigma(g) = g$  if and only if  $\sigma$  is even. Hence,  $\sigma(g) = -g$  if and only if  $\sigma$  is odd.

**8.20.** Let  $\sigma, \tau \in S_n$ . Show that  $\text{sgn}(\tau \circ \sigma) = (\text{sgn } \tau)(\text{sgn } \sigma)$ .

Thus, the product of two even or two odd permutations is even, and the product of an odd and an even permutation is odd.

Using [Problem 8.19](#), we have

$$\text{sgn}(\tau \circ \sigma) g = (\tau \circ \sigma)(g) = \tau(\sigma(g)) = \tau((\text{sgn } \sigma)g) = (\text{sgn } \tau)(\text{sgn } \sigma)g$$

Accordingly,  $\text{sgn}(\tau \circ \sigma) = (\text{sgn } \tau)(\text{sgn } \sigma)$ .

**8.21.** Consider the permutation  $\sigma = j_1 j_2 \dots j_n$ . Show that  $\text{sgn } \sigma^{-1} = \text{sgn } \sigma$  and, for scalars  $a_{ij}$ , show that

$$a_{j_1 1} a_{j_2 2} \cdots a_{j_n n} = a_{1 k_1} a_{2 k_2} \cdots a_{n k_n}$$

where  $\sigma^{-1} = k_1 k_2 \dots k_n$ .

We have  $\sigma^{-1} \circ \sigma = \varepsilon$ , the identity permutation. Because  $\varepsilon$  is even,  $\sigma^{-1}$  and  $\sigma$  are both even or both odd. Hence  $\text{sgn } \sigma^{-1} = \text{sgn } \sigma$ .

Because  $\sigma = j_1 j_2 \dots j_n$  is a permutation,  $a_{j_1 1} a_{j_2 2} \cdots a_{j_n n} = a_{1 k_1} a_{2 k_2} \cdots a_{n k_n}$ . Then  $k_1, k_2, \dots, k_n$  have the property that

$$\sigma(k_1) = 1, \quad \sigma(k_2) = 2, \quad \dots, \quad \sigma(k_n) = n$$

Let  $\iota = k_1 k_2 \dots k_n$ . Then, for  $i = 1, \dots, n$ ,

$$(\sigma \circ \tau)(i) = \sigma(\tau(i)) = \sigma(k_i) = i$$

Thus,  $\sigma \circ \iota = \varepsilon$ , the identity permutation. Hence,  $\iota = \sigma^{-1}$ .

## Proofs of Theorems

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**8.22.** Prove [Theorem 8.1](#):  $|A^T| = |A|$ .

If  $A = [a_{ij}]$ , then  $A^T = [b_{ij}]$  with  $b_{ij} = a_{ji}$ . Hence,

$$|A^T| = \sum_{\sigma \in S_n} (\text{sgn } \sigma) b_{1\sigma(1)} b_{2\sigma(2)} \cdots b_{n\sigma(n)} = \sum_{\sigma \in S_n} (\text{sgn } \sigma) a_{\sigma(1),1} a_{\sigma(2),2} \cdots a_{\sigma(n),n}$$

Let  $\iota = \sigma^{-1}$ . By [Problem 8.21](#)  $\text{sgn } \iota = \text{sgn } \sigma$ , and  $a_{\sigma(1),1} a_{\sigma(2),2} \cdots a_{\sigma(n),n} = a_{1\iota(1)} a_{2\iota(2)} \cdots a_{n\iota(n)}$ . Hence,

$$|A^T| = \sum_{\sigma \in S_n} (\text{sgn } \tau) a_{1\iota(1)} a_{2\iota(2)} \cdots a_{n\iota(n)}$$

However, as  $\iota$  runs through all the elements of  $S_n$ ,  $\iota = \sigma^{-1}$  also runs through all the elements of  $S_n$ . Thus,  $|A^T| = |A|$ .

**8.23.** Prove [Theorem 8.3\(i\)](#): If two rows (columns) of  $A$  are interchanged, then  $|B| = -|A|$ .

We prove the theorem for the case that two columns are interchanged. Let  $\iota$  be the transposition that interchanges the two numbers corresponding to the two columns of  $A$  that are interchanged. If  $A = [a_{ij}]$  and  $B = [b_{ij}]$ , then  $b_{ij} = a_{\iota(j)}$ . Hence, for any permutation  $\sigma$ ,

$$b_{1\sigma(1)}b_{2\sigma(2)}\cdots b_{n\sigma(n)} = a_{1(\tau\circ\sigma)(1)}a_{2(\tau\circ\sigma)(2)}\cdots a_{n(\tau\circ\sigma)(n)}$$

Thus,

$$|B| = \sum_{\sigma \in S_n} (\text{sgn } \sigma) b_{1\sigma(1)} b_{2\sigma(2)} \cdots b_{n\sigma(n)} = \sum_{\sigma \in S_n} (\text{sgn } \sigma) a_{1(\tau\circ\sigma)(1)} a_{2(\tau\circ\sigma)(2)} \cdots a_{n(\tau\circ\sigma)(n)}$$

Because the transposition  $\iota$  is an odd permutation,  $\text{sgn}(\iota \circ \sigma) = (\text{sgn } \iota)(\text{sgn } \sigma) = -\text{sgn } \sigma$ . Accordingly,  $\text{sgn } \sigma = -\text{sgn}(\iota \circ \sigma)$ , and so

$$|B| = - \sum_{\sigma \in S_n} [\text{sgn}(\tau \circ \sigma)] a_{1(\tau\circ\sigma)(1)} a_{2(\tau\circ\sigma)(2)} \cdots a_{n(\tau\circ\sigma)(n)}$$

But as  $\sigma$  runs through all the elements of  $S_n$ ,  $\iota \circ \sigma$  also runs through all the elements of  $S_n$ . Hence,  $|B| = -|A|$ .

#### 8.24. Prove [Theorem 8.2](#).

- (i) If  $A$  has a row (column) of zeros, then  $|A| = 0$ .
- (ii) If  $A$  has two identical rows (columns), then  $|A| = 0$ .
- (iii) If  $A$  is triangular, then  $|A| = \text{product of diagonal elements}$ . Thus,  $|I| = 1$ .
- (i) Each term in  $|A|$  contains a factor from every row, and so from the row of zeros. Thus, each term of  $|A|$  is zero, and so  $|A| = 0$ .
- (ii) Suppose  $1 + 1 \neq 0$  in  $K$ . If we interchange the two identical rows of  $A$ , we still obtain the matrix  $A$ . Hence, by [Problem 8.23](#),  $|A| = -|A|$ , and so  $|A| = 0$ .

Now suppose  $1 + 1 = 0$  in  $K$ . Then  $\text{sgn } \sigma = 1$  for every  $\sigma \in S_n$ . Because  $A$  has two identical rows, we can arrange the terms of  $A$  into pairs of equal terms. Because each pair is 0, the determinant of  $A$  is zero.

(iii) Suppose  $A = [a_{ij}]$  is lower triangular; that is, the entries above the diagonal are all zero:  $a_{ij} = 0$  whenever  $i < j$ . Consider a term  $t$  of the determinant of  $A$ :

$$t = (\text{sgn } \sigma) a_{1i_1} a_{2i_2} \cdots a_{ni_n}, \quad \text{where} \quad \sigma = i_1 i_2 \cdots i_n$$

Suppose  $i_1 \neq 1$ . Then  $1 < i_1$  and so  $a_{1i_1} = 0$  hence,  $t = 0$ . That is, each term for which  $i_1 \neq 1$  is zero.

Now suppose  $i_1 = 1$  but  $i_2 \neq 2$ . Then  $2 < i_2$ , and so  $a_{2i_2} = 0$ ; hence,  $t = 0$ . Thus, each term for which  $i_1 \neq 1$  or  $i_2 \neq 2$  is zero.

Similarly, we obtain that each term for which  $i_1 \neq 1$  or  $i_2 \neq 2$  or ... or  $i_n$  is zero. Accordingly,  $|A| = a_{11} a_{22} \cdots a_{nn} = \text{product of diagonal elements}$ .

**8.25.** Prove [Theorem 8.3](#):  $B$  is obtained from  $A$  by an elementary operation.

(i) If two rows (columns) of  $A$  were interchanged, then  $|B| = -|A|$ .

(ii) If a row (column) of  $A$  were multiplied by a scalar  $k$ , then  $|B| = k|A|$ .

(iii) If a multiple of a row (column) of  $A$  were added to another row (column) of  $A$ , then  $|B| = |A|$ .

(i) This result was proved in [Problem 8.23](#).

(ii) If the  $j$ th row of  $A$  is multiplied by  $k$ , then every term in  $|A|$  is multiplied by  $k$ , and so  $|B| = k|A|$ . That is,

$$|B| = \sum_{\sigma} (\text{sgn } \sigma) a_{1i_1} a_{2i_2} \cdots (ka_{ji_j}) \cdots a_{ni_n} = k \sum_{\sigma} (\text{sgn } \sigma) a_{1i_1} a_{2i_2} \cdots a_{ni_n} = k|A|$$

(iii) Suppose  $c$  times the  $k$ th row is added to the  $j$ th row of  $A$ . Using the symbol  $\wedge$  to denote the  $j$ th position

in a determinant term, we have

$$\begin{aligned} |B| &= \sum_{\sigma} (\text{sgn } \sigma) a_{1i_1} a_{2i_2} \cdots (\widehat{ca_{ki_k} + a_{ji_j}}) \cdots a_{ni_n} \\ &= c \sum_{\sigma} (\text{sgn } \sigma) a_{1i_1} a_{2i_2} \cdots \widehat{a_{ki_k}} \cdots a_{ni_n} + \sum_{\sigma} (\text{sgn } \sigma) a_{1i_1} a_{2i_2} \cdots a_{ji_j} \cdots a_{ni_n} \end{aligned}$$

The first sum is the determinant of a matrix whose  $k$ th and  $j$ th rows are identical. Accordingly, by [Theorem 8.2\(ii\)](#), the sum is zero. The second sum is the determinant of  $A$ . Thus,  $|B| = c \cdot 0 + |A| = |A|$ .

**8.26.** Prove Lemma 8.6: Let  $E$  be an elementary matrix. Then  $|EA| = |E| |A|$ .

Consider the elementary row operations: (i) Multiply a row by a constant  $k \neq 0$ ,  
(ii) Interchange two rows, (iii) Add a multiple of one row to another.

Let  $E_1, E_2, E_3$  be the corresponding elementary matrices. That is,  $E_1, E_2, E_3$  are obtained by applying the above operations to the identity matrix  $I$ . By [Problem 8.25](#),

$$|E_1| = k|I| = k, \quad |E_2| = -|I| = -1, \quad |E_3| = |I| = 1$$

Recall (Theorem 3.11) that  $E_i A$  is identical to the matrix obtained by applying the corresponding operation to  $A$ . Thus, by [Theorem 8.3](#), we obtain the following which proves our lemma:

$$|E_1 A| = k|A| = |E_1| |A|, \quad |E_2 A| = -|A| = |E_2| |A|, \quad |E_3 A| = |A| = 1|A| = |E_3| |A|$$

**8.27.** Suppose  $B$  is row equivalent to a square matrix  $A$ . Prove that  $|B| = 0$  if and only if  $|A| = 0$ .

By [Theorem 8.3](#), the effect of an elementary row operation is to change the sign of the determinant or

to multiply the determinant by a nonzero scalar.  
Hence,  $|B| = 0$  if and only if  $|A| = 0$ .

**8.28.** Prove [Theorem 8.5](#): Let  $A$  be an  $n$ -square matrix. Then the following are equivalent:

- (i)  $A$  is invertible,      (ii)  $AX = 0$  has only the zero solution,      (iii)  $\det(A) \neq 0$ .

The proof is by the Gaussian algorithm. If  $A$  is invertible, it is row equivalent to  $I$ . But  $|I| \neq 0$ . Hence, by [Problem 8.27](#),  $|A| \neq 0$ . If  $A$  is not invertible, it is row equivalent to a matrix with a zero row. Hence,  $\det(A) = 0$ . Thus, (i) and (iii) are equivalent.

If  $AX = 0$  has only the solution  $X = 0$ , then  $A$  is row equivalent to  $I$  and  $A$  is invertible. Conversely, if  $A$  is invertible with inverse  $A^{-1}$ , then

$$X = IX = (A^{-1}A)X = A^{-1}(AX) = A^{-1}0 = 0$$

is the only solution of  $AX = 0$ . Thus, (i) and (ii) are equivalent.

**8.29.** Prove [Theorem 8.4](#):  $|AB| = |A| |B|$ .

If  $A$  is singular, then  $AB$  is also singular, and so  $|AB| = 0 = |A| |B|$ . On the other hand, if  $A$  is nonsingular, then  $A = E_n \cdots E_2 E_1$ , a product of elementary matrices. Then, Lemma 8.6 and induction yields

$$|AB| = |E_n \cdots E_2 E_1 B| = |E_n| \cdots |E_2| |E_1| |B| = |A| |B|$$

**8.30.** Suppose  $P$  is invertible. Prove that  $|P^{-1}| = |P|^{-1}$ .

$$P^{-1}P = I. \text{ Hence, } 1 = |I| = |P^{-1}P| = |P^{-1}| |P|, \text{ and so } |P^{-1}| = |P|^{-1}.$$

**8.31.** Prove [Theorem 8.7](#): Suppose  $A$  and  $B$  are similar matrices. Then  $|A| = |B|$ .

Because  $A$  and  $B$  are similar, there exists an invertible matrix  $P$  such that  $B = P^{-1}AP$ . Therefore, using [Problem 8.30](#), we get  $|B| = |P^{-1}AP| = |P^{-1}| |A| |P| = |A| |P^{-1}| |P| = |A|$ .

We remark that although the matrices  $P^{-1}$  and  $A$  may not commute, their determinants  $|P^{-1}|$  and  $|A|$  do commute, because they are scalars in the field  $K$ .

**8.32.** Prove [Theorem 8.8](#) (Laplace): Let  $A = [a_{ij}]$ , and let  $A_{ij}$  denote the cofactor of  $a_{ij}$ . Then, for any  $i$  or  $j$

$$|A| = a_{i1}A_{i1} + \cdots + a_{in}A_{in} \quad \text{and} \quad |A| = a_{1j}A_{1j} + \cdots + a_{nj}A_{nj}$$

Because  $|A| = |A^T|$ , we need only prove one of the expansions, say, the first one in terms of rows of  $A$ . Each term in  $|A|$  contains one and only one entry of the  $i$ th row ( $a_{i1}, a_{i2}, \dots, a_{in}$ ) of  $A$ . Hence, we can write  $|A|$  in the form

$$|A| = a_{i1}A_{i1}^* + a_{i2}A_{i2}^* + \cdots + a_{in}A_{in}^*$$

(Note that  $A_{ij}^*$  is a sum of terms involving no entry of the  $i$ th row of  $A$ .) Thus, the theorem is proved if we can show that

$$A_{ij}^* = A_{ij} = (-1)^{i+j}|M_{ij}|$$

where  $M_{ij}$  is the matrix obtained by deleting the row and column containing the entry  $a_{ij}$ . (Historically, the expression  $A_{ij}^*$  was defined as the cofactor of  $a_{ij}$ , and so the theorem reduces to showing that the two definitions of the cofactor are equivalent.)

First we consider the case that  $i = n, j = n$ . Then the sum of terms in  $|A|$  containing  $a_{nn}$  is

$$a_{nn}A_{nn}^* = a_{nn} \sum_{\sigma} (\text{sgn } \sigma) a_{1\sigma(1)} a_{2\sigma(2)} \cdots a_{n-1,\sigma(n-1)}$$

where we sum over all permutations  $\sigma \in S_n$  for which  $\sigma(n) = n$ . However, this is equivalent (Prove!) to summing over all permutations of  $\{1, \dots, n-1\}$ . Thus,  $A_{nn}^* = |M_{nn}| = (-1)^{n+n} |M_{nn}|$ .

Now we consider any  $i$  and  $j$ . We interchange the  $i$ th row with each succeeding row until it is last, and we interchange the  $j$ th column with each succeeding column until it is last. Note that the determinant  $|M_{ij}|$  is not affected, because the relative positions of the other rows and columns are not affected by these interchanges. However, the “sign” of  $|A|$  and of  $A_{ij}^*$  is changed  $n-i-1$  and then  $n-j$  times. Accordingly,

$$A_{ij}^* = (-1)^{n-i+n-j} |M_{ij}| = (-1)^{i+j} |M_{ij}|$$

**8.33.** Let  $A = [a_{ij}]$  and let  $B$  be the matrix obtained from  $A$  by replacing the  $i$ th row of  $A$  by the row vector  $(b_{i1}, \dots, b_{in})$ . Show that

$$|B| = b_{i1}A_{i1} + b_{i2}A_{i2} + \cdots + b_{in}A_{in}$$

Furthermore, show that, for  $j \neq i$ ,

$$a_{j1}A_{i1} + a_{j2}A_{i2} + \cdots + a_{jn}A_{in} = 0 \quad \text{and} \quad a_{1j}A_{1i} + a_{2j}A_{2i} + \cdots + a_{nj}A_{ni} = 0$$

Let  $B = [b_{ij}]$ . By [Theorem 8.8](#),

$$|B| = b_{i1}B_{i1} + b_{i2}B_{i2} + \cdots + b_{in}B_{in}$$

Because  $B_{ij}$  does not depend on the  $i$ th row of  $B$ , we get  $B_{ij} = A_{ij}$  for  $j = 1, \dots, n$ . Hence,

$$|B| = b_{i1}A_{i1} + b_{i2}A_{i2} + \cdots + b_{in}A_{in}$$

Now let  $A'$  be obtained from  $A$  by replacing the  $i$ th row of  $A$  by the  $j$ th row of  $A$ . Because  $A'$  has two identical rows,  $|A'| = 0$ . Thus, by the above result,

$$|A'| = a_{j1}A_{i1} + a_{j2}A_{i2} + \cdots + a_{jn}A_{in} = 0$$

Using  $|A^T| = |A|$ , we also obtain that  $a_{1j}A_{1i} + a_{2j}A_{2i} + \cdots + a_{nj}A_{ni} = 0$ .

**8.34.** Prove [Theorem 8.9](#):  $A(\text{adj } A) = (\text{adj } A)A = |A| I$ .

Let  $A = [a_{ij}]$  and let  $A(\text{adj } A) = [b_{ij}]$ . The  $i$ th row of  $A$  is

$$(a_{i1}, a_{i2}, \dots, a_{in}) \tag{1}$$

Because  $\text{adj } A$  is the transpose of the matrix of cofactors, the  $j$ th column of  $\text{adj } A$  is the transpose of the cofactors of the  $j$ th row of  $A$ :

$$(A_{j1}, A_{j2}, \dots, A_{jn})^T \tag{2}$$

Now  $b_{ij}$ , the  $ij$  entry in  $A(\text{adj } A)$ , is obtained by multiplying expressions (1) and (2):

$$b_{ij} = a_{i1}A_{j1} + a_{i2}A_{j2} + \cdots + a_{in}A_{jn}$$

By [Theorem 8.8](#) and [Problem 8.33](#),

$$b_{ij} = \begin{cases} |A| & \text{if } i = j \\ 0 & \text{if } i \neq j \end{cases}$$

Accordingly,  $A(\text{adj } A)$  is the diagonal matrix with each diagonal element  $|A|$ . In other words,  $A(\text{adj } A) = |A|I$ . Similarly,  $(\text{adj } A)A = |A|I$ .

**8.35.** Prove [Theorem 8.10](#) (Cramer's rule): The (square) system  $AX = B$  has a unique solution if and only if  $D \neq 0$ . In this case,  $x_i = N_i/D$  for each  $i$ .

By previous results,  $AX = B$  has a unique solution if and only if  $A$  is invertible, and  $A$  is invertible if and only if  $D = |A| \neq 0$ .

Now suppose  $D \neq 0$ . By [Theorem 8.9](#),  $A^{-1} = (1/D)(\text{adj } A)$ . Multiplying  $AX = B$  by  $A^{-1}$ , we obtain

$$X = A^{-1}AX = (1/D)(\text{adj } A)B \quad (1)$$

Note that the  $i$ th row of  $(1/D)(\text{adj } A)$  is  $(1/D)(A_{1i}, A_{2i}, \dots, A_{ni})$ . If  $B = (b_1, b_2, \dots, b_n)^T$ , then, by (1),

$$x_i = (1/D)(b_1A_{1i} + b_2A_{2i} + \dots + b_nA_{ni})$$

However, as in [Problem 8.33](#),  $b_1A_{1i} + b_2A_{2i} + \dots + b_nA_{ni} = N_i$ , the determinant of the matrix obtained by replacing the  $i$ th column of  $A$  by the column vector  $B$ . Thus,  $x_i = (1/D)N_i$ , as required.

**8.36.** Prove [Theorem 8.12](#): Suppose  $M$  is an upper (lower) triangular block matrix with diagonal blocks  $A_1, A_2, \dots, A_n$ . Then

$$\det(M) = \det(A_1) \det(A_2) \cdots \det(A_n)$$

We need only prove the theorem for  $n = 2$ —that is, when  $M$  is a square matrix of the form  $M = \begin{bmatrix} A & C \\ 0 & B \end{bmatrix}$ . The proof of the general theorem follows easily by induction.

Suppose  $A = [a_{ij}]$  is  $r$ -square,  $B = [b_{ij}]$  is  $s$ -square, and  $M = [m_{ij}]$  is  $n$ -square, where  $n = r + s$ . By

definition,

$$\det(M) = \sum_{\sigma \in S_n} (\operatorname{sgn} \sigma) m_{1\sigma(1)} m_{2\sigma(2)} \cdots m_{n\sigma(n)}$$

If  $i > r$  and  $j < r$ , then  $m_{ij} = 0$ . Thus, we need only consider those permutations  $\sigma$  such that

$$\sigma\{r+1, r+2, \dots, r+s\} = \{r+1, r+2, \dots, r+s\} \quad \text{and} \quad \sigma\{1, 2, \dots, r\} = \{1, 2, \dots, r\}$$

Let  $\sigma_1(k) = \sigma(k)$  for  $k \leq r$ , and let  $\sigma_2(k) = \sigma(r+k) - r$  for  $k \leq s$ . Then

$$(\operatorname{sgn} \sigma) m_{1\sigma(1)} m_{2\sigma(2)} \cdots m_{n\sigma(n)} = (\operatorname{sgn} \sigma_1) a_{1\sigma_1(1)} a_{2\sigma_1(2)} \cdots a_{r\sigma_1(r)} (\operatorname{sgn} \sigma_2) b_{1\sigma_2(1)} b_{2\sigma_2(2)} \cdots b_{s\sigma_2(s)}$$

which implies  $\det(M) = \det(A) \det(B)$ .

**8.37.** Prove [Theorem 8.15](#): There exists a unique function  $D : \mathbf{M} \rightarrow K$  such that

(i)  $D$  is multilinear, (ii)  $D$  is alternating, (iii)  $D(I) = 1$ .

This function  $D$  is the determinant function; that is,  $D(A) = |A|$ .

Let  $D$  be the determinant function,  $D(A) = |A|$ . We must show that  $D$  satisfies (i), (ii), and (iii), and that  $D$  is the only function satisfying (i), (ii), and (iii).

By [Theorem 8.2](#),  $D$  satisfies (ii) and (iii). Hence, we show that it is multilinear. Suppose the  $i$ th row of  $A = [a_{ij}]$  has the form  $(b_{i1} + b_{i2} + c_{i2}, \dots, b_{in} + c_{in})$ . Then

$$\begin{aligned} D(A) &= D(A_1, \dots, B_i + C_i, \dots, A_n) \\ &= \sum_{\sigma \in S_n} (\operatorname{sgn} \sigma) a_{1\sigma(1)} \cdots a_{i-1,\sigma(i-1)} (b_{i\sigma(i)} + c_{i\sigma(i)}) \cdots a_{n\sigma(n)} \\ &= \sum_{\sigma \in S_n} (\operatorname{sgn} \sigma) a_{1\sigma(1)} \cdots b_{i\sigma(i)} \cdots a_{n\sigma(n)} + \sum_{\sigma \in S_n} (\operatorname{sgn} \sigma) a_{1\sigma(1)} \cdots c_{i\sigma(i)} \cdots a_{n\sigma(n)} \\ &= D(A_1, \dots, B_i, \dots, A_n) + D(A_1, \dots, C_i, \dots, A_n) \end{aligned}$$

Also, by [Theorem 8.3\(ii\)](#),

$$D(A_1, \dots, kA_i, \dots, A_n) = kD(A_1, \dots, A_i, \dots, A_n)$$

Thus,  $D$  is multilinear— $D$  satisfies (i).

We next must prove the uniqueness of  $D$ . Suppose  $D$  satisfies (i), (ii), and (iii). If  $\{e_1, \dots, e_n\}$  is the usual basis of  $K^n$ , then, by (iii),  $D(e_1, e_2, \dots, e_n) = D(I) = 1$ . Using (ii), we also have that

$$D(e_{i_1}, e_{i_2}, \dots, e_{i_n}) = \text{sgn } \sigma, \quad \text{where} \quad \sigma = i_1 i_2 \cdots i_n \quad (1)$$

Now suppose  $A = [a_{ij}]$ . Observe that the  $k$ th row  $A_k$  of  $A$  is

$$A_k = (a_{k1}, a_{k2}, \dots, a_{kn}) = a_{k1}e_1 + a_{k2}e_2 + \cdots + a_{kn}e_n$$

Thus,

$$D(A) = D(a_{11}e_1 + \cdots + a_{1n}e_n, a_{21}e_1 + \cdots + a_{2n}e_n, \dots, a_{n1}e_1 + \cdots + a_{nn}e_n)$$

Using the multilinearity of  $D$ , we can write  $D(A)$  as a sum of terms of the form

$$\begin{aligned} D(A) &= \sum D(a_{1i_1}e_{i_1}, a_{2i_2}e_{i_2}, \dots, a_{ni_n}e_{i_n}) \\ &= \sum (a_{1i_1}a_{2i_2} \cdots a_{ni_n}) D(e_{i_1}, e_{i_2}, \dots, e_{i_n}) \end{aligned} \quad (2)$$

where the sum is summed over all sequences  $i_1 i_2 \dots i_n$ , where  $i_k \in \{1, \dots, n\}$ . If two of the indices are equal, say  $i_j = i_k$  but  $j \neq k$ , then, by (ii),

$$D(e_{i_1}, e_{i_2}, \dots, e_{i_n}) = 0$$

Accordingly, the sum in (2) need only be summed over all permutations  $\sigma = i_1 i_2 \dots i_n$ . Using (1), we finally have that

$$\begin{aligned}
 D(A) &= \sum_{\sigma} (a_{1i_1} a_{2i_2} \cdots a_{ni_n}) D(e_{i_1}, e_{i_2}, \dots, e_{i_n}) \\
 &= \sum_{\sigma} (\operatorname{sgn} \sigma) a_{1i_1} a_{2i_2} \cdots a_{ni_n}, \quad \text{where} \quad \sigma = i_1 i_2 \cdots i_n
 \end{aligned}$$

Hence,  $D$  is the determinant function, and so the theorem is proved.

## SUPPLEMENTARY PROBLEMS

### Computation of Determinants

**8.38.** Evaluate:

$$(a) \begin{vmatrix} 2 & 6 \\ 4 & 1 \end{vmatrix}, \quad (b) \begin{vmatrix} 5 & 1 \\ 3 & -2 \end{vmatrix}, \quad (c) \begin{vmatrix} -2 & 8 \\ -5 & -3 \end{vmatrix}, \quad (d) \begin{vmatrix} 4 & 9 \\ 1 & -3 \end{vmatrix}, \quad (e) \begin{vmatrix} a+b & a \\ b & a+b \end{vmatrix}$$

**8.39.** Find all  $t$  such that

$$(a) \begin{vmatrix} t-4 & 3 \\ 2 & t-9 \end{vmatrix} = 0, \quad (b) \begin{vmatrix} t-1 & 4 \\ 3 & t-2 \end{vmatrix} = 0$$

**8.40.** Compute the determinant of each of the following matrices:

$$(a) \begin{bmatrix} 2 & 1 & 1 \\ 0 & 5 & -2 \\ 1 & -3 & 4 \end{bmatrix}, \quad (b) \begin{bmatrix} 3 & -2 & -4 \\ 2 & 5 & -1 \\ 0 & 6 & 1 \end{bmatrix}, \quad (c) \begin{bmatrix} -2 & -1 & 4 \\ 6 & -3 & -2 \\ 4 & 1 & 2 \end{bmatrix}, \quad (d) \begin{bmatrix} 7 & 6 & 5 \\ 1 & 2 & 1 \\ 3 & -2 & 1 \end{bmatrix}$$

**8.41.** Find the determinant of each of the following matrices:

$$(a) \begin{bmatrix} 1 & 2 & 2 & 3 \\ 1 & 0 & -2 & 0 \\ 3 & -1 & 1 & -2 \\ 4 & -3 & 0 & 2 \end{bmatrix}, \quad (b) \begin{bmatrix} 2 & 1 & 3 & 2 \\ 3 & 0 & 1 & -2 \\ 1 & -1 & 4 & 3 \\ 2 & 2 & -1 & 1 \end{bmatrix}$$

**8.42.** Evaluate:

$$(a) \begin{vmatrix} 2 & -1 & 3 & -4 \\ 2 & 1 & -2 & 1 \\ 3 & 3 & -5 & 4 \\ 5 & 2 & -1 & 4 \end{vmatrix}, \quad (b) \begin{vmatrix} 2 & -1 & 4 & -3 \\ -1 & 1 & 0 & 2 \\ 3 & 2 & 3 & -1 \\ 1 & -2 & 2 & -3 \end{vmatrix}, \quad (c) \begin{vmatrix} 1 & -2 & 3 & -1 \\ 1 & 1 & -2 & 0 \\ 2 & 0 & 4 & -5 \\ 1 & 4 & 4 & -6 \end{vmatrix}$$

**8.43.** Evaluate each of the following determinants:

$$(a) \begin{vmatrix} 1 & 2 & -1 & 3 & 1 \\ 2 & -1 & 1 & -2 & 3 \\ 3 & 1 & 0 & 2 & -1 \\ 5 & 1 & 2 & -3 & 4 \\ -2 & 3 & -1 & 1 & -2 \end{vmatrix}, \quad (b) \begin{vmatrix} 1 & 3 & 5 & 7 & 9 \\ 2 & 4 & 2 & 4 & 2 \\ 0 & 0 & 1 & 2 & 3 \\ 0 & 0 & 5 & 6 & 2 \\ 0 & 0 & 2 & 3 & 1 \end{vmatrix}, \quad (c) \begin{vmatrix} 1 & 2 & 3 & 4 & 5 \\ 5 & 4 & 3 & 2 & 1 \\ 0 & 0 & 6 & 5 & 1 \\ 0 & 0 & 0 & 7 & 4 \\ 0 & 0 & 0 & 2 & 3 \end{vmatrix}$$

## **Cofactors, Classical Adjoints, Inverses**

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**8.44.** Find  $\det(A)$ ,  $\text{adj } A$ , and  $A^{-1}$ , where

$$(a) \quad A = \begin{bmatrix} 1 & 1 & 0 \\ 1 & 1 & 1 \\ 0 & 2 & 1 \end{bmatrix}, \quad (b) \quad A = \begin{bmatrix} 1 & 2 & 2 \\ 3 & 1 & 0 \\ 1 & 1 & 1 \end{bmatrix}$$

**8.45.** Find the classical adjoint of each matrix in [Problem 8.41](#).

**8.46.** Let

$$A = \begin{bmatrix} a & b \\ c & d \end{bmatrix}. \quad (a) \text{ Find } \text{adj } A, \quad (b) \text{ Show that } \text{adj}(\text{adj } A) = A, \quad (c) \text{ When does } A = \text{adj } A?$$

**8.47.** Show that if  $A$  is diagonal (triangular) then  $\text{adj } A$  is diagonal (triangular).

**8.48.** Suppose  $A = [a_{ij}]$  is triangular. Show that

(a)  $A$  is invertible if and only if each diagonal element  $a_{ii} \neq 0$ .

- (b) The diagonal elements of  $A^{-1}$  (if it exists) are  $a_{ii}^{-1}$ , the reciprocals of the diagonal elements of  $A$ .

## Minors, Principal Minors

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**8.49.** Let  $A = \begin{bmatrix} 1 & 2 & 3 & 2 \\ 1 & 0 & -2 & 3 \\ 3 & -1 & 2 & 5 \\ 4 & -3 & 0 & -1 \end{bmatrix}$  and  $B = \begin{bmatrix} 1 & 3 & -1 & 5 \\ 2 & -3 & 1 & 4 \\ 0 & -5 & 2 & 1 \\ 3 & 0 & 5 & -2 \end{bmatrix}$ .

Find the minor and the signed minor corresponding to the following submatrices:

- (a)  $A(1,4; 3,4)$ , (b)  $B(1,4; 3,4)$ , (c)  $A(2,3; 2,4)$ , (d)  $B(2,3; 2,4)$ .

- 8.50.** For  $k = 1, 2, 3$ , find the sum  $S_k$  of all principal minors of order  $k$  for

(a)  $A = \begin{bmatrix} 1 & 3 & 2 \\ 2 & -4 & 3 \\ 5 & -2 & 1 \end{bmatrix}$ , (b)  $B = \begin{bmatrix} 1 & 5 & -4 \\ 2 & 6 & 1 \\ 3 & -2 & 0 \end{bmatrix}$ , (c)  $C = \begin{bmatrix} 1 & -4 & 3 \\ 2 & 1 & 5 \\ 4 & -7 & 11 \end{bmatrix}$

- 8.51.** For  $k = 1, 2, 3, 4$ , find the sum  $S_k$  of all principal minors of order  $k$  for

(a)  $A = \begin{bmatrix} 1 & 2 & 3 & -1 \\ 1 & -2 & 0 & 5 \\ 0 & 1 & -2 & 2 \\ 4 & 0 & -1 & -3 \end{bmatrix}$ , (b)  $B = \begin{bmatrix} 1 & 2 & 1 & 2 \\ 0 & 1 & 2 & 3 \\ 1 & 3 & 0 & 4 \\ 2 & 7 & 4 & 5 \end{bmatrix}$

## Determinants and Linear Equations

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- 8.52.** Solve the following systems by determinants:

(a)  $\begin{cases} 3x + 5y = 8 \\ 4x - 2y = 1 \end{cases}$ , (b)  $\begin{cases} 2x - 3y = -1 \\ 4x + 7y = -1 \end{cases}$ , (c)  $\begin{cases} ax - 2by = c \\ 3ax - 5by = 2c \end{cases} \quad (ab \neq 0)$

- 8.53.** Solve the following systems by determinants:

$$(a) \begin{cases} 2x - 5y + 2z = 2 \\ x + 2y - 4z = 5 \\ 3x - 4y - 6z = 1 \end{cases}, \quad (b) \begin{cases} 2z + 3 = y + 3x \\ x - 3z = 2y + 1 \\ 3y + z = 2 - 2x \end{cases}$$

**8.54.** Prove [Theorem 8.11](#): The system  $AX = 0$  has a nonzero solution if and only if  $D = |A| = 0$ .

## Permutations

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**8.55.** Find the parity of the permutations  $\sigma = 32154$ ,  $\iota = 13524$ ,  $\pi = 42531$  in  $S_5$ .

**8.56.** For the permutations in [Problem 8.55](#), find

$$(a) \tau \circ \sigma, (b) \pi \circ \sigma, (c) \sigma^{-1}, (d) \tau^{-1}.$$

**8.57.** Let  $\iota \in S_n$ . Show that  $\iota \circ \sigma$  runs through  $S_n$  as  $\sigma$  runs through  $S_n$ , that is,  $S_n = \{\iota \circ \sigma : \sigma \in S_n\}$ .

**8.58.** Let  $\sigma \in S_n$  have the property that  $\sigma(n) = n$ . Let  $\sigma^* \in S_{n-1}$  be defined by  $\sigma^*(x) = \sigma(x)$ .

(a) Show that  $\text{sgn } \sigma^* = \text{sgn } \sigma$ ,

(b) Show that as  $\sigma$  runs through  $S_n$ , where  $\sigma(n) = n$ ,  $\sigma^*$  runs through  $S_{n-1}$ ; that is,

$$S_{n-1} = \{\sigma^* : \sigma \in S_n, \sigma(n) = n\}.$$

**8.59.** Consider a permutation  $\sigma = j_1 j_2 \dots j_n$ . Let  $\{e_i\}$  be the usual basis of  $K^n$ , and let  $A$  be the matrix whose  $i$ th row is  $e_{j_i}$ . [i.e.,  $A = (e_{j_1}, e_{j_2}, \dots, e_{j_n})$ ]. Show that  $|A| = \text{sgn } \sigma$ .

## Determinant of Linear Operators

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**8.60.** Find the determinant of each of the following linear transformations:

- (a)  $T: \mathbf{R}^2 \rightarrow \mathbf{R}^2$  defined by  $T(x, y) = (2x - 9y, 3x - 5y)$ ,
- (b)  $T: \mathbf{R}^3 \rightarrow \mathbf{R}^3$  defined by  $T(x, y, z) = (3x - 2z, 5y + 7z, x + y + z)$ ,
- (c)  $T: \mathbf{R}^3 \rightarrow \mathbf{R}^2$  defined by  $T(x, y, z) = (2x + 7y - 4z, 4x - 6y + 2z)$ .

**8.61.** Let  $\mathbf{D}: V \rightarrow V$  be the differential operator; that is,  $\mathbf{D}(f(t)) = df/dt$ . Find  $\det(\mathbf{D})$  if  $V$  is the vector space of functions with the following bases: (a)  $\{1, t, \dots, t^5\}$ , (b)  $\{e^t, e^{2t}, e^{3t}\}$ , (c)  $\{\sin t, \cos t\}$ .

**8.62.** Prove [Theorem 8.13](#): Let  $F$  and  $G$  be linear operators on a vector space  $V$ . Then

- (i)  $\det(F \circ G) = \det(F) \det(G)$ , (ii)  $F$  is invertible if and only if  $\det(F) \neq 0$ .

**8.63.** Prove (a)  $\det(\mathbf{1}_V) = 1$ , where  $\mathbf{1}_V$  is the identity operator, (b)  $\det(T^{-1}) = \det(T)^{-1}$  when  $T$  is invertible.

## Miscellaneous Problems

**8.64.** Find the volume  $V(S)$  of the parallelepiped  $S$  in  $\mathbf{R}^3$  determined by the following vectors:

- (a)  $u_1 = (1, 2, -3)$ ,  $u_2 = (3, 4, -1)$ ,  $u_3 = (2, -1, 5)$ ,
- (b)  $u_1 = (1, 1, 3)$ ,  $u_2 = (1, -2, -4)$ ,  $u_3 = (4, 1, 5)$ .

**8.65.** Find the volume  $V(S)$  of the parallelepiped  $S$  in  $\mathbf{R}^4$  determined by the following vectors:

$$u_1 = (1, -2, 5, -1), \quad u_2 = (2, 1, -2, 1), \quad u_3 = (3, 0, 1, -2), \quad u_4 = (1, -1, 4, -1)$$

**8.66.** Let  $V$  be the space of  $2 \times 2$  matrices  $M = \begin{bmatrix} a & b \\ c & d \end{bmatrix}$  over  $\mathbf{R}$ . Determine whether  $D: V \rightarrow \mathbf{R}$  is 2-linear (with respect to the rows), where

- (a)  $D(M) = a + d$ ,      (c)  $D(M) = ac - bd$ ,      (e)  $D(M) = 0$   
 (b)  $D(M) = ad$ ,      (d)  $D(M) = ab - cd$ ,      (f)  $D(M) = 1$

**8.67.** Let  $A$  be an  $n$ -square matrix. Prove  $|kA| = k^n|A|$ .

**8.68.** Let  $A, B, C, D$  be commuting  $n$ -square matrices.

Consider the  $2n$ -square block matrix  $M = \begin{bmatrix} A & B \\ C & D \end{bmatrix}$ . Prove that  $|M| = |A||D| - |B||C|$ . Show that the result may not be true if the matrices do not commute.

**8.69.** Suppose  $A$  is orthogonal; that is,  $A^T A = I$ . Show that  $\det(A) = \pm 1$ .

**8.70.** Let  $V$  be the space of  $m$ -square matrices viewed as  $m$ -tuples of row vectors. Suppose  $D: V \rightarrow K$  is  $m$ -linear and alternating. Show that

- (a)  $D(\dots, A, \dots, B, \dots) = -D(\dots, B, \dots, A, \dots)$ ; sign changed when two rows are interchanged.  
 (b) If  $A_1, A_2, \dots, A_m$  are linearly dependent, then  $D(A_1, A_2, \dots, A_m) = 0$ .

**8.71.** Let  $V$  be the space of  $m$ -square matrices (as above), and suppose  $D: V \rightarrow K$ . Show that the following weaker statement is equivalent to  $D$  being alternating:

$$D(A_1, A_2, \dots, A_n) = 0 \quad \text{whenever} \quad A_i = A_{i+1} \text{ for some } i$$

Let  $V$  be the space of  $n$ -square matrices over  $K$ . Suppose  $B \in V$  is invertible and so  $\det(B) \neq 0$ . Define  $D: V \rightarrow K$  by  $D(A) = \det(AB)/\det(B)$ , where  $A \in V$ . Hence,

$$D(A_1, A_2, \dots, A_n) = \det(A_1 B, A_2 B, \dots, A_n B)/\det(B)$$

where  $A_i$  is the  $i$ th row of  $A$ , and so  $A_i B$  is the  $i$ th row of  $AB$ . Show that  $D$  is multilinear and alternating, and that  $D(I) = 1$ . (This method is used by some texts to prove that  $|AB| = |A||B|$ .)

- 8.72.** Show that  $g = g(x_1, \dots, x_n) = (-1)^n V_{n-1}(x)$  where  $g = g(x_i)$  is the difference product in [Problem 8.19](#),  $x = x_n$ , and  $V_{n-1}$  is the Vandermonde determinant defined by

$$V_{n-1}(x) \equiv \begin{vmatrix} 1 & 1 & \dots & 1 & 1 \\ x_1 & x_2 & \dots & x_{n-1} & x \\ x_1^2 & x_2^2 & \dots & x_{n-1}^2 & x^2 \\ \dots & \dots & \dots & \dots & \dots \\ x_1^{n-1} & x_2^{n-1} & \dots & x_{n-1}^{n-1} & x^{n-1} \end{vmatrix}$$

- 8.73.** Let  $A$  be any matrix. Show that the signs of a minor  $A[I, J]$  and its complementary minor  $A[I', J']$  are equal.
- 8.74.** Let  $A$  be an  $n$ -square matrix. The *determinantal rank* of  $A$  is the order of the largest square submatrix of  $A$  (obtained by deleting rows and columns of  $A$ ) whose determinant is not zero. Show that the determinantal rank of  $A$  is equal to its rank—the maximum number of linearly independent rows (or columns).

## ANSWERS TO SUPPLEMENTARY PROBLEMS

*Notation:*  $M = [R_1; R_2; \dots]$  denotes a matrix with rows  $R_1, R_2, \dots$

### 8.38.

- (a)  $-22$ , (b)  $-13$ , (c)  $46$ , (d)  $-21$ , (e)  $a^2 + ab + b^2$

**8.39.** (a) 3, 10; (b) 5, -2

**8.40.** (a) 21, (b) -11, (c) 100, (d) 0

**8.41.** (a) -131, (b) -55

**8.42.** (a) 33, (b) 0, (c) 45

**8.43.** (a) -32, (b) -14, (c) -468

**8.44.**

(a)  $|A| = -2$ ,  $\text{adj } A = [-1, -1, 1; -1, 1, -1; 2, -2, 0]$ ,

(b)  $|A| = -1$ ,  $\text{adj } A = [1, 0, -2; -3, -1, 6; 2, 1, -5]$ . Also,  $A^{-1} = (\text{adj } A)/|A|$

**8.45.**

(a)  $[-16, -29, -26, -2; -30, -38, -16, 29; -8, 51, -13, -1; -13, 1, 28, -18]$ ,

(b)  $[21, -14, -17, -19; -44, 11, 33, 11; -29, 1, 13, 21; 17, 7, -19, -18]$

**8.46.** (a)  $\text{adj } A = [d, -b; -c, a]$ , (c)  $A = kI$

**8.49.**

(a) -3, -3, (b) -23, -23, (c) 3, -3, (d) 17, -17

**8.50.** (a) -2, -17, 73, (b) 7, 10, 105, (c) 13, 54, 0

**8.51.** (a) -6, 13, 62, -219; (b) 7, -37, 30, 20

**8.52.**

(a)  $x = \frac{21}{26}, y = \frac{29}{26}$ ; (b)  $x = -\frac{5}{13}, y = \frac{1}{13}$ ; (c)  $x = -\frac{c}{a}, y = -\frac{c}{b}$

**8.53.** (a)  $x = 5, y = 2, z = 1$ , (b) Because  $D = 0$ , the system cannot be solved by determinants.

**8.55.** (a)  $\text{sgn } \sigma = 1, \text{sgn } \tau = -1, \text{sgn } \pi = -1$

**8.56.**

(a)  $\tau \circ \sigma = 53142$ , (b)  $\pi \circ \sigma = 52413$ , (c)  $\sigma^{-1} = 32154$ , (d)  $\tau^{-1} = 14253$

**8.60.** (a)  $\det(T) = 17$ , (b)  $\det(T) = 4$ , (c) not defined

**8.61.** (a) 0, (b) 6, (c) 1

**8.64.** (a) 18, (b) 0

**8.65.** 17

**8.66.** (a) no, (b) yes, (c) yes, (d) no, (e) yes, (f ) no

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# CHAPTER 9

## ***Diagonalization: Eigenvalues and Eigenvectors***

### **9.1 Introduction**

---

The ideas in this chapter can be discussed from two points of view.

#### **Matrix Point of View**

---

Suppose an  $n$ -square matrix  $A$  is given. The matrix  $A$  is said to be *diagonalizable* if there exists a nonsingular matrix  $P$  such that

$$B = P^{-1}AP$$

is diagonal. This chapter discusses the diagonalization of a matrix  $A$ . In particular, an algorithm is given to find the matrix  $P$  when it exists.

#### **Linear Operator Point of View**

---

Suppose a linear operator  $T: V \rightarrow V$  is given. The linear operator  $T$  is said to be *diagonalizable* if there exists a basis  $S$  of  $V$  such that the matrix representation of  $T$  relative to the basis  $S$  is a diagonal matrix  $D$ . This chapter discusses conditions under which the linear operator  $T$  is diagonalizable.

## **Equivalence of the Two Points of View**

---

The above two concepts are essentially the same. Specifically, a square matrix  $A$  may be viewed as a linear operator  $F$  defined by

$$F(X) = AX$$

where  $X$  is a column vector, and  $B = P^{-1}AP$  represents  $F$  relative to a new coordinate system (basis)  $S$  whose elements are the columns of  $P$ . On the other hand, any linear operator  $T$  can be represented by a matrix  $A$  relative to one basis and, when a second basis is chosen,  $T$  is represented by the matrix

$$B = P^{-1}AP$$

where  $P$  is the change-of-basis matrix.

Most theorems will be stated in two ways: one in terms of matrices  $A$  and again in terms of linear mappings  $T$ .

## **Role of Underlying Field $K$**

---

The underlying number field  $K$  did not play any special role in our previous discussions on vector spaces and linear mappings. However, the diagonalization of a matrix  $A$  or a linear operator  $T$  will depend on the roots of a polynomial  $\Delta(t)$  over  $K$ , and these roots do depend on  $K$ . For example, suppose  $\Delta(t) = t^2 + 1$ . Then  $\Delta(t)$  has no roots if  $K = \mathbf{R}$ , the real field; but  $\Delta(t)$  has roots  $\pm i$  if  $K = \mathbf{C}$ , the complex field.

Furthermore, finding the roots of a polynomial with degree greater than two is a subject unto itself (frequently discussed in numerical analysis courses). Accordingly, our examples will usually lead to those polynomials  $\Delta(t)$  whose roots can be easily determined.

## 9.2 Polynomials of Matrices

---

Consider a polynomial  $f(t) = a_n t^n + \cdots + a_1 t + a_0$  over a field  $K$ . Recall (Section 2.8) that if  $A$  is any square matrix, then we define

$$f(A) = a_n A^n + \cdots + a_1 A + a_0 I$$

where  $I$  is the identity matrix. In particular, we say that  $A$  is a *root* of  $f(t)$  if  $f(A) = 0$ , the zero matrix.

**EXAMPLE 9.1** Let  $A = \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix}$ . Then  $A^2 = \begin{bmatrix} 7 & 10 \\ 15 & 22 \end{bmatrix}$ . Let

$$f(t) = 2t^2 - 3t + 5 \quad \text{and} \quad g(t) = t^2 - 5t - 2$$

Then

$$f(A) = 2A^2 - 3A + 5I = \begin{bmatrix} 14 & 20 \\ 30 & 44 \end{bmatrix} + \begin{bmatrix} -3 & -6 \\ -9 & -12 \end{bmatrix} + \begin{bmatrix} 5 & 0 \\ 0 & 5 \end{bmatrix} = \begin{bmatrix} 16 & 14 \\ 21 & 37 \end{bmatrix}$$

and

$$g(A) = A^2 - 5A - 2I = \begin{bmatrix} 7 & 10 \\ 15 & 22 \end{bmatrix} + \begin{bmatrix} -5 & -10 \\ -15 & -20 \end{bmatrix} + \begin{bmatrix} -2 & 0 \\ 0 & -2 \end{bmatrix} = \begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix}$$

Thus,  $A$  is a zero of  $g(t)$ .

The following theorem (proved in Problem 9.7) applies.

**THEOREM 9.1:** Let  $f$  and  $g$  be polynomials. For any square matrix  $A$  and scalar  $k$ ,

$$(i) (f + g)(A) = f(A) + g(A)$$

$$(ii) (fg)(A) = f(A)g(A)$$

$$(iii) (kf)(A) = kf(A)$$

$$(iv) f(A)g(A) = g(A)f(A).$$

Observe that (iv) tells us that any two polynomials in  $A$  commute.

## **Matrices and Linear Operators**

---

Now suppose that  $T: V \rightarrow V$  is a linear operator on a vector space  $V$ . Powers of  $T$  are defined by the composition operation:

$$T^2 = T \circ T, \quad T^3 = T^2 \circ T, \quad \dots$$

Also, for any polynomial  $f(t) = a_n t^n + \dots + a_1 t + a_0$ , we define  $f(T)$  in the same way as we did for matrices:

$$f(T) = a_n T^n + \dots + a_1 T + a_0 I$$

where  $I$  is now the identity mapping. We also say that  $T$  is a *zero* or *root* of  $f(t)$  if  $f(T) = 0$ , the zero mapping. We note that the relations in [Theorem 9.1](#) hold for linear operators as they do for matrices.

**Remark:** Suppose  $A$  is a matrix representation of a linear operator  $T$ . Then  $f(A)$  is the matrix representation of  $f(T)$ , and, in particular,  $f(T) = 0$  if and only if  $f(A) = 0$ .

## 9.3 Characteristic Polynomial, Cayley-Hamilton Theorem

---

Let  $A = [a_{ij}]$  be an  $n$ -square matrix. The matrix  $M = A - tI_n$ , where  $I_n$  is the  $n$ -square identity matrix and  $t$  is an indeterminate, may be obtained by subtracting  $t$  down the diagonal of  $A$ . The negative of  $M$  is the matrix  $tI_n - A$ , and its determinant

$$\Delta(t) = \det(tI_n - A) = (-1)^n \det(A - tI_n)$$

which is a polynomial in  $t$  of degree  $n$  and is called the *characteristic polynomial* of  $A$ .

We state an important theorem in linear algebra (proved in Problem 9.8).

**THEOREM 9.2:** (Cayley-Hamilton) Every matrix  $A$  is a root of its characteristic polynomial.

**Remark:** Suppose  $A = [a_{ij}]$  is a triangular matrix. Then  $tI - A$  is a triangular matrix with diagonal entries  $t - a_{ii}$ ; hence,

$$\Delta(t) = \det(tI - A) = (t - a_{11})(t - a_{22}) \cdots (t - a_{nn})$$

Observe that the roots of  $\Delta(t)$  are the diagonal elements of  $A$ .

**EXAMPLE 9.2** Let  $A = \begin{bmatrix} 1 & 3 \\ 4 & 5 \end{bmatrix}$ . Its characteristic polynomial is

$$\Delta(t) = |tI - A| = \begin{vmatrix} t-1 & -3 \\ -4 & t-5 \end{vmatrix} = (t-1)(t-5) - 12 = t^2 - 6t - 7$$

As expected from the Cayley-Hamilton theorem,  $A$  is a root of  $\Delta(t)$ ; that is,

$$\Delta(A) = A^2 - 6A - 7I = \begin{bmatrix} 13 & 18 \\ 24 & 37 \end{bmatrix} + \begin{bmatrix} -6 & -18 \\ -24 & -30 \end{bmatrix} + \begin{bmatrix} -7 & 0 \\ 0 & -7 \end{bmatrix} = \begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix}$$

Now suppose  $A$  and  $B$  are similar matrices, say  $B = P^{-1}AP$ , where  $P$  is invertible. We show that  $A$  and  $B$  have the same characteristic polynomial. Using  $tI = P^{-1}tIP$ , we have

$$\begin{aligned} \Delta_B(t) &= \det(tI - B) = \det(tI - P^{-1}AP) = \det(P^{-1}tIP - P^{-1}AP) \\ &= \det[P^{-1}(tI - A)P] = \det(P^{-1}) \det(tI - A) \det(P) \end{aligned}$$

Using the fact that determinants are scalars and commute and that  $\det(P^{-1}) \det(P) = 1$ , we finally obtain

$$\Delta_B(t) = \det(tI - A) = \Delta_A(t)$$

Thus, we have proved the following theorem.

**THEOREM 9.3:** Similar matrices have the same characteristic polynomial.

## **Characteristic Polynomials of Degrees 2 and 3**

There are simple formulas for the characteristic polynomials of matrices of orders 2 and 3.

(a) Suppose  $A = \begin{bmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{bmatrix}$ . Then

$$\Delta(t) = t^2 - (a_{11} + a_{22})t + \det(A) = t^2 - \operatorname{tr}(A)t + \det(A)$$

Here  $\operatorname{tr}(A)$  denotes the trace of  $A$ —that is, the sum of the diagonal elements of  $A$ .

(b) Suppose  $A = \begin{bmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{bmatrix}$ . Then

$$\Delta(t) = t^3 - \operatorname{tr}(A) t^2 + (A_{11} + A_{22} + A_{33})t - \det(A)$$

(Here  $A_{11}$ ,  $A_{22}$ ,  $A_{33}$  denote, respectively, the cofactors of  $a_{11}$ ,  $a_{22}$ ,  $a_{33}$ .)

**EXAMPLE 9.3** Find the characteristic polynomial of each of the following matrices:

(a)  $A = \begin{bmatrix} 5 & 3 \\ 2 & 10 \end{bmatrix}$ , (b)  $B = \begin{bmatrix} 7 & -1 \\ 6 & 2 \end{bmatrix}$ , (c)  $C = \begin{bmatrix} 5 & -2 \\ 4 & -4 \end{bmatrix}$ .

(a) We have  $\operatorname{tr}(A) = 5 + 10 = 15$  and  $|A| = 50 - 6 = 44$ ; hence,  $\Delta(t) = t^2 - 15t + 44$ .

(b) We have  $\operatorname{tr}(B) = 7 + 2 = 9$  and  $|B| = 14 + 6 = 20$ ; hence,  $\Delta(t) = t^2 - 9t + 20$ .

(c) We have  $\operatorname{tr}(C) = 5 - 4 = 1$  and  $|C| = 20 + 8 = 12$ ; hence,  $\Delta(t) = t^2 - t - 12$ .

**EXAMPLE 9.4** Find the characteristic polynomial of

$$A = \begin{bmatrix} 1 & 1 & 2 \\ 0 & 3 & 2 \\ 1 & 3 & 9 \end{bmatrix}.$$

We have  $\operatorname{tr}(A) = 1 + 3 + 9 = 13$ . The cofactors of the diagonal elements are as follows:

$$A_{11} = \begin{vmatrix} 3 & 2 \\ 3 & 9 \end{vmatrix} = 21, \quad A_{22} = \begin{vmatrix} 1 & 2 \\ 1 & 9 \end{vmatrix} = 7, \quad A_{33} = \begin{vmatrix} 1 & 1 \\ 0 & 3 \end{vmatrix} = 3$$

Thus,  $A_{11} + A_{22} + A_{33} = 31$ . Also,  $|A| = 27 + 2 + 0 - 6 - 6 - 0 = 17$ . Accordingly,

$$\Delta(t) = t^3 - 13t^2 + 31t - 17$$

**Remark:** The coefficients of the characteristic polynomial  $\Delta(t)$  of the 3-square matrix  $A$  are, with alternating signs, as follows:

$$S_1 = \text{tr}(A), \quad S_2 = A_{11} + A_{22} + A_{33}, \quad S_3 = \det(A)$$

We note that each  $S_k$  is the sum of all principal minors of  $A$  of order  $k$ .

The next theorem, whose proof lies beyond the scope of this text, tells us that this result is true in general.

**THEOREM 9.4:** Let  $A$  be an  $n$ -square matrix. Then its characteristic polynomial is

$$\Delta(t) = t^n - S_1 t^{n-1} + S_2 t^{n-2} + \cdots + (-1)^n S_n$$

where  $S_k$  is the sum of the principal minors of order  $k$ .

## **Characteristic Polynomial of a Linear Operator**

Now suppose  $T: V \rightarrow V$  is a linear operator on a vector space  $V$  of finite dimension. We define the *characteristic polynomial*  $\Delta(t)$  of  $T$  to be the characteristic polynomial of any matrix representation of  $T$ . Recall that if  $A$  and  $B$  are matrix representations of  $T$ , then  $B = P^{-1}AP$ , where  $P$  is a change-of-basis matrix. Thus,  $A$  and  $B$  are similar, and by [Theorem 9.3](#),  $A$  and  $B$  have the same characteristic polynomial. Accordingly, the characteristic polynomial of  $T$  is independent of the particular basis in which the matrix representation of  $T$  is computed.

Because  $f(T) = 0$  if and only if  $f(A) = 0$ , where  $f(t)$  is any polynomial and  $A$  is any matrix representation of  $T$ , we have the following analogous theorem for linear operators.

**THEOREM 9.2':** (Cayley–Hamilton) A linear operator  $T$  is a zero of its characteristic polynomial.

## 9.4 Diagonalization, Eigenvalues and Eigenvectors

---

Let  $A$  be any  $n$ -square matrix. Then  $A$  can be represented by (or is similar to) a diagonal matrix  $D = \text{diag}(k_1, k_2, \dots, k_n)$  if and only if there exists a basis  $S$  consisting of (column) vectors  $u_1, u_2, \dots, u_n$  such that

$$\begin{array}{rcl} Au_1 & = & k_1 u_1 \\ Au_2 & = & k_2 u_2 \\ & \dots\dots\dots & \\ Au_n & = & k_n u_n \end{array}$$

In such a case,  $A$  is said to be *diagonalizable*. Furthermore,  $D = P^{-1}AP$ , where  $P$  is the nonsingular matrix whose columns are, respectively, the basis vectors  $u_1, u_2, \dots, u_n$ .

The above observation leads us to the following definition.

**DEFINITION:** Let  $A$  be any square matrix. A scalar  $\lambda$  is called an *eigenvalue* of  $A$  if there exists a nonzero (column) vector  $v$  such that

$$Av = \lambda v$$

Any vector satisfying this relation is called an *eigenvector* of  $A$  *belonging* to the eigenvalue  $\lambda$ .

We note that each scalar multiple  $kv$  of an eigenvector  $v$  belonging to  $\lambda$  is also such an eigenvector, because

$$A(kv) = k(Av) = k(\lambda v) = \lambda(kv)$$

The set  $E_\lambda$  of all such eigenvectors is a subspace of  $V$  (Problem 9.19), called the *eigenspace* of  $\lambda$ . (If  $\dim E_\lambda = 1$ , then  $E_\lambda$  is called an *eigenline* and  $\lambda$  is called a *scaling factor*.)

The terms *characteristic value* and *characteristic vector* (or *proper value* and *proper vector*) are sometimes used instead of eigenvalue and eigenvector.

The above observation and definitions give us the following theorem.

**THEOREM 9.5:** An  $n$ -square matrix  $A$  is similar to a diagonal matrix  $D$  if and only if  $A$  has  $n$  linearly independent eigenvectors. In this case, the diagonal elements of  $D$  are the corresponding eigenvalues and  $D = P^{-1}AP$ , where  $P$  is the matrix whose columns are the eigenvectors.

Suppose a matrix  $A$  can be diagonalized as above, say  $P^{-1}AP = D$ , where  $D$  is diagonal. Then  $A$  has the extremely useful *diagonal factorization*:

$$A = PDP^{-1}$$

Using this factorization, the algebra of  $A$  reduces to the algebra of the diagonal matrix  $D$ , which can be easily calculated. Specifically, suppose  $D = \text{diag}(k_1, k_2, \dots, k_n)$ . Then

$$A^m = (PDP^{-1})^m = PD^mP^{-1} = P \operatorname{diag}(k_1^m, \dots, k_n^m)P^{-1}$$

More generally, for any polynomial  $f(t)$ ,

$$f(A) = f(PDP^{-1}) = Pf(D)P^{-1} = P \operatorname{diag}(f(k_1), f(k_2), \dots, f(k_n))P^{-1}$$

Furthermore, if the diagonal entries of  $D$  are nonnegative, let

$$B = P \operatorname{diag}(\sqrt{k_1}, \sqrt{k_2}, \dots, \sqrt{k_n}) P^{-1}$$

Then  $B$  is a *nonnegative square root* of  $A$ ; that is,  $B^2 = A$  and the eigenvalues of  $B$  are nonnegative.

**EXAMPLE 9.5** Let  $A = \begin{bmatrix} 3 & 1 \\ 2 & 2 \end{bmatrix}$  and  $v_1 = \begin{bmatrix} 1 \\ -2 \end{bmatrix}$  and  $v_2 = \begin{bmatrix} 1 \\ 1 \end{bmatrix}$ .

Then

$$Av_1 = \begin{bmatrix} 3 & 1 \\ 2 & 2 \end{bmatrix} \begin{bmatrix} 1 \\ -2 \end{bmatrix} = \begin{bmatrix} 1 \\ -2 \end{bmatrix} = v_1 \quad \text{and} \quad Av_2 = \begin{bmatrix} 3 & 1 \\ 2 & 2 \end{bmatrix} \begin{bmatrix} 1 \\ 1 \end{bmatrix} = \begin{bmatrix} 4 \\ 4 \end{bmatrix} = 4v_2$$

Thus,  $v_1$  and  $v_2$  are eigenvectors of  $A$  belonging, respectively, to the eigenvalues  $\lambda_1 = 1$  and  $\lambda_2 = 4$ . Observe that  $v_1$  and  $v_2$  are linearly independent and hence form a basis of  $\mathbf{R}^2$ . Accordingly,  $A$  is diagonalizable. Furthermore, let  $P$  be the matrix whose columns are the eigenvectors  $v_1$  and  $v_2$ . That is, let

$$P = \begin{bmatrix} 1 & 1 \\ -2 & 1 \end{bmatrix}, \quad \text{and so} \quad P^{-1} = \begin{bmatrix} \frac{1}{3} & -\frac{1}{3} \\ \frac{2}{3} & \frac{1}{3} \end{bmatrix}$$

Then  $A$  is similar to the diagonal matrix

$$D = P^{-1}AP = \begin{bmatrix} \frac{1}{3} & -\frac{1}{3} \\ \frac{2}{3} & \frac{1}{3} \end{bmatrix} \begin{bmatrix} 3 & 1 \\ 2 & 2 \end{bmatrix} \begin{bmatrix} 1 & 1 \\ -2 & 1 \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ 0 & 4 \end{bmatrix}$$

As expected, the diagonal elements 1 and 4 in  $D$  are the eigenvalues corresponding, respectively, to the eigenvectors  $v_1$  and  $v_2$ , which are the columns of  $P$ . In particular,  $A$  has the factorization

$$A = PDP^{-1} = \begin{bmatrix} 1 & 1 \\ -2 & 1 \end{bmatrix} \begin{bmatrix} 1 & 0 \\ 0 & 4 \end{bmatrix} \begin{bmatrix} \frac{1}{3} & -\frac{1}{3} \\ \frac{2}{3} & \frac{1}{3} \end{bmatrix}$$

Accordingly,

$$A^4 = \begin{bmatrix} 1 & 1 \\ -2 & 1 \end{bmatrix} \begin{bmatrix} 1 & 0 \\ 0 & 256 \end{bmatrix} \begin{bmatrix} \frac{1}{3} & -\frac{1}{3} \\ \frac{2}{3} & \frac{1}{3} \end{bmatrix} = \begin{bmatrix} 171 & 85 \\ 170 & 86 \end{bmatrix}$$

Moreover, suppose  $f(t) = t^3 - 5t^2 + 3t + 6$ ; hence,  $f(1) = 5$  and  $f(4) = 2$ . Then

$$f(A) = Pf(D)P^{-1} = \begin{bmatrix} 1 & 1 \\ -2 & 1 \end{bmatrix} \begin{bmatrix} 5 & 0 \\ 0 & 2 \end{bmatrix} \begin{bmatrix} \frac{1}{3} & -\frac{1}{3} \\ \frac{2}{3} & \frac{1}{3} \end{bmatrix} = \begin{bmatrix} 3 & -1 \\ -2 & 4 \end{bmatrix}$$

Last, we obtain a “positive square root” of  $A$ . Specifically, using  $\sqrt{1} = 1$  and  $\sqrt{4} = 2$ , we obtain the matrix

$$B = P\sqrt{D}P^{-1} = \begin{bmatrix} 1 & 1 \\ -2 & 1 \end{bmatrix} \begin{bmatrix} 1 & 0 \\ 0 & 2 \end{bmatrix} \begin{bmatrix} \frac{1}{3} & -\frac{1}{3} \\ \frac{2}{3} & \frac{1}{3} \end{bmatrix} = \begin{bmatrix} \frac{5}{3} & \frac{1}{3} \\ \frac{2}{3} & \frac{4}{3} \end{bmatrix}$$

where  $B^2 = A$  and where  $B$  has positive eigenvalues 1 and 2.

**Remark:** Throughout this chapter, we use the following fact:

$$\text{If } P = \begin{bmatrix} a & b \\ c & d \end{bmatrix}, \text{ then } P^{-1} = \begin{bmatrix} d/|P| & -b/|P| \\ -c/|P| & a/|P| \end{bmatrix}.$$

That is,  $P^{-1}$  is obtained by interchanging the diagonal elements  $a$  and  $d$  of  $P$ , taking the negatives of the nondiagonal elements  $b$  and  $c$ , and dividing each element by the determinant  $|P|$ .

## **Properties of Eigenvalues and Eigenvectors**

---

[Example 9.5](#) indicates the advantages of a diagonal representation (factorization) of a square matrix. In the following theorem (proved in Problem 9.20), we list properties that help us to find such a representation.

**THEOREM 9.6:** Let  $A$  be a square matrix. Then the following are equivalent.

- (i) A scalar  $\lambda$  is an eigenvalue of  $A$ .
- (ii) The matrix  $M = A - \lambda I$  is singular.
- (iii) The scalar  $\lambda$  is a root of the characteristic polynomial  $\Delta(t)$  of  $A$ .

The eigenspace  $E_\lambda$  of an eigenvalue  $\lambda$  is the solution space of the homogeneous system  $MX = 0$ , where  $M = A - \lambda I$ ; that is,  $M$  is obtained by subtracting  $\lambda$  down the diagonal of  $A$ .

Some matrices have no eigenvalues and hence no eigenvectors. However, using [Theorem 9.6](#) and the Fundamental Theorem of Algebra (every polynomial over the complex field  $\mathbf{C}$  has a root), we obtain the following result.

**THEOREM 9.7:** Let  $A$  be a square matrix over the complex field  $\mathbf{C}$ . Then  $A$  has at least one

eigenvalue.

The following theorems will be used subsequently. (The theorem equivalent to [Theorem 9.8](#) for linear operators is proved in Problem 9.21, and [Theorem 9.9](#) is proved in Problem 9.22.)

**THEOREM 9.8:** Suppose  $v_1, v_2, \dots, v_n$  are nonzero eigenvectors of a matrix  $A$  belonging to distinct eigenvalues  $\lambda_1, \lambda_2, \dots, \lambda_n$ . Then  $v_1, v_2, \dots, v_n$  are linearly independent.

**THEOREM 9.9:** Suppose the characteristic polynomial  $\Delta(t)$  of an  $n$ -square matrix  $A$  is a product of  $n$  distinct factors, say,  $\Delta(t) = (t - a_1)(t - a_2)\cdots(t - a_n)$ . Then  $A$  is similar to the diagonal matrix  $D = \text{diag}(a_1, a_2, \dots, a_n)$ .

If  $\lambda$  is an eigenvalue of a matrix  $A$ , then the *algebraic multiplicity* of  $\lambda$  is defined to be the multiplicity of  $\lambda$  as a root of the characteristic polynomial of  $A$ , and the *geometric multiplicity* of  $\lambda$  is defined to be the dimension of its eigenspace,  $\dim E_\lambda$ . The following theorem (whose equivalent for linear operators is proved in Problem 9.23) holds.

**THEOREM 9.10:** The geometric multiplicity of an eigenvalue  $\lambda$  of a matrix  $A$  does not exceed its algebraic multiplicity.

## **Diagonalization of Linear Operators**

---

Consider a linear operator  $T: V \rightarrow V$ . Then  $T$  is said to be *diagonalizable* if it can be represented by a diagonal matrix

D. Thus,  $T$  is diagonalizable if and only if there exists a basis  $S = \{u_1, u_2, \dots, u_n\}$  of  $V$  for which

$$\begin{array}{rcl} T(u_1) & = & k_1 u_1 \\ T(u_2) & = & k_2 u_2 \\ & \dots\dots\dots & \\ T(u_n) & = & k_n u_n \end{array}$$

In such a case,  $T$  is represented by the diagonal matrix

$$D = \text{diag}(k_1, k_2, \dots, k_n)$$

relative to the basis  $S$ .

The above observation leads us to the following definitions and theorems, which are analogous to the definitions and theorems for matrices discussed above.

**DEFINITION:** Let  $T$  be a linear operator. A scalar  $\lambda$  is called an *eigenvalue* of  $T$  if there exists a nonzero vector  $u$  such that  $T(u) = \lambda u$ . Every vector satisfying this relation is called an *eigenvector* of  $T$  *belonging* to the eigenvalue  $\lambda$ .

The set  $E_\lambda$  of all eigenvectors belonging to an eigenvalue  $\lambda$  is a subspace of  $V$ , called the *eigenspace* of  $\lambda$ .

(Alternatively,  $\lambda$  is an eigenvalue of  $T$  if  $\lambda I - T$  is singular, and, in this case,  $E_\lambda$  is the kernel of  $\lambda I - T$ .) The *algebraic* and *geometric multiplicities* of an eigenvalue  $\lambda$  of a linear operator  $T$  are defined in the same way as those of an eigenvalue of a matrix  $A$ .

The following theorems apply to a linear operator  $T$  on a vector space  $V$  of finite dimension.

**THEOREM 9.5':**  $T$  can be represented by a diagonal matrix  $D$  if and only if there exists a basis  $S$  of  $V$  consisting of eigenvectors of  $T$ . In this case, the diagonal elements of  $D$  are the corresponding eigenvalues.

**THEOREM 9.6':** Let  $T$  be a linear operator. Then the following are equivalent:

- (i) A scalar  $\lambda$  is an eigenvalue of  $T$ .
- (ii) The linear operator  $\lambda I - T$  is singular.
- (iii) The scalar  $\lambda$  is a root of the characteristic polynomial  $\Delta(t)$  of  $T$ .

**THEOREM 9.7':** Suppose  $V$  is a complex vector space. Then  $T$  has at least one eigenvalue.

**THEOREM 9.8':** Suppose  $v_1, v_2, \dots, v_n$  are nonzero eigenvectors of a linear operator  $T$  belonging to distinct eigenvalues  $\lambda_1, \lambda_2, \dots, \lambda_n$ . Then  $v_1, v_2, \dots, v_n$  are linearly independent.

**THEOREM 9.9':** Suppose the characteristic polynomial  $\Delta(t)$  of  $T$  is a product of  $n$  distinct factors, say,  $\Delta(t) = (t - a_1)(t - a_2) \cdots (t - a_n)$ . Then  $T$  can be represented by the diagonal matrix  $D = \text{diag}(a_1, a_2, \dots, a_n)$ .

**THEOREM 9.10':** The geometric multiplicity of an eigenvalue  $\lambda$  of  $T$  does not exceed its algebraic multiplicity.

**Remark:** The following theorem reduces the investigation of the diagonalization of a linear operator  $T$  to the diagonalization of a matrix  $A$ .

**THEOREM 9.11:** Suppose  $A$  is a matrix representation of  $T$ . Then  $T$  is diagonalizable if and only if  $A$  is diagonalizable.

## 9.5 Computing Eigenvalues and Eigenvectors, Diagonalizing Matrices

---

This section gives an algorithm for computing eigenvalues and eigenvectors for a given square matrix  $A$  and for determining whether or not a nonsingular matrix  $P$  exists such that  $P^{-1}AP$  is diagonal.

**ALGORITHM 9.1:** (Diagonalization Algorithm) The input is an  $n$ -square matrix  $A$ .

**Step 1.** Find the characteristic polynomial  $\Delta(t)$  of  $A$ .

**Step 2.** Find the roots of  $\Delta(t)$  to obtain the eigenvalues of  $A$ .

**Step 3.** Repeat (a) and (b) for each eigenvalue  $\lambda$  of  $A$ .

(a) Form the matrix  $M = A - \lambda I$  by subtracting  $\lambda$  down the diagonal of  $A$ .

(b) Find a basis for the solution space of the homogeneous system  $MX = 0$ . (These basis vectors are linearly independent eigenvectors of  $A$  belonging to  $\lambda$ .)

**Step 4.** Consider the collection  $S = \{v_1, v_2, \dots, v_m\}$  of all eigenvectors obtained in Step 3.

- (a) If  $m \neq n$ , then  $A$  is not diagonalizable.
- (b) If  $m = n$ , then  $A$  is diagonalizable. Specifically, let  $P$  be the matrix whose columns are the eigenvectors  $v_1, v_2, \dots, v_n$ . Then

$$D = P^{-1}AP = \text{diag}(\lambda_1, \lambda_2, \dots, \lambda_n)$$

where  $\lambda_i$  is the eigenvalue corresponding to the eigenvector  $v_i$ .

**EXAMPLE 9.6** The diagonalization algorithm is applied to

$$A = \begin{bmatrix} 4 & 2 \\ 3 & -1 \end{bmatrix},$$

(1) The characteristic polynomial  $\Delta(t)$  of  $A$  is computed:

$$\text{tr}(A) = 4 - 1 = 3, \quad |A| = -4 - 6 = -10, \quad \text{so} \quad \Delta(t) = t^2 - 3t - 10 = (t - 5)(t + 2)$$

(2) Set  $\Delta(t) = t^2 - 3t - 10 = (t - 5)(t + 2) = 0$ . The roots  $\lambda_1 = 5$  and  $\lambda_2 = -2$  are the eigenvalues of  $A$ .

(3) (i) We find an eigenvector  $v_1$  of  $A$  belonging to the eigenvalue  $\lambda_1 = 5$ . Subtract  $\lambda_1 = 5$  down the diagonal of  $A$  to obtain the matrix

$$M = \begin{bmatrix} -1 & 2 \\ 3 & -6 \end{bmatrix} \text{ and the homogeneous system } \begin{array}{l} -x + 2y = 0 \\ 3x - 6y = 0 \end{array} \quad \text{or} \quad -x + 2y = 0$$

The system has only one independent solution, for example,  
 $v_1 = (2, 1)$ .

(ii) We find an eigenvector  $v_2$  of  $A$  belonging to the eigenvalue  $\lambda_2 = -2$ . Subtract  $-2$  (or add 2) down the diagonal of  $A$  to obtain the matrix

$$M = \begin{bmatrix} 6 & 2 \\ 3 & 1 \end{bmatrix} \text{ and the homogeneous system } \begin{matrix} 6x + 2y = 0 \\ 3x + y = 0 \end{matrix} \text{ or } 3x + y = 0$$

The system has only one independent solution, for example,  
 $v_2 = (-1, 3)$ .

(4) Let  $P$  be the matrix whose columns are the eigenvectors  $v_1$  and  $v_2$ . Then  $P = \begin{bmatrix} 2 & -1 \\ 1 & 3 \end{bmatrix}$ . Thus  $D = P^{-1}AP$  is the diagonal matrix whose diagonal entries are the corresponding eigenvalues:

$$D = P^{-1}AP = \begin{bmatrix} 3/7 & 1/7 \\ -1/7 & 2/7 \end{bmatrix} \begin{bmatrix} 4 & 2 \\ 3 & -1 \end{bmatrix} \begin{bmatrix} 2 & -1 \\ 1 & 3 \end{bmatrix} = \begin{bmatrix} 5 & 0 \\ 0 & -2 \end{bmatrix}$$

**Remark:** Find a  $2 \times 2$  matrix  $A$  with eigenvalues  $\lambda_1 = 2$  and  $\lambda_2 = 3$  and corresponding eigenvectors  $v_1 = (1, 3)$  and  $v_2 = (1, 4)$ . We know that  $P^{-1}AP = D$  where  $P = \begin{bmatrix} 1 & 1 \\ 3 & 4 \end{bmatrix}$  and  $D = \begin{bmatrix} 2 & 0 \\ 0 & 3 \end{bmatrix}$ .

$$[\text{Here } P^{-1} = \begin{bmatrix} 4 & -1 \\ -3 & 1 \end{bmatrix}.] \text{ Thus } A = PDP^{-1} = \begin{bmatrix} 1 & 1 \\ 3 & 4 \end{bmatrix} \begin{bmatrix} 2 & 0 \\ 0 & 3 \end{bmatrix} \begin{bmatrix} 4 & -1 \\ -3 & 1 \end{bmatrix} = \begin{bmatrix} -1 & 1 \\ -12 & 6 \end{bmatrix}.$$

**EXAMPLE 9.7** Consider the matrix  $B = \begin{bmatrix} 5 & -1 \\ 1 & 3 \end{bmatrix}$ . We have

$$\text{tr}(B) = 5 + 3 = 8, \quad |B| = 15 + 1 = 16; \quad \text{so} \quad \Delta(t) = t^2 - 8t + 16 = (t - 4)^2$$

Accordingly,  $\lambda = 4$  is the only eigenvalue of  $B$ .

Subtract  $\lambda = 4$  down the diagonal of  $B$  to obtain the matrix

$$M = \begin{bmatrix} 1 & -1 \\ 1 & -1 \end{bmatrix} \text{ and the homogeneous system } \begin{matrix} x - y = 0 \\ x - y = 0 \end{matrix} \text{ or } x - y = 0$$

The system has only one independent solution; for example,  $x = 1$ ,  $y = 1$ . Thus,  $v = (1, 1)$  and its multiples are the only eigenvectors of  $B$ . Accordingly,  $B$  is not diagonalizable, because there does not exist a basis consisting of eigenvectors of  $B$ .

**EXAMPLE 9.8** Consider the matrix  $A = \begin{bmatrix} 3 & -5 \\ 2 & -3 \end{bmatrix}$ . Here  $\text{tr}(A) = 3 - 3 = 0$  and  $|A| = 9 + 10 = 1$ . Thus,  $\Delta(t) = t^2 + 1$  is the characteristic polynomial of  $A$ . We consider two cases:

- (a)  $A$  is a matrix over the real field  $\mathbf{R}$ . Then  $\Delta(t)$  has no (real) roots. Thus,  $A$  has no eigenvalues and no eigenvectors, and so  $A$  is not diagonalizable.
- (b)  $A$  is a matrix over the complex field  $\mathbf{C}$ . Then  $\Delta(t) = (t - i)(t + i)$  has two roots,  $i$  and  $-i$ . Thus,  $A$  has two distinct eigenvalues  $i$  and  $-i$ , and hence,  $A$  has two independent eigenvectors. Accordingly there exists a nonsingular matrix  $P$  over the complex field  $\mathbf{C}$  for which

$$P^{-1}AP = \begin{bmatrix} i & 0 \\ 0 & -i \end{bmatrix}$$

Therefore,  $A$  is diagonalizable (over  $\mathbf{C}$ ).

## 9.6 Diagonalizing Real Symmetric Matrices and Quadratic Forms

---

There are many real matrices  $A$  that are not diagonalizable. In fact, some real matrices may not have any (real) eigenvalues. However, if  $A$  is a real symmetric matrix, then these problems do not exist. Namely, we have the following theorems.

**THEOREM 9.12:** Let  $A$  be a real symmetric matrix. Then each root  $\lambda$  of its characteristic polynomial is real.

**THEOREM 9.13:** Let  $A$  be a real symmetric matrix. Suppose  $u$  and  $v$  are eigenvectors of  $A$  belonging to distinct eigenvalues  $\lambda_1$  and  $\lambda_2$ . Then  $u$  and  $v$  are orthogonal, that is,  $\langle u, v \rangle = 0$ .

The above two theorems give us the following fundamental result.

**THEOREM 9.14:** Let  $A$  be a real symmetric matrix. Then there exists an orthogonal matrix  $P$  such that  $D = P^{-1}AP$  is diagonal.

The orthogonal matrix  $P$  is obtained by normalizing a basis of orthogonal eigenvectors of  $A$  as illustrated below. In such a case, we say that  $A$  is “orthogonally diagonalizable.”

**EXAMPLE 9.9** Let  $A = \begin{bmatrix} 2 & -2 \\ -2 & 5 \end{bmatrix}$ , a real symmetric matrix. Find an orthogonal matrix  $P$  such that  $P^{-1}AP$  is diagonal.

First we find the characteristic polynomial  $\Delta(t)$  of  $A$ . We have

$$\operatorname{tr}(A) = 2 + 5 = 7, \quad |A| = 10 - 4 = 6; \quad \text{so} \quad \Delta(t) = t^2 - 7t + 6 = (t - 6)(t - 1)$$

Accordingly,  $\lambda_1 = 6$  and  $\lambda_2 = 1$  are the eigenvalues of  $A$ .

(a) Subtracting  $\lambda_1 = 6$  down the diagonal of  $A$  yields the matrix

$$M = \begin{bmatrix} -4 & -2 \\ -2 & -1 \end{bmatrix} \quad \text{and the homogeneous system} \quad \begin{array}{l} -4x - 2y = 0 \\ -2x - y = 0 \end{array} \quad \text{or} \quad 2x + y = 0$$

A nonzero solution is  $u_1 = (1, -2)$ .

(b) Subtracting  $\lambda_2 = 1$  down the diagonal of  $A$  yields the matrix

$$M = \begin{bmatrix} 1 & -2 \\ -2 & 4 \end{bmatrix} \quad \text{and the homogeneous system} \quad x - 2y = 0$$

(The second equation drops out, because it is a multiple of the first equation.) A nonzero solution is  $u_2 = (2, 1)$ .

As expected from [Theorem 9.13](#),  $u_1$  and  $u_2$  are orthogonal. Normalizing  $u_1$  and  $u_2$  yields the orthonormal vectors

$$\hat{u}_1 = (1/\sqrt{5}, -2/\sqrt{5}) \quad \text{and} \quad \hat{u}_2 = (2/\sqrt{5}, 1/\sqrt{5})$$

Finally, let  $P$  be the matrix whose columns are

$$P = \begin{bmatrix} 1/\sqrt{5} & 2/\sqrt{5} \\ -2/\sqrt{5} & 1/\sqrt{5} \end{bmatrix} \quad \text{and} \quad P^{-1}AP = \begin{bmatrix} 6 & 0 \\ 0 & 1 \end{bmatrix} \quad \text{and}$$

$$q(x_1, x_2, \dots, x_n) = \sum_i c_i x_i^2 + \sum_{i < j} d_{ij} x_i x_j, \quad \text{where} \quad c_i, d_{ij} \in \mathbf{R},$$

respectively. Then

$$a_{ij} = a_{ji} = \frac{1}{2} d_{ij}$$

As expected, the diagonal entries of  $P^{-1}AP$  are the eigenvalues corresponding to the columns of  $P$ .

The procedure in the above [Example 9.9](#) is formalized in the following algorithm, which finds an orthogonal matrix  $P$  such that  $P^{-1}AP$  is diagonal.

**ALGORITHM 9.2:** (Orthogonal Diagonalization Algorithm)  
The input is a real symmetric matrix  $A$ .

**Step 1.** Find the characteristic polynomial  $\Delta(t)$  of  $A$ .

**Step 2.** Find the eigenvalues of  $A$ , which are the roots of  $\Delta(t)$ .

**Step 3.** For each eigenvalue  $\lambda$  of  $A$  in Step 2, find an orthogonal basis of its eigenspace.

**Step 4.** Normalize all eigenvectors in Step 3, which then forms an orthonormal basis of  $\mathbf{R}^n$ .

**Step 5.** Let  $P$  be the matrix whose columns are the normalized eigenvectors in Step 4.

### **Application to Quadratic Forms**

---

Let  $q$  be a real polynomial in variables  $x_1, x_2, \dots, x_n$  such that every term in  $q$  has degree two; that is,

$$q(x_1, x_2, \dots, x_n) = \sum_i c_i x_i^2 + \sum_{i < j} d_{ij} x_i x_j, \quad \text{where} \quad c_i, d_{ij} \in \mathbf{R}$$

Then  $q$  is called a *quadratic form*. If there are no cross-product terms  $x_i x_j$  (i.e., all  $d_{ij} = 0$ ), then  $q$  is said to be *diagonal*.

The above quadratic form  $q$  determines a real symmetric matrix  $A = [a_{ij}]$ , where  $a_{ii} = c_i$  and  $a_{ij} = a_{ji} = \frac{1}{2}d_{ij}$ . Namely,  $q$  can be written in the matrix form

$$q(X) = X^T A X$$

where  $X = [x_1, x_2, \dots, x_n]^T$  is the column vector of the variables. Furthermore, suppose  $X = PY$  is a linear

substitution of the variables. Then substitution in the quadratic form yields

$$q(Y) = (PY)^T A (PY) = Y^T (P^T A P) Y$$

Thus,  $P^T A P$  is the matrix representation of  $q$  in the new variables.

We seek an orthogonal matrix  $P$  such that the *orthogonal substitution*  $X = PY$  yields a diagonal quadratic form for which  $P^T A P$  is diagonal. Because  $P$  is orthogonal,  $P^T = P^{-1}$ , and hence,  $P^T A P = P^{-1} A P$ . The above theory yields such an orthogonal matrix  $P$ .

**EXAMPLE 9.10** Consider the quadratic form

$$q(x, y) = 2x^2 - 4xy + 5y^2 = X^T A X, \quad \text{where} \quad A = \begin{bmatrix} 2 & -2 \\ -2 & 5 \end{bmatrix} \quad \text{and} \quad X = \begin{bmatrix} x \\ y \end{bmatrix}$$

By [Example 9.9](#),

$$P^{-1} A P = \begin{bmatrix} 6 & 0 \\ 0 & 1 \end{bmatrix} = P^T A P, \quad \text{where} \quad P = \begin{bmatrix} 1/\sqrt{5} & 2/\sqrt{5} \\ -2/\sqrt{5} & 1/\sqrt{5} \end{bmatrix}$$

Let  $Y = [s, t]^T$ : Then matrix  $P$  corresponds to the following linear orthogonal substitution  $x = PY$  of the variables  $x$  and  $y$  in terms of the variables  $s$  and  $t$ :

$$x = \frac{1}{\sqrt{5}}s + \frac{2}{\sqrt{5}}t, \quad y = -\frac{2}{\sqrt{5}}s + \frac{1}{\sqrt{5}}t$$

This substitution in  $q(x, y)$  yields the diagonal quadratic form  $q(s, t) = 6s^2 + t^2$ .

## 9.7 Minimal Polynomial

---

Let  $A$  be any square matrix. Let  $J(A)$  denote the collection of all polynomials  $f(t)$  for which  $A$  is a root—that is, for which  $f(A) = 0$ . The set  $J(A)$  is not empty, because the Cayley-Hamilton [Theorem 9.1](#) tells us that the characteristic polynomial  $\Delta_A(t)$  of  $A$  belongs to  $J(A)$ . Let  $m(t)$  denote the monic polynomial of lowest degree in  $J(A)$ . (Such a polynomial  $m(t)$  exists and is unique.) We call  $m(t)$  the *minimal polynomial* of the matrix  $A$ .

**Remark:** A polynomial  $f(t) \neq 0$  is *monic* if its leading coefficient equals one.

The following theorem (proved in [Problem 9.33](#)) holds.

**THEOREM 9.15:** The minimal polynomial  $m(t)$  of a matrix (linear operator)  $A$  divides every polynomial that has  $A$  as a zero. In particular,  $m(t)$  divides the characteristic polynomial  $\Delta(t)$  of  $A$ .

There is an even stronger relationship between  $m(t)$  and  $\Delta(t)$ .

**THEOREM 9.16:** The characteristic polynomial  $\Delta(t)$  and the minimal polynomial  $m(t)$  of a matrix  $A$  have the same irreducible factors.

This theorem (proved in [Problem 9.35](#)) does not say that  $m(t) = \Delta(t)$ , only that any irreducible factor of one must divide the other. In particular, because a linear factor is irreducible,  $m(t)$  and  $\Delta(t)$  have the same linear factors. Hence, they have the same roots. Thus, we have the following theorem.

**THEOREM 9.17:** A scalar  $\lambda$  is an eigenvalue of the matrix  $A$  if and only if  $\lambda$  is a root of the minimal polynomial of  $A$ .

**EXAMPLE 9.11** Find the minimal polynomial  $m(t)$  of

$$A = \begin{bmatrix} 2 & 2 & -5 \\ 3 & 7 & -15 \\ 1 & 2 & -4 \end{bmatrix}.$$

First find the characteristic polynomial  $\Delta(t)$  of  $A$ . We have

$$\text{tr}(A) = 5, \quad A_{11} + A_{22} + A_{33} = 2 - 3 + 8 = 7, \quad \text{and} \quad |A| = 3$$

Hence,

$$\Delta(t) = t^3 - 5t^2 + 7t - 3 = (t - 1)^2(t - 3)$$

The minimal polynomial  $m(t)$  must divide  $\Delta(t)$ . Also, each irreducible factor of  $\Delta(t)$  (i.e.,  $t - 1$  and  $t - 3$ ) must also be a factor of  $m(t)$ . Thus,  $m(t)$  is exactly one of the following:

$$f(t) = (t - 3)(t - 1) \quad \text{or} \quad g(t) = (t - 3)(t - 1)^2$$

We know, by the Cayley-Hamilton theorem, that  $g(A) = \Delta(A) = 0$ . Hence, we need only test  $f(t)$ . We have

$$f(A) = (A - I)(A - 3I) = \begin{bmatrix} 1 & 2 & -5 \\ 3 & 6 & -15 \\ 1 & 2 & -5 \end{bmatrix} \begin{bmatrix} -1 & 2 & -5 \\ 3 & 4 & -15 \\ 1 & 2 & -7 \end{bmatrix} = \begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}$$

Thus,  $f(t) = m(t) = (t - 1)(t - 3) = t^2 - 4t + 3$  is the minimal polynomial of  $A$ .

**EXAMPLE 9.12**

(a) Consider the following two  $r$ -square matrices, where  $a \neq 0$ :

$$J(\lambda, r) = \begin{bmatrix} \lambda & 1 & 0 & \dots & 0 & 0 \\ 0 & \lambda & 1 & \dots & 0 & 0 \\ \dots & \dots & \dots & \dots & \dots & \dots \\ 0 & 0 & 0 & \dots & \lambda & 1 \\ 0 & 0 & 0 & \dots & 0 & \lambda \end{bmatrix} \quad \text{and} \quad A = \begin{bmatrix} \lambda & a & 0 & \dots & 0 & 0 \\ 0 & \lambda & a & \dots & 0 & 0 \\ \dots & \dots & \dots & \dots & \dots & \dots \\ 0 & 0 & 0 & \dots & \lambda & a \\ 0 & 0 & 0 & \dots & 0 & \lambda \end{bmatrix}$$

The first matrix, called a Jordan Block, has  $\lambda$ 's on the diagonal, 1's on the *superdiagonal* (consisting of the entries above the diagonal entries), and 0's elsewhere. The second matrix  $A$  has  $\lambda$ 's on the diagonal,  $a$ 's on the superdiagonal, and 0's elsewhere. [Thus,  $A$  is a generalization of  $J(\lambda, r)$ .] One can show that

$$f(t) = (t - \lambda)^r$$

is both the characteristic and minimal polynomial of both  $J(\lambda, r)$  and  $A$ .

(b) Consider an arbitrary monic polynomial:

$$f(t) = t^n + a_{n-1}t^{n-1} + \dots + a_1t + a_0$$

Let  $C(f)$  be the  $n$ -square matrix with 1's on the *subdiagonal* (consisting of the entries below the diagonal entries), the negatives of the coefficients in the last column, and 0's elsewhere as follows:

$$C(f) = \begin{bmatrix} 0 & 0 & \dots & 0 & -a_0 \\ 1 & 0 & \dots & 0 & -a_1 \\ 0 & 1 & \dots & 0 & -a_2 \\ \dots & \dots & \dots & \dots & \dots \\ 0 & 0 & \dots & 1 & -a_{n-1} \end{bmatrix}$$

Then  $C(f)$  is called the *companion matrix* of the polynomial  $f(t)$ . Moreover, the minimal polynomial  $m(t)$  and the characteristic polynomial  $\Delta(t)$  of the companion matrix  $C(f)$  are both equal to the original polynomial  $f(t)$ .

## **Minimal Polynomial of a Linear Operator**

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The *minimal polynomial*  $m(t)$  of a linear operator  $T$  is defined to be the monic polynomial of lowest degree for which  $T$  is a root. However, for any polynomial  $f(t)$ , we have

$$f(T) = 0 \quad \text{if and only if} \quad f(A) = 0$$

where  $A$  is any matrix representation of  $T$ . Accordingly,  $T$  and  $A$  have the same minimal polynomials. Thus, the above theorems on the minimal polynomial of a matrix also hold for the minimal polynomial of a linear operator. That is, we have the following theorems.

**THEOREM 9.15':** The minimal polynomial  $m(t)$  of a linear operator  $T$  divides every polynomial that has  $T$  as a root. In particular,  $m(t)$  divides the characteristic polynomial  $\Delta(t)$  of  $T$ .

**THEOREM 9.16':** The characteristic and minimal polynomials of a linear operator  $T$  have the same irreducible factors.

**THEOREM 9.17':** A scalar  $\lambda$  is an eigenvalue of a linear operator  $T$  if and only if  $\lambda$  is a root of the minimal polynomial  $m(t)$  of  $T$ .

## 9.8 Characteristic and Minimal Polynomials of Block Matrices

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This section discusses the relationship of the characteristic polynomial and the minimal polynomial to certain (square) block matrices.

### Characteristic Polynomial and Block Triangular Matrices

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Suppose  $M$  is a block triangular matrix, say  $M = \begin{bmatrix} A_1 & B \\ 0 & A_2 \end{bmatrix}$ , where  $A_1$  and  $A_2$  are square matrices. Then  $tI - M$  is also a block triangular matrix, with diagonal blocks  $tI - A_1$  and  $tI - A_2$ . Thus,

$$|tI - M| = \begin{vmatrix} tI - A_1 & -B \\ 0 & tI - A_2 \end{vmatrix} = |tI - A_1| |tI - A_2|$$

That is, the characteristic polynomial of  $M$  is the product of the characteristic polynomials of the diagonal blocks  $A_1$  and  $A_2$ .

By induction, we obtain the following useful result.

**THEOREM 9.18:** Suppose  $M$  is a block triangular matrix with diagonal blocks  $A_1, A_2, \dots, A_r$ . Then the characteristic polynomial of  $M$  is the product of the characteristic polynomials of the diagonal blocks  $A_i$ ; that is,

$$\Delta_M(t) = \Delta_{A_1}(t) \Delta_{A_2}(t) \dots \Delta_{A_r}(t)$$

**EXAMPLE 9.13** Consider the matrix  $M = \begin{bmatrix} 9 & -1 & 1 & 5 & 7 \\ 8 & -3 & 1 & 2 & -4 \\ 0 & 0 & 1 & 3 & 6 \\ 0 & 0 & 1 & -1 & 8 \end{bmatrix}$ .

Then  $M$  is a block triangular matrix with diagonal blocks  $A = \begin{bmatrix} 9 & -1 \\ 8 & 3 \end{bmatrix}$  and  $B = \begin{bmatrix} 3 & 6 \\ -1 & 8 \end{bmatrix}$ . Here

$$\begin{aligned} \operatorname{tr}(A) &= 9 + 3 = 12, & \det(A) &= 27 + 8 = 35, & \text{and so } \Delta_A(t) &= t^2 - 12t + 35 = (t - 5)(t - 7) \\ \operatorname{tr}(B) &= 3 + 8 = 11, & \det(B) &= 24 + 6 = 30, & \text{and so } \Delta_B(t) &= t^2 - 11t + 30 = (t - 5)(t - 6) \end{aligned}$$

Accordingly, the characteristic polynomial of  $M$  is the product

$$\Delta_M(t) = \Delta_A(t)\Delta_B(t) = (t - 5)^2(t - 6)(t - 7)$$

## Minimal Polynomial and Block Diagonal Matrices

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The following theorem (proved in [Problem 9.36](#)) holds.

**THEOREM 9.19:** Suppose  $M$  is a block diagonal matrix with diagonal blocks  $A_1, A_2, \dots, A_r$ . Then the minimal polynomial of  $M$  is equal to the least common multiple (LCM) of the minimal polynomials of the diagonal blocks  $A_i$ .

**Remark:** We emphasize that this theorem applies to block diagonal matrices, whereas the analogous [Theorem 9.18](#) on characteristic polynomials applies to block triangular matrices.

**EXAMPLE 9.14** Find the characteristic polynomial  $\Delta(t)$  and the minimal polynomial  $m(t)$  of the block diagonal matrix:

$$M = \begin{bmatrix} 2 & 5 & 0 & 0 & 0 \\ 0 & 2 & 0 & 0 & 0 \\ 0 & 0 & 4 & 2 & 0 \\ 0 & 0 & 3 & 5 & 0 \\ 0 & 0 & 0 & 0 & 7 \end{bmatrix} = \text{diag}(A_1, A_2, A_3), \text{ where } A_1 = \begin{bmatrix} 2 & 5 \\ 0 & 2 \end{bmatrix}, A_2 = \begin{bmatrix} 4 & 2 \\ 3 & 5 \end{bmatrix}, A_3 = [7]$$

Then  $\Delta(t)$  is the product of the characterization polynomials  $\Delta_1(t)$ ,  $\Delta_2(t)$ ,  $\Delta_3(t)$  of  $A_1$ ,  $A_2$ ,  $A_3$ , respectively. One can show that

$$\Delta_1(t) = (t - 2)^2, \quad \Delta_2(t) = (t - 2)(t - 7), \quad \Delta_3(t) = t - 7$$

Thus,  $\Delta(t) = (t - 2)^3(t - 7)^2$ . [As expected,  $\deg \Delta(t) = 5$ .]

The minimal polynomials  $m_1(t)$ ,  $m_2(t)$ ,  $m_3(t)$  of the diagonal blocks  $A_1$ ,  $A_2$ ,  $A_3$ , respectively, are equal to the characteristic polynomials; that is,

$$m_1(t) = (t - 2)^2, \quad m_2(t) = (t - 2)(t - 7), \quad m_3(t) = t - 7$$

But  $m(t)$  is equal to the least common multiple of  $m_1(t)$ ,  $m_2(t)$ ,  $m_3(t)$ . Thus,  $m(t) = (t - 2)^2(t - 7)$ .

## SOLVED PROBLEMS

### Polynomials of Matrices, Characteristic Polynomials

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**9.1.** Let  $A = \begin{bmatrix} 1 & -2 \\ 4 & 5 \end{bmatrix}$ . Find  $f(A)$ , where

$$(a) \quad f(t) = t^2 - 3t + 7, \quad (b) \quad f(t) = t^2 - 6t + 13$$

$$\text{First find } A^2 = \begin{bmatrix} 1 & -2 \\ 4 & 5 \end{bmatrix} \begin{bmatrix} 1 & -2 \\ 4 & 5 \end{bmatrix} = \begin{bmatrix} -7 & -12 \\ 24 & 17 \end{bmatrix}. \text{ Then}$$

$$\begin{aligned} \text{(a)} \quad f(A) &= A^2 - 3A + 7I = \begin{bmatrix} -7 & -12 \\ 24 & 17 \end{bmatrix} + \begin{bmatrix} -3 & 6 \\ -12 & -15 \end{bmatrix} + \begin{bmatrix} 7 & 0 \\ 0 & 7 \end{bmatrix} = \begin{bmatrix} -3 & -6 \\ 12 & 9 \end{bmatrix} \\ \text{(b)} \quad f(A) &= A^2 - 6A + 13I = \begin{bmatrix} -7 & -12 \\ 24 & 17 \end{bmatrix} + \begin{bmatrix} -6 & 12 \\ -24 & -30 \end{bmatrix} + \begin{bmatrix} 13 & 0 \\ 0 & 13 \end{bmatrix} = \begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix} \end{aligned}$$

[Thus,  $A$  is a root of  $f(t)$ .]

**9.2.** Find the characteristic polynomial  $\Delta(t)$  of each of the following matrices:

$$\text{(a)} \quad A = \begin{bmatrix} 2 & 5 \\ 4 & 1 \end{bmatrix}, \text{(b)} \quad B = \begin{bmatrix} 7 & -3 \\ 5 & -2 \end{bmatrix}, \text{(c)} \quad C = \begin{bmatrix} 3 & -2 \\ 9 & -3 \end{bmatrix}$$

Use the formula  $\Delta(t) = t^2 - \text{tr}(M)t + |M|$  for a  $2 \times 2$  matrix  $M$ :

$$\text{(a)} \quad \text{tr}(A) = 2 + 1 = 3, |A| = 2 - 20 = -18, \text{ so } \Delta(t) = t^2 - 3t - 18$$

$$\text{(b)} \quad \text{tr}(B) = 7 - 2 = 5, |B| = 14 + 15 = 1, \text{ so } \Delta(t) = t^2 - 5t + 1$$

$$\text{(c)} \quad \text{tr}(C) = 3 - 3 = 0, |C| = 9 + 18 = 9, \text{ so } \Delta(t) = t^2 + 9$$

**9.3.** Find the characteristic polynomial  $\Delta(t)$  of each of the following matrices:

$$\text{(a)} \quad A = \begin{bmatrix} 1 & 2 & 3 \\ 3 & 0 & 4 \\ 6 & 4 & 5 \end{bmatrix}, \text{(b)} \quad B = \begin{bmatrix} 1 & 6 & -2 \\ -3 & 2 & 0 \\ 0 & 3 & -4 \end{bmatrix}$$

Use the formula  $\Delta(t) = t^3 - \text{tr}(A)t^2 + (A_{11} + A_{22} + A_{33})t - |A|$ , where  $A_{ij}$  is the cofactor of  $a_{ij}$  in the  $3 \times 3$  matrix  $A = [a_{ij}]$ .

$$\text{(a)} \quad \text{tr}(A) = 1 + 0 + 5 = 6,$$

$$A_{11} = \begin{vmatrix} 0 & 4 \\ 4 & 5 \end{vmatrix} = -16, \quad A_{22} = \begin{vmatrix} 1 & 3 \\ 6 & 5 \end{vmatrix} = -13, \quad A_{33} = \begin{vmatrix} 1 & 2 \\ 3 & 0 \end{vmatrix} = -6$$

$$A_{11} + A_{22} + A_{33} = -35, \quad \text{and} \quad |A| = 48 + 36 - 16 - 30 = 38$$

Thus,

$$\Delta(t) = t^3 - 6t^2 - 35t - 38$$

$$(b) \operatorname{tr}(B) = 1 + 2 - 4 = -1$$

$$B_{11} = \begin{vmatrix} 2 & 0 \\ 3 & -4 \end{vmatrix} = -8, \quad B_{22} = \begin{vmatrix} 1 & -2 \\ 0 & -4 \end{vmatrix} = -4, \quad B_{33} = \begin{vmatrix} 1 & 6 \\ -3 & 2 \end{vmatrix} = 20$$

$$B_{11} + B_{22} + B_{33} = 8, \quad \text{and} \quad |B| = -8 + 18 - 72 = -62$$

Thus,

$$\Delta(t) = t^3 + t^2 - 8t + 62$$

**9.4.** Find the characteristic polynomial  $\Delta(t)$  of each of the following matrices:

$$(a) A = \begin{bmatrix} 2 & 5 & 1 & 1 \\ 1 & 4 & 2 & 2 \\ 0 & 0 & 6 & -5 \\ 0 & 0 & 2 & 3 \end{bmatrix}, \quad (b) B = \begin{bmatrix} 1 & 1 & 2 & 2 \\ 0 & 3 & 3 & 4 \\ 0 & 0 & 5 & 5 \\ 0 & 0 & 0 & 6 \end{bmatrix}$$

(a)  $A$  is block triangular with diagonal blocks

$$A_1 = \begin{bmatrix} 2 & 5 \\ 1 & 4 \end{bmatrix} \quad \text{and} \quad A_2 = \begin{bmatrix} 6 & -5 \\ 2 & 3 \end{bmatrix}$$

$$\text{Thus,} \quad \Delta(t) = \Delta_{A_1}(t)\Delta_{A_2}(t) = (t^2 - 6t + 3)(t^2 - 9t + 28)$$

(b) Because  $B$  is triangular,  $\Delta(t) = (t - 1)(t - 3)(t - 5)(t - 6)$ .

**9.5.** Find the characteristic polynomial  $\Delta(t)$  of each of the following linear operators:

(a)  $F: \mathbf{R}^2 \rightarrow \mathbf{R}^2$  defined by  $F(x, y) = (3x + 5y; 2x - 7y)$ .

(b)  $\mathbf{D}: V \rightarrow V$  defined by  $\mathbf{D}(f) = df/dt$ , where  $V$  is the space of functions with basis  $S = \{\sin t, \cos t\}$ .

The characteristic polynomial  $\Delta(t)$  of a linear operator is equal to the characteristic polynomial of any matrix  $A$  that represents the linear operator.

(a) Find the matrix  $A$  that represents  $T$  relative to the usual basis of  $\mathbf{R}^2$ . We have

$$A = \begin{bmatrix} 3 & 5 \\ 2 & -7 \end{bmatrix}, \quad \text{so} \quad \Delta(t) = t^2 - \text{tr}(A) t + |A| = t^2 + 4t - 31$$

(b) Find the matrix  $A$  representing the differential operator  $\mathbf{D}$  relative to the basis  $S$ . We have

$$\begin{aligned} \mathbf{D}(\sin t) &= \cos t = 0(\sin t) + 1(\cos t) \\ \mathbf{D}(\cos t) &= -\sin t = -1(\sin t) + 0(\cos t) \end{aligned} \quad \text{and so} \quad A = \begin{bmatrix} 0 & -1 \\ 1 & 0 \end{bmatrix}$$

$$\text{Therefore,} \quad \Delta(t) = t^2 - \text{tr}(A) t + |A| = t^2 + 1$$

**9.6.** Show that a matrix  $A$  and its transpose  $A^T$  have the same characteristic polynomial.

By the transpose operation,  $(tI - A)^T = tI^T - A^T = tI - A^T$ . Because a matrix and its transpose have the same determinant,

$$\Delta_A(t) = |tI - A| = |(tI - A)^T| = |tI - A^T| = \Delta_{A^T}(t)$$

**9.7.** Prove [Theorem 9.1](#): Let  $f$  and  $g$  be polynomials. For any square matrix  $A$  and scalar  $k$ ,

$$(i) \quad (f + g)(A) = f(A) + g(A),$$

$$(ii) (fg)(A) = f(A)g(A),$$

$$(iii) (kf)(A) = kf(A),$$

$$(iv) f(A)g(A) = g(A)f(A).$$

Suppose  $f = a_n t^n + \cdots + a_1 t + a_0$  and  $g = b_m t^m + \cdots + b_1 t + b_0$ . Then, by definition,

$$f(A) = a_n A^n + \cdots + a_1 A + a_0 I \quad \text{and} \quad g(A) = b_m A^m + \cdots + b_1 A + b_0 I$$

(i) Suppose  $m \leq n$  and let  $b_i = 0$  if  $i > m$ . Then

$$f + g = (a_n + b_n)t^n + \cdots + (a_1 + b_1)t + (a_0 + b_0)$$

Hence,

$$\begin{aligned} (f + g)(A) &= (a_n + b_n)A^n + \cdots + (a_1 + b_1)A + (a_0 + b_0)I \\ &= a_n A^n + b_n A^n + \cdots + a_1 A + b_1 A + a_0 I + b_0 I = f(A) + g(A) \end{aligned}$$

(ii) By definition,  $fg = c_{n+m}t^{n+m} + \cdots + c_1 t + c_0 = \sum_{k=0}^{n+m} c_k t^k$ ,

where

$$c_k = a_0 b_k + a_1 b_{k-1} + \cdots + a_k b_0 = \sum_{i=0}^k a_i b_{k-i}$$

Hence,  $(fg)(A) = \sum_{k=0}^{n+m} c_k A^k$  and

$$f(A)g(A) = \left( \sum_{i=0}^n a_i A^i \right) \left( \sum_{j=0}^m b_j A^j \right) = \sum_{i=0}^n \sum_{j=0}^m a_i b_j A^{i+j} = \sum_{k=0}^{n+m} c_k A^k = (fg)(A)$$

(iii) By definition,  $kf = ka_n t^n + \cdots + ka_1 t + ka_0$ , and so

$$(kf)(A) = ka_n A^n + \cdots + ka_1 A + ka_0 I = k(a_n A^n + \cdots + a_1 A + a_0 I) = kf(A)$$

(iv) By (ii),  $g(A)f(A) = (gf)(A) = (fg)(A) = f(A)g(A)$ .

**9.8.** Prove the Cayley-Hamilton [Theorem 9.2](#): Every matrix  $A$  is a root of its characteristic polynomial  $\Delta(t)$ .

Let  $A$  be an arbitrary  $n$ -square matrix and let  $\Delta(t)$  be its characteristic polynomial, say,

$$\Delta(t) = |tI - A| = t^n + a_{n-1}t^{n-1} + \cdots + a_1t + a_0$$

Now let  $B(t)$  denote the classical adjoint of the matrix  $tI - A$ . The elements of  $B(t)$  are cofactors of the matrix  $tI - A$  and hence are polynomials in  $t$  of degree not exceeding  $n - 1$ . Thus,

$$B(t) = B_{n-1}t^{n-1} + \cdots + B_1t + B_0$$

where the  $B_i$  are  $n$ -square matrices over  $K$  which are independent of  $t$ . By the fundamental property of the classical adjoint ([Theorem 8.9](#)),  $(tI - A)B(t) = |tI - A|I$ , or

$$(tI - A)(B_{n-1}t^{n-1} + \cdots + B_1t + B_0) = (t^n + a_{n-1}t^{n-1} + \cdots + a_1t + a_0)I$$

Removing the parentheses and equating corresponding powers of  $t$  yields

$$B_{n-1} = I, \quad B_{n-2} - AB_{n-1} = a_{n-1}I, \quad \dots, \quad B_0 - AB_1 = a_1I, \quad -AB_0 = a_0I$$

Multiplying the above equations by  $A^n$ ,  $A^{n-1}$ , ...,  $A$ ,  $I$ , respectively, yields

$$A^n B_{n-1} = A^n I, \quad A^{n-1} B_{n-2} - A^n B_{n-1} = a_{n-1} A^{n-1}, \quad \dots, \quad AB_0 - A^2 B_1 = a_1 A, \quad -AB_0 = a_0 I$$

Adding the above matrix equations yields 0 on the left-hand side and  $\Delta(A)$  on the right-hand side; that is,

$$0 = A^n + a_{n-1}A^{n-1} + \cdots + a_1A + a_0I$$

Therefore,  $\Delta(A) = 0$ , which is the Cayley–Hamilton theorem.

## **Eigenvalues and Eigenvectors of $2 \times 2$ Matrices**

**9.9.** Let  $A = \begin{bmatrix} 3 & -4 \\ 2 & -6 \end{bmatrix}$ .

- (a) Find all eigenvalues and corresponding eigenvectors.
- (b) Find matrices  $P$  and  $D$  such that  $P$  is nonsingular and  $D = P^{-1}AP$  is diagonal.
- (a) First find the characteristic polynomial  $\Delta(t)$  of  $A$ :

$$\Delta(t) = t^2 - \operatorname{tr}(A)t + |A| = t^2 + 3t - 10 = (t - 2)(t + 5)$$

The roots  $\lambda = 2$  and  $\lambda = 5$  of  $\Delta(t)$  are the eigenvalues of  $A$ . We find corresponding eigenvectors.

- (i) Subtract  $\lambda = 2$  down the diagonal of  $A$  to obtain the matrix  $M = A - 2I$ , where the corresponding homogeneous system  $MX = 0$  yields the eigenvectors corresponding to  $\lambda = 2$ . We have

$$M = \begin{bmatrix} 1 & -4 \\ 2 & -8 \end{bmatrix}, \quad \text{corresponding to} \quad \begin{array}{l} x - 4y = 0 \\ 2x - 8y = 0 \end{array} \quad \text{or} \quad x - 4y = 0$$

The system has only one free variable, and  $v_1 = (4, 1)$  is a nonzero solution. Thus,  $v_1 = (4, 1)$  is an eigenvector belonging to (and spanning the eigenspace of)  $\lambda = 2$ .

- (ii) Subtract  $\lambda = -5$  (or, equivalently, add 5) down the diagonal of  $A$  to obtain

$$M = \begin{bmatrix} 8 & -4 \\ 2 & -1 \end{bmatrix}, \quad \text{corresponding to} \quad \begin{array}{l} 8x - 4y = 0 \\ 2x - y = 0 \end{array} \quad \text{or} \quad 2x - y = 0$$

The system has only one free variable, and  $v_2 = (1, 2)$  is a nonzero solution. Thus,  $v_2 = (1, 2)$  is an eigenvector belonging to  $\lambda = 5$ .

(b) Let  $P$  be the matrix whose columns are  $v_1$  and  $v_2$ . Then

$$P = \begin{bmatrix} 4 & 1 \\ 1 & 2 \end{bmatrix} \quad \text{and} \quad D = P^{-1}AP = \begin{bmatrix} 2 & 0 \\ 0 & -5 \end{bmatrix}$$

Note that  $D$  is the diagonal matrix whose diagonal entries are the eigenvalues of  $A$  corresponding to the eigenvectors appearing in  $P$ .

**Remark:** Here  $P$  is the change-of-basis matrix from the usual basis of  $\mathbf{R}^2$  to the basis  $S = \{v_1, v_2\}$ , and  $D$  is the matrix that represents (the matrix function)  $A$  relative to the new basis  $S$ .

**9.10.** Let  $A = \begin{bmatrix} 2 & 2 \\ 1 & 3 \end{bmatrix}$ .

(a) Find all eigenvalues and corresponding eigenvectors.

(b) Find a nonsingular matrix  $P$  such that  $D = P^{-1}AP$  is diagonal, and  $P^{-1}$ .

(c) Find  $A^6$  and  $f(A)$ , where  $t^4 - 3t^3 - 6t^2 + 7t + 3$ .

(d) Find a “real cube root” of  $B$ —that is, a matrix  $B$  such that  $B^3 = A$  and  $B$  has real eigenvalues.

(a) First find the characteristic polynomial  $\Delta(t)$  of  $A$ :

$$\Delta(t) = t^2 - \text{tr}(A)t + |A| = t^2 - 5t + 4 = (t - 1)(t - 4)$$

The roots  $\lambda = 1$  and  $\lambda = 4$  of  $\Delta(t)$  are the eigenvalues of  $A$ . We find corresponding eigenvectors.

- (i) Subtract  $\lambda = 1$  down the diagonal of  $A$  to obtain the matrix  $M = A - \lambda I$ , where the corresponding homogeneous system  $MX = 0$  yields the eigenvectors belonging to  $\lambda = 1$ . We have

$$M = \begin{bmatrix} 1 & 2 \\ 1 & 2 \end{bmatrix}, \quad \text{corresponding to} \quad \begin{array}{l} x + 2y = 0 \\ x + 2y = 0 \end{array} \quad \text{or} \quad x + 2y = 0$$

The system has only one independent solution; for example,  $x = 2, y = 1$ . Thus,  $v_1 = (2, -1)$  is an eigenvector belonging to (and spanning the eigenspace of)  $\lambda = 1$ .

- (ii) Subtract  $\lambda = 4$  down the diagonal of  $A$  to obtain

$$M = \begin{bmatrix} -2 & 2 \\ 1 & -1 \end{bmatrix}, \quad \text{corresponding to} \quad \begin{array}{l} -2x + 2y = 0 \\ x - y = 0 \end{array} \quad \text{or} \quad x - y = 0$$

The system has only one independent solution; for example,  $x = 1, y = 1$ . Thus,  $v_2 = (1, 1)$  is an eigenvector belonging to  $\lambda = 4$ .

- (b) Let  $P$  be the matrix whose columns are  $v_1$  and  $v_2$ . Then

$$P = \begin{bmatrix} 2 & 1 \\ -1 & 1 \end{bmatrix} \quad \text{and} \quad D = P^{-1}AP = \begin{bmatrix} 1 & 0 \\ 0 & 4 \end{bmatrix}, \quad \text{where} \quad P^{-1} = \begin{bmatrix} \frac{1}{3} & -\frac{1}{3} \\ \frac{1}{3} & \frac{2}{3} \end{bmatrix}$$

- (c) Using the diagonal factorization  $A = PDP^{-1}$ , and  $1^6 = 1$  and  $4^6 = 4096$ , we get

$$A^6 = PD^6P^{-1} = \begin{bmatrix} 2 & 1 \\ -1 & 1 \end{bmatrix} \begin{bmatrix} 1 & 0 \\ 0 & 4096 \end{bmatrix} \begin{bmatrix} \frac{1}{3} & -\frac{1}{3} \\ \frac{1}{3} & \frac{2}{3} \end{bmatrix} = \begin{bmatrix} 1366 & 2230 \\ 1365 & 2731 \end{bmatrix}$$

Also,  $f(1) = 2$  and  $f(4) = 1$ . Hence,

$$f(A) = Pf(D)P^{-1} = \begin{bmatrix} 2 & 1 \\ -1 & 1 \end{bmatrix} \begin{bmatrix} 2 & 0 \\ 0 & -1 \end{bmatrix} \begin{bmatrix} \frac{1}{3} & -\frac{1}{3} \\ \frac{1}{3} & \frac{2}{3} \end{bmatrix} = \begin{bmatrix} 1 & 2 \\ -1 & 0 \end{bmatrix}$$

(d) Here  $\begin{bmatrix} 1 & 0 \\ 0 & \sqrt[3]{4} \end{bmatrix}$  is the real cube root of  $D$ . Hence the real cube root of  $A$  is

$$B = P\sqrt[3]{D}P^{-1} = \begin{bmatrix} 2 & 1 \\ -1 & 1 \end{bmatrix} \begin{bmatrix} 1 & 0 \\ 0 & \sqrt[3]{4} \end{bmatrix} \begin{bmatrix} \frac{1}{3} & -\frac{1}{3} \\ \frac{1}{3} & \frac{2}{3} \end{bmatrix} = \frac{1}{3} \begin{bmatrix} 2 + \sqrt[3]{4} & -2 + 2\sqrt[3]{4} \\ -1 + \sqrt[3]{4} & 1 + 2\sqrt[3]{4} \end{bmatrix}$$

**9.11.** Each of the following real matrices defines a linear transformation on  $\mathbf{R}^2$ :

$$(a) A = \begin{bmatrix} 5 & 6 \\ 3 & -2 \end{bmatrix}, (b) B = \begin{bmatrix} 1 & -1 \\ 2 & -1 \end{bmatrix}, (c) C = \begin{bmatrix} 5 & -1 \\ 1 & 3 \end{bmatrix}$$

Find, for each matrix, all eigenvalues and a maximum set  $S$  of linearly independent eigenvectors. Which of these linear operators are diagonalizable—that is, which can be represented by a diagonal matrix?

(a) First find  $\Delta(t) = t^2 - 3t - 28 = (t - 7)(t + 4)$ . The roots  $\lambda = 7$  and  $\lambda = -4$  are the eigenvalues of  $A$ . We find corresponding eigenvectors.

(i) Subtract  $\lambda = 7$  down the diagonal of  $A$  to obtain

$$M = \begin{bmatrix} -2 & 6 \\ 3 & -9 \end{bmatrix}, \quad \text{corresponding to} \quad \begin{array}{l} -2x + 6y = 0 \\ 3x - 9y = 0 \end{array} \quad \text{or} \quad x - 3y = 0$$

Here  $v_1 = (3, 1)$  is a nonzero solution.

(ii) Subtract  $\lambda = -4$  (or add 4) down the diagonal of  $A$  to obtain

$$M = \begin{bmatrix} 9 & 6 \\ 3 & 2 \end{bmatrix}, \quad \text{corresponding to} \quad \begin{array}{l} 9x + 6y = 0 \\ 3x + 2y = 0 \end{array} \quad \text{or} \quad 3x + 2y = 0$$

Here  $v_2 = (2, -3)$  is a nonzero solution.

Then  $S = \{v_1, v_2\} = \{(3, 1), (2, -3)\}$  is a maximal set of linearly independent eigenvectors. Because  $S$  is a basis of  $\mathbf{R}^2$ ,  $A$  is diagonalizable. Using the basis  $S$ ,  $A$  is represented by the diagonal matrix  $D = \text{diag}(7, -4)$ .

(b) First find the characteristic polynomial  $\Delta(t) = t^2 + 1$ . There are no real roots. Thus  $B$ , a real matrix representing a linear transformation on  $\mathbf{R}^2$ , has no eigenvalues and no eigenvectors. Hence, in particular,  $B$  is not diagonalizable.

(c) First find  $\Delta(t) = t^2 - 8t + 16 = (t - 4)^2$ . Thus,  $\lambda = 4$  is the only eigenvalue of  $C$ . Subtract  $\lambda = 4$  down the diagonal of  $C$  to obtain

$$M = \begin{bmatrix} 1 & -1 \\ 1 & -1 \end{bmatrix}, \quad \text{corresponding to} \quad x - y = 0$$

The homogeneous system has only one independent solution; for example,  $x = 1, y = 1$ .

Thus,  $v = (1, 1)$  is an eigenvector of  $C$ .

Furthermore, as there are no other eigenvalues, the singleton set  $S = \{v\} = \{(1, 1)\}$  is a maximal set of linearly independent eigenvectors of  $C$ . Furthermore, because  $S$  is not a basis of  $\mathbf{R}^2$ ,  $C$  is not diagonalizable.

**9.12.** Suppose the matrix  $B$  in [Problem 9.11](#) represents a linear operator on complex space  $\mathbf{C}^2$ . Show that, in this case,  $B$  is diagonalizable by finding a basis  $S$  of  $\mathbf{C}^2$  consisting of eigenvectors of  $B$ .

The characteristic polynomial of  $B$  is still  $\Delta(t) = t^2 + 1$ . As a polynomial over  $\mathbf{C}$ ,  $\Delta(t)$  *does* factor; specifically,  $\Delta(t) = (t - i)(t + i)$ . Thus,  $\lambda = i$  and  $\lambda = -i$  are the eigenvalues of  $B$ .

(i) Subtract  $\lambda = i$  down the diagonal of  $B$  to obtain the homogeneous system

$$\begin{array}{rcl} (1 - i)x - & y = 0 \\ 2x + (-1 - i)y = 0 \end{array} \quad \text{or} \quad (1 - i)x - y = 0$$

The system has only one independent solution; for example,  $x = 1$ ,  $y = 1 - i$ . Thus,  $v_1 = (1, 1 - i)$  is an eigenvector that spans the eigenspace of  $\lambda = i$ .

(ii) Subtract  $\lambda = -i$  (or add  $i$ ) down the diagonal of  $B$  to obtain the homogeneous system

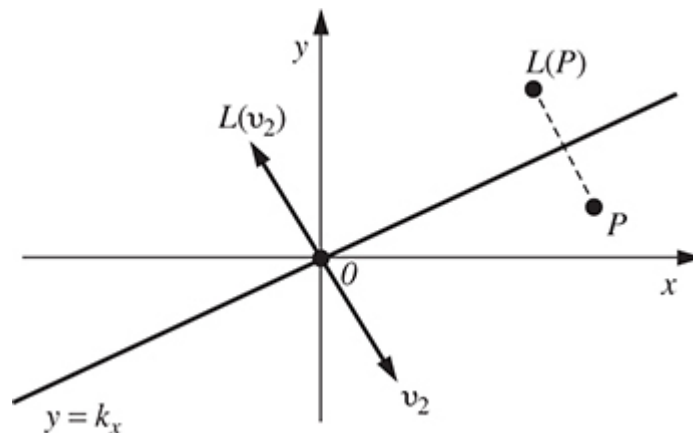
$$\begin{array}{rcl} (1 + i)x - & y = 0 \\ 2x + (-1 + i)y = 0 \end{array} \quad \text{or} \quad (1 + i)x - y = 0$$

The system has only one independent solution; for example,  $x = 1$ ,  $y = 1 + i$ . Thus,  $v_2 = (1, 1 + i)$  is an eigenvector that spans the eigenspace of  $\lambda = -i$ .

As a complex matrix,  $B$  is diagonalizable. Specifically,  $S = \{v_1, v_2\} = \{(1, 1 - i), (1, 1 + i)\}$  is a basis of  $\mathbf{C}^2$  consisting of eigenvectors of  $B$ . Using this basis  $S$ ,  $B$  is represented by the diagonal matrix  $D = \text{diag}(i, -i)$ .

**9.13.** Let  $L$  be the linear transformation on  $\mathbf{R}^2$  that reflects each point  $P$  across the line  $y = kx$ , where  $k > 0$ . (See Fig. 9-1.)

- (a) Show that  $v_1 = (k, 1)$  and  $v_2 = (1, -k)$  are eigenvectors of  $L$ .
- (b) Show that  $L$  is diagonalizable, and find a diagonal representation  $D$ .



**Figure 9-1**

(a) The vector  $v_1 = (k, 1)$  lies on the line  $y = kx$ , and hence is left fixed by  $L$ ; that is,  $L(v_1) = v_1$ . Thus,  $v_1$  is an eigenvector of  $L$  belonging to the eigenvalue  $\lambda_1 = 1$ .

The vector  $v_2 = (1, -k)$  is perpendicular to the line  $y = kx$ , and hence,  $L$  reflects  $v_2$  into its negative; that is,  $L(v_2) = -v_2$ . Thus,  $v_2$  is an eigenvector of  $L$  belonging to the eigenvalue  $\lambda_2 = -1$ .

(b) Here  $S = \{v_1, v_2\}$  is a basis of  $\mathbf{R}^2$  consisting of eigenvectors of  $L$ . Thus,  $L$  is diagonalizable, with the

diagonal representation  $D = \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix}$  (relative to the basis  $S$ ).

## Eigenvalues and Eigenvectors

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**9.14.** Let  $A = \begin{bmatrix} 4 & 1 & -1 \\ 2 & 5 & -2 \\ 1 & 1 & 2 \end{bmatrix}$ . (a) Find all eigenvalues of  $A$ .

(b) Find a maximum set  $S$  of linearly independent eigenvectors of  $A$ .

(c) Is  $A$  diagonalizable? If yes, find  $P$  such that  $D = P^{-1}AP$  is diagonal.

(a) First find the characteristic polynomial  $\Delta(t)$  of  $A$ . We have

$$\text{tr}(A) = 4 + 5 + 2 = 11 \quad \text{and} \quad |A| = 40 - 2 - 2 + 5 + 8 - 4 = 45$$

Also, find each cofactor  $A_{ji}$  of  $a_{ij}$  in  $A$ :

$$A_{11} = \begin{vmatrix} 5 & -2 \\ 1 & 2 \end{vmatrix} = 12, \quad A_{22} = \begin{vmatrix} 4 & -1 \\ 1 & 2 \end{vmatrix} = 9, \quad A_{33} = \begin{vmatrix} 4 & 1 \\ 2 & 5 \end{vmatrix} = 18$$

Hence,

$$\Delta(t) = t^3 - \text{tr}(A)t^2 + (A_{11} + A_{22} + A_{33})t - |A| = t^3 - 11t^2 + 39t - 45$$

Assuming  $\Delta t$  has a rational root, it must be among  $\pm 1, \pm 3, \pm 5, \pm 9, \pm 15, \pm 45$ . Testing, by synthetic division, we get

$$\begin{array}{r|rrrr} 3 & 1 & -11 & 39 & -45 \\ & & 3 & -24 & 45 \\ \hline & 1 & -8 & 15 & 0 \end{array}$$

Thus,  $t = 3$  is a root of  $\Delta(t)$ . Also,  $t - 3$  is a factor and  $t^2 - 8t + 15$  is a factor. Hence,

$$\Delta(t) = (t - 3)(t^2 - 8t + 15) = (t - 3)(t - 5)(t - 3) = (t - 3)^2(t - 5)$$

Accordingly,  $\lambda = 3$  and  $\lambda = 5$  are eigenvalues of  $A$ .

(b) Find linearly independent eigenvectors for each eigenvalue of  $A$ .

(i) Subtract  $\lambda = 3$  down the diagonal of  $A$  to obtain the matrix

$$M = \begin{bmatrix} 1 & 1 & -1 \\ 2 & 2 & -2 \\ 1 & 1 & -1 \end{bmatrix}, \quad \text{corresponding to} \quad x + y - z = 0$$

Here  $u = (1, -1, 0)$  and  $v = (1, 0, 1)$  are linearly independent solutions.

(ii) Subtract  $\lambda = 5$  down the diagonal of  $A$  to obtain the matrix

$$M = \begin{bmatrix} -1 & 1 & -1 \\ 2 & 0 & -2 \\ 1 & 1 & -3 \end{bmatrix}, \quad \text{corresponding to} \quad \begin{array}{l} -x + y - z = 0 \\ 2x - 2z = 0 \\ x + y - 3z = 0 \end{array} \quad \text{or} \quad \begin{array}{l} x - z = 0 \\ y - 2z = 0 \end{array}$$

Only  $z$  is a free variable. Here  $w = (1, 2, 1)$  is a solution.

Thus,  $S = \{u, v, w\} = \{(1, -1, 0), (1, 0, 1), (1, 2, 1)\}$  is a maximal set of linearly independent eigenvectors of  $A$ .

**Remark:** The vectors  $u$  and  $v$  were chosen so that they were independent solutions of the system  $x + y - z = 0$ . On the other hand,  $w$  is automatically independent of  $u$  and  $v$  because  $w$  belongs to a different eigenvalue of  $A$ . Thus, the three vectors are linearly independent.

(c)  $A$  is diagonalizable, because it has three linearly independent eigenvectors. Let  $P$  be the matrix with columns  $u, v, w$ . Then

$$P = \begin{bmatrix} 1 & 1 & 1 \\ -1 & 0 & 2 \\ 0 & 1 & 1 \end{bmatrix} \quad \text{and} \quad D = P^{-1}AP = \begin{bmatrix} 3 & & \\ & 3 & \\ & & 5 \end{bmatrix}$$

**9.15.** Repeat [Problem 9.14](#) for the matrix

$$B = \begin{bmatrix} 3 & -1 & 1 \\ 7 & -5 & 1 \\ 6 & -6 & 2 \end{bmatrix}.$$

(a) First find the characteristic polynomial  $\Delta(t)$  of  $B$ .  
We have

$$\text{tr}(B) = 0, \quad |B| = -16, \quad B_{11} = -4, \quad B_{22} = 0, \quad B_{33} = -8, \quad \text{so} \quad \sum_i B_{ii} = -12$$

Therefore,  $\Delta(t) = t^3 - 12t + 16 = (t - 2)^2(t + 4)$ .  
Thus,  $\lambda_1 = 2$  and  $\lambda_2 = -4$  are the eigenvalues of  $B$ .

(b) Find a basis for the eigenspace of each eigenvalue of  $B$ .

(i) Subtract  $\lambda_1 = 2$  down the diagonal of  $B$  to obtain

$$M = \begin{bmatrix} 1 & -1 & 1 \\ 7 & -7 & 1 \\ 6 & -6 & 0 \end{bmatrix}, \quad \text{corresponding to} \quad \begin{array}{l} x - y + z = 0 \\ 7x - 7y + z = 0 \\ 6x - 6y = 0 \end{array} \quad \text{or} \quad \begin{array}{l} x - y + z = 0 \\ z = 0 \end{array}$$

The system has only one independent solution;  
for example,  $x = 1, y = 1, z = 0$ . Thus,  $u = (1, 1, 0)$  forms a basis for the eigenspace of  $\lambda_1 = 2$ .

(ii) Subtract  $\lambda_2 = -4$  (or add 4) down the diagonal of  $B$  to obtain

$$M = \begin{bmatrix} 7 & -1 & 1 \\ 7 & -1 & 1 \\ 6 & -6 & 6 \end{bmatrix}, \quad \text{corresponding to} \quad \begin{array}{l} 7x - y + z = 0 \\ 7x - y + z = 0 \\ 6x - 6y + 6z = 0 \end{array} \quad \text{or} \quad \begin{array}{l} x - y + z = 0 \\ 6y - 6z = 0 \end{array}$$

The system has only one independent solution; for example,  $x = 0$ ,  $y = 1$ ,  $z = 1$ . Thus,  $v = (0, 1, 1)$  forms a basis for the eigenspace of  $\lambda_2 = -4$ .

Thus  $S = \{u, v\}$  is a maximal set of linearly independent eigenvectors of  $B$ .

(c) Because  $B$  has at most two linearly independent eigenvectors,  $B$  is not similar to a diagonal matrix; that is,  $B$  is not diagonalizable.

**9.16.** Find the algebraic and geometric multiplicities of the eigenvalue  $\lambda_1 = 2$  of the matrix  $B$  in [Problem 9.15](#).

The algebraic multiplicity of  $\lambda_1 = 2$  is 2, because  $t - 2$  appears with exponent 2 in  $\Delta(t)$ . However, the geometric multiplicity of  $\lambda_1 = 2$  is 1, because  $\dim E_{\lambda_1} = 1$  (where  $E_{\lambda_1}$  is the eigenspace of  $\lambda_1$ ).

**9.17.** Let  $T: \mathbf{R}^3 \rightarrow \mathbf{R}^3$  be defined by  $T(x, y, z) = (2x + y - 2z, 2x + 3y - 4z, x + y - z)$ . Find all eigenvalues of  $T$ , and find a basis of each eigenspace. Is  $T$  diagonalizable? If so, find the basis  $S$  of  $\mathbf{R}^3$  that diagonalizes  $T$ , and find its diagonal representation  $D$ .

First find the matrix  $A$  that represents  $T$  relative to the usual basis of  $\mathbf{R}^3$  by writing down the coefficients of  $x, y, z$  as rows, and then find the characteristic polynomial of  $A$  (and  $T$ ). We have

$$A = [T] = \begin{bmatrix} 2 & 1 & -2 \\ 2 & 3 & -4 \\ 1 & 1 & -1 \end{bmatrix} \quad \text{and} \quad \begin{array}{l} \text{tr}(A) = 4, \quad |A| = 2 \\ A_{11} = 1, \quad A_{22} = 0, \quad A_{33} = 4 \\ \sum_i A_{ii} = 5 \end{array}$$

Therefore,  $\Delta(t) = t^3 - 4t^2 + 5t - 2 = (t - 1)^2(t - 2)$ , and so  $\lambda = 1$  and  $\lambda = 2$  are the eigenvalues of  $A$  (and  $T$ ). We next find linearly independent eigenvectors for each eigenvalue of  $A$ .

(i) Subtract  $\lambda = 1$  down the diagonal of  $A$  to obtain the matrix

$$M = \begin{bmatrix} 1 & 1 & -2 \\ 2 & 2 & -4 \\ 1 & 1 & -2 \end{bmatrix}, \quad \text{corresponding to} \quad x + y - 2z = 0$$

Here  $y$  and  $z$  are free variables, and so there are two linearly independent eigenvectors belonging to  $\lambda = 1$ . For example,  $u = (1, -1, 0)$  and  $v = (2, 0, 1)$  are two such eigenvectors.

(ii) Subtract  $\lambda = 2$  down the diagonal of  $A$  to obtain

$$M = \begin{bmatrix} 0 & 1 & -2 \\ 2 & 1 & -4 \\ 1 & 1 & -3 \end{bmatrix}, \quad \text{corresponding to} \quad \begin{array}{l} y - 2z = 0 \\ 2x + y - 4z = 0 \\ x + y - 3z = 0 \end{array} \quad \text{or} \quad \begin{array}{l} x + y - 3z = 0 \\ y - 2z = 0 \end{array}$$

Only  $z$  is a free variable. Here  $w = (1, 2, 1)$  is a solution.

Thus,  $T$  is diagonalizable, because it has three independent eigenvectors. Specifically, choosing

$$S = \{u, v, w\} = \{(1, -1, 0), (2, 0, 1), (1, 2, 1)\}$$

as a basis,  $T$  is represented by the diagonal matrix  $D = \text{diag}(1, 1, 2)$ .

**9.18.** Prove the following for a linear operator (matrix)  $T$ :

(a) The scalar 0 is an eigenvalue of  $T$  if and only if  $T$  is singular.

(b) If  $\lambda$  is an eigenvalue of  $T$ , where  $T$  is invertible, then  $\lambda^{-1}$  is an eigenvalue of  $T^{-1}$ .

(a) We have that 0 is an eigenvalue of  $T$  if and only if there is a vector  $v \neq 0$  such that  $T(v) = 0 \cdot v$ —that is, if and only if  $T$  is singular.

(b) Because  $T$  is invertible, it is nonsingular; hence, by (a),  $\lambda \neq 0$ . By definition of an eigenvalue, there exists  $v \neq 0$  such that  $T(v) = \lambda v$ . Applying  $T^{-1}$  to both sides, we obtain

$$v = T^{-1}(\lambda v) = \lambda T^{-1}(v), \quad \text{and so} \quad T^{-1}(v) = \lambda^{-1}v$$

Therefore,  $\lambda^{-1}$  is an eigenvalue of  $T^{-1}$ .

**9.19.** Let  $\lambda$  be an eigenvalue of a linear operator  $T: V \rightarrow V$ , and let  $E_\lambda$  consists of all the eigenvectors belonging to  $\lambda$  (called the *eigenspace* of  $\lambda$ ). Prove that  $E_\lambda$  is a subspace of  $V$ . That is, prove

(a) If  $u \in E_\lambda$ , then  $ku \in E_\lambda$  for any scalar  $k$ . (b) If  $u, v \in E_\lambda$ , then  $u + v \in E_\lambda$ .

(a) Because  $u \in E_\lambda$ , we have  $T(u) = \lambda u$ . Then  $T(ku) = kT(u) = k(\lambda u) = \lambda(ku)$ , and so  $ku \in E_\lambda$ : (We view the zero vector  $0 \in V$  as an “eigenvector” of  $\lambda$  in order for  $E_\lambda$  to be a subspace of  $V$ .)

(b) As  $u, v \in E_\lambda$ , we have  $T(u) = \lambda u$  and  $T(v) = \lambda v$ . Then  $T(u + v) = T(u) + T(v) = \lambda u + \lambda v = \lambda(u + v)$ ; and so  $u + v \in E_\lambda$ .

**9.20.** Prove [Theorem 9.6](#): The following are equivalent: (i) The scalar  $\lambda$  is an eigenvalue of  $A$ .

(ii) The matrix  $\lambda I - A$  is singular.

(iii) The scalar  $\lambda$  is a root of the characteristic polynomial  $\Delta(t)$  of  $A$ .

The scalar  $\lambda$  is an eigenvalue of  $A$  if and only if there exists a nonzero vector  $v$  such that

$$Av = \lambda v \quad \text{or} \quad (\lambda I)v - Av = 0 \quad \text{or} \quad (\lambda I - A)v = 0$$

or  $\lambda I - A$  is singular. In such a case,  $\lambda$  is a root of  $\Delta(t) = |\lambda I - A|$ . Also,  $v$  is in the eigenspace  $E_\lambda$  of  $\lambda$  if and only if the above relations hold. Hence,  $v$  is a solution of  $(\lambda I - A)X = 0$ .

**9.21.** Prove [Theorem 9.8'](#): Suppose  $v_1, v_2, \dots, v_n$  are nonzero eigenvectors of  $T$  belonging to distinct eigenvalues  $\lambda_1, \lambda_2, \dots, \lambda_n$ . Then  $v_1, v_2, \dots, v_n$  are linearly independent.

Suppose the theorem is not true. Let  $v_1, v_2, \dots, v_s$  be a minimal set of vectors for which the theorem is not true. We have  $s > 1$ , because  $v_1 \neq 0$ . Also, by the minimality condition,  $v_2, \dots, v_s$  are linearly independent. Thus,  $v_1$  is a linear combination of  $v_2, \dots, v_s$ , say,

$$v_1 = a_2 v_2 + a_3 v_3 + \cdots + a_s v_s \tag{1}$$

(where some  $a_k \neq 0$ ). Applying  $T$  to (1) and using the linearity of  $T$  yields

$$T(v_1) = T(a_2 v_2 + a_3 v_3 + \cdots + a_s v_s) = a_2 T(v_2) + a_3 T(v_3) + \cdots + a_s T(v_s) \tag{2}$$

Because  $v_j$  is an eigenvector of  $T$  belonging to  $\lambda_j$ , we have  $T(v_j) = \lambda_j v_j$ . Substituting in (2) yields

$$\lambda_1 v_1 = a_2 \lambda_2 v_2 + a_3 \lambda_3 v_3 + \cdots + a_s \lambda_s v_s \tag{3}$$

Multiplying (1) by  $\lambda_1$  yields

$$\lambda_1 v_1 = a_2 \lambda_1 v_2 + a_3 \lambda_1 v_3 + \cdots + a_s \lambda_1 v_s \quad (4)$$

Setting the right-hand sides of (3) and (4) equal to each other, or subtracting (3) from (4) yields

$$a_2(\lambda_1 - \lambda_2)v_2 + a_3(\lambda_1 - \lambda_3)v_3 + \cdots + a_s(\lambda_1 - \lambda_s)v_s = 0 \quad (5)$$

Because  $v_2, v_3, \dots, v_s$  are linearly independent, the coefficients in (5) must all be zero. That is,

$$a_2(\lambda_1 - \lambda_2) = 0, \quad a_3(\lambda_1 - \lambda_3) = 0, \quad \dots, \quad a_s(\lambda_1 - \lambda_s) = 0$$

However, the  $\lambda_i$  are distinct. Hence  $\lambda_1 - \lambda_j \neq 0$  for  $j > 1$ . Hence,  $a_2 = 0, a_3 = 0, \dots, a_s = 0$ . This contradicts the fact that some  $a_k \neq 0$ . The theorem is proved.

**9.22.** Prove [Theorem 9.9](#). Suppose  $\Delta(t) = (t - a_1)(t - a_2) \cdots (t - a_n)$  is the characteristic polynomial of an  $n$ -square matrix  $A$ , and suppose the  $n$  roots  $a_i$  are distinct. Then  $A$  is similar to the diagonal matrix  $D = \text{diag}(a_1, a_2, \dots, a_n)$ .

Let  $v_1, v_2, \dots, v_n$  be (nonzero) eigenvectors corresponding to the eigenvalues  $a_i$ . Then the  $n$  eigenvectors  $v_i$  are linearly independent ([Theorem 9.8](#)), and hence form a basis of  $K^n$ . Accordingly,  $A$  is diagonalizable (i.e.,  $A$  is similar to a diagonal matrix  $D$ ), and the diagonal elements of  $D$  are the eigenvalues  $a_i$ .

**9.23.** Prove [Theorem 9.10'](#): The geometric multiplicity of an eigenvalue  $\lambda$  of  $T$  does not exceed its algebraic multiplicity.

Suppose the geometric multiplicity of  $\lambda$  is  $r$ . Then its eigenspace  $E_\lambda$  contains  $r$  linearly independent eigenvectors  $v_1, \dots, v_r$ . Extend the set  $\{v_i\}$  to a basis of  $V$ , say,  $\{v_1, \dots, v_r, w_1, \dots, w_s\}$ . We have

$$T(v_1) = \lambda v_1, \quad T(v_2) = \lambda v_2, \quad \dots, \quad T(v_r) = \lambda v_r,$$

$$T(w_1) = a_{11}v_1 + \dots + a_{1r}v_r + b_{11}w_1 + \dots + b_{1s}w_s$$

$$T(w_2) = a_{21}v_1 + \dots + a_{2r}v_r + b_{21}w_1 + \dots + b_{2s}w_s$$

.....

$$T(w_s) = a_{s1}v_1 + \dots + a_{sr}v_r + b_{s1}w_1 + \dots + b_{ss}w_s$$

Then  $M = \begin{bmatrix} \lambda I_r & A \\ 0 & B \end{bmatrix}$  is the matrix of  $T$  in the above basis, where  $A = [a_{ij}]^T$  and  $B = [b_{ij}]^T$ :

Because  $M$  is block diagonal, the characteristic polynomial  $(t - \lambda)^r$  of the block  $\lambda I_r$  must divide the characteristic polynomial of  $M$  and hence of  $T$ . Thus, the algebraic multiplicity of  $\lambda$  for  $T$  is at least  $r$ , as required.

## Diagonalizing Real Symmetric Matrices and Quadratic Forms

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**9.24.** Let  $A = \begin{bmatrix} 7 & 3 \\ 3 & -1 \end{bmatrix}$ . Find an orthogonal matrix  $P$  such that  $D = P^{-1}AP$  is diagonal.

First find the characteristic polynomial  $\Delta(t)$  of  $A$ . We have

$$\Delta(t) = t^2 - \text{tr}(A)t + |A| = t^2 - 6t - 16 = (t - 8)(t + 2)$$

Thus, the eigenvalues of  $A$  are  $\lambda = 8$  and  $\lambda = -2$ . We next find corresponding eigenvectors.

Subtract  $\lambda = 8$  down the diagonal of  $A$  to obtain the matrix

$$M = \begin{bmatrix} -1 & 3 \\ 3 & -9 \end{bmatrix}, \quad \text{corresponding to} \quad \begin{array}{l} -x + 3y = 0 \\ 3x - 9y = 0 \end{array} \quad \text{or} \quad x - 3y = 0$$

A nonzero solution is  $u_1 = (3, 1)$ .

Subtract  $\lambda = -2$  (or add 2) down the diagonal of  $A$  to obtain the matrix

$$M = \begin{bmatrix} 9 & 3 \\ 3 & 1 \end{bmatrix}, \quad \text{corresponding to} \quad \begin{array}{l} 9x + 3y = 0 \\ 3x + y = 0 \end{array} \quad \text{or} \quad 3x + y = 0$$

A nonzero solution is  $u_2 = (1, -3)$ .

As expected, because  $A$  is symmetric, the eigenvectors  $u_1$  and  $u_2$  are orthogonal. Normalize  $u_1$  and  $u_2$  to obtain, respectively, the unit vectors

$$\hat{u}_1 = (3/\sqrt{10}, 1/\sqrt{10}) \quad \text{and} \quad \hat{u}_2 = (1/\sqrt{10}, -3/\sqrt{10}).$$

Finally, let  $P$  be the matrix whose columns are the unit vectors  $\hat{u}_1$  and  $\hat{u}_2$ , respectively. Then

$$P = \begin{bmatrix} 3/\sqrt{10} & 1/\sqrt{10} \\ 1/\sqrt{10} & -3/\sqrt{10} \end{bmatrix} \quad \text{and} \quad D = P^{-1}AP = \begin{bmatrix} 8 & 0 \\ 0 & -2 \end{bmatrix}$$

As expected, the diagonal entries in  $D$  are the eigenvalues of  $A$ .

**9.25.** Let  $B = \begin{bmatrix} 11 & -8 & 4 \\ -8 & -1 & -2 \\ 4 & -2 & -4 \end{bmatrix}$ . (a) Find all eigenvalues of  $B$ .

(b) Find a maximal set  $S$  of nonzero orthogonal eigenvectors of  $B$ .

(c) Find an orthogonal matrix  $P$  such that  $D = P^{-1}BP$  is diagonal.

(a) First find the characteristic polynomial of  $B$ . We have

$$\operatorname{tr}(B) = 6, \quad |B| = 400, \quad B_{11} = 0, \quad B_{22} = -60, \quad B_{33} = -75, \quad \text{so} \quad \sum B_{ii} = -135$$

Hence,  $\Delta(t) = t^3 - 6t^2 - 135t - 400$ . If  $\Delta(t)$  has an integer root it must divide 400. Testing  $t = -5$ , by synthetic division, yields

$$\begin{array}{r|rrrr} -5 & 1 & -6 & -135 & -400 \\ & & -5 & +55 & +400 \\ \hline & 1 & -11 & -80 & 0 \end{array}$$

Thus,  $t + 5$  is a factor of  $\Delta(t)$ , and  $t^2 - 11t - 80$  is a factor. Thus,

$$\Delta(t) = (t + 5)(t^2 - 11t - 80) = (t + 5)^2(t - 16)$$

The eigenvalues of  $B$  are  $\lambda = 5$  (multiplicity 2), and  $\lambda = 16$  (multiplicity 1).

(b) Find an orthogonal basis for each eigenspace. Subtract  $\lambda = 5$  (or, add 5) down the diagonal of  $B$  to obtain the homogeneous system

$$16x - 8y + 4z = 0, \quad -8x + 4y - 2z = 0, \quad 4x - 2y + z = 0$$

That is,  $4x - 2y + z = 0$ . The system has two independent solutions. One solution is  $v_1 = (0, 1, 2)$ . We seek a second solution  $v_2 = (a, b, c)$ , which is orthogonal to  $v_1$ , such that

$$4a - 2b + c = 0, \quad \text{and also} \quad b - 2c = 0$$

One such solution is  $v_2 = (-5, -8, 4)$ .

Subtract  $\lambda = 16$  down the diagonal of  $B$  to obtain the homogeneous system

$$-5x - 8y + 4z = 0, \quad -8x - 17y - 2z = 0, \quad 4x - 2y - 20z = 0$$

This system yields a nonzero solution  $v_3 = (4, -2, 1)$ . (As expected from [Theorem 9.13](#), the eigenvector  $v_3$  is orthogonal to  $v_1$  and  $v_2$ .)

Then  $v_1, v_2, v_3$  form a maximal set of nonzero orthogonal eigenvectors of  $B$ .

(c) Normalize  $v_1, v_2, v_3$  to obtain the orthonormal basis:

$$\hat{v}_1 = v_1/\sqrt{5}, \quad \hat{v}_2 = v_2/\sqrt{105}, \quad \hat{v}_3 = v_3/\sqrt{21}$$

Then  $P$  is the matrix whose columns are  $\hat{v}_1, \hat{v}_2, \hat{v}_3$ . Thus,

$$P = \begin{bmatrix} 0 & -5/\sqrt{105} & 4/\sqrt{21} \\ 1/\sqrt{5} & -8/\sqrt{105} & -2/\sqrt{21} \\ 2/\sqrt{5} & 4/\sqrt{105} & 1/\sqrt{21} \end{bmatrix} \quad \text{and} \quad D = P^{-1}BP = \begin{bmatrix} -5 & & \\ & -5 & \\ & & 16 \end{bmatrix}$$

**9.26.** Let  $q(x, y) = x^2 + 6xy - 7y^2$ . Find an orthogonal substitution that diagonalizes  $q$ .

Find the symmetric matrix  $A$  that represents  $q$  and its characteristic polynomial  $\Delta(t)$ . We have

$$A = \begin{bmatrix} 1 & 3 \\ 3 & -7 \end{bmatrix} \quad \text{and} \quad \Delta(t) = t^2 + 6t - 16 = (t - 2)(t + 8)$$

The eigenvalues of  $A$  are  $\lambda = 2$  and  $\lambda = 8$ . Thus, using  $s$  and  $t$  as new variables, a diagonal form of  $q$  is

$$q(s, t) = 2s^2 - 8t^2$$

The corresponding orthogonal substitution is obtained by finding an orthogonal set of eigenvectors of  $A$ .

(i) Subtract  $\lambda = 2$  down the diagonal of  $A$  to obtain the matrix

$$M = \begin{bmatrix} -1 & 3 \\ 3 & -9 \end{bmatrix}, \quad \text{corresponding to} \quad \begin{array}{l} -x + 3y = 0 \\ 3x - 9y = 0 \end{array} \quad \text{or} \quad -x + 3y = 0$$

A nonzero solution is  $u_1 = (3, 1)$ .

(ii) Subtract  $\lambda = -8$  (or add 8) down the diagonal of  $A$  to obtain the matrix

$$M = \begin{bmatrix} 9 & 3 \\ 3 & 1 \end{bmatrix}, \quad \text{corresponding to} \quad \begin{array}{l} 9x + 3y = 0 \\ 3x + y = 0 \end{array} \quad \text{or} \quad 3x + y = 0$$

A nonzero solution is  $u_2 = (-1, 3)$ .

As expected, because  $A$  is symmetric, the eigenvectors  $u_1$  and  $u_2$  are orthogonal.

Now normalize  $u_1$  and  $u_2$  to obtain, respectively, the unit vectors

$$\hat{u}_1 = (3/\sqrt{10}, 1/\sqrt{10}) \quad \text{and} \quad \hat{u}_2 = (-1/\sqrt{10}, 3/\sqrt{10}).$$

Finally, let  $P$  be the matrix whose columns are the unit vectors  $\hat{u}_1$  and  $\hat{u}_2$ , respectively, and then  $[x, y]^T = P[s, t]^T$  is the required orthogonal change of coordinates. That is,

$$P = \begin{bmatrix} 3/\sqrt{10} & -1/\sqrt{10} \\ 1/\sqrt{10} & 3/\sqrt{10} \end{bmatrix} \quad \text{and} \quad x = \frac{3s - t}{\sqrt{10}}, \quad y = \frac{s + 3t}{\sqrt{10}}$$

One can also express  $s$  and  $t$  in terms of  $x$  and  $y$  by using  $P^{-1} = P^T$ . That is,

$$s = \frac{3x + y}{\sqrt{10}}, \quad t = \frac{-x + 3y}{\sqrt{10}}$$

## Minimal Polynomial

---

**9.27.** Let  $A = \begin{bmatrix} 4 & -2 & 2 \\ 6 & -3 & 4 \\ 3 & -2 & 3 \end{bmatrix}$  and  $B = \begin{bmatrix} 3 & -2 & 2 \\ 4 & -4 & 6 \\ 2 & -3 & 5 \end{bmatrix}$ . The characteristic polynomial of both matrices is  $\Delta(t) = (t - 2)(t - 1)^2$ . Find the minimal polynomial  $m(t)$  of each matrix.

The minimal polynomial  $m(t)$  must divide  $\Delta(t)$ . Also, each factor of  $\Delta(t)$  (i.e.,  $t - 2$  and  $t - 1$ ) must also be a factor of  $m(t)$ . Thus,  $m(t)$  must be exactly one of the following:

$$f(t) = (t - 2)(t - 1) \quad \text{or} \quad g(t) = (t - 2)(t - 1)^2$$

(a) By the Cayley-Hamilton theorem,  $g(A) = \Delta(A) = 0$ , so we need only test  $f(t)$ . We have

$$f(A) = (A - 2I)(A - I) = \begin{bmatrix} 2 & -2 & 2 \\ 6 & -5 & 4 \\ 3 & -2 & 1 \end{bmatrix} \begin{bmatrix} 3 & -2 & 2 \\ 6 & -4 & 4 \\ 3 & -2 & 2 \end{bmatrix} = \begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}$$

Thus,  $m(t) = f(t) = (t - 2)(t - 1) = t^2 - 3t + 2$  is the minimal polynomial of  $A$ .

(b) Again  $g(B) = \Delta(B) = 0$ , so we need only test  $f(t)$ . We get

$$f(B) = (B - 2I)(B - I) = \begin{bmatrix} 1 & -2 & 2 \\ 4 & -6 & 6 \\ 2 & -3 & 3 \end{bmatrix} \begin{bmatrix} 2 & -2 & 2 \\ 4 & -5 & 6 \\ 2 & -3 & 4 \end{bmatrix} = \begin{bmatrix} -2 & 2 & -2 \\ -4 & 4 & -4 \\ -2 & 2 & -2 \end{bmatrix} \neq 0$$

Thus,  $m(t) \neq f(t)$ . Accordingly,  $m(t) = g(t) = (t - 2)(t - 1)^2$  is the minimal polynomial of  $B$ . [We emphasize that we do not need to compute  $g(B)$ ; we know  $g(B) = 0$  from the Cayley-Hamilton theorem.]

**9.28.** Find the minimal polynomial  $m(t)$  of each of the following matrices:

(a)

$$A = \begin{bmatrix} 5 & 1 \\ 3 & 7 \end{bmatrix}, \text{ (b) } B = \begin{bmatrix} 1 & 2 & 3 \\ 0 & 2 & 3 \\ 0 & 0 & 3 \end{bmatrix}, \text{ (c) } C = \begin{bmatrix} 4 & -1 \\ 1 & 2 \end{bmatrix}$$

(a) The characteristic polynomial of  $A$  is  $\Delta(t) = t^2 - 12t + 32 = (t - 4)(t - 8)$ . Because  $\Delta(t)$  has distinct factors, the minimal polynomial  $m(t) = \Delta(t) = t^2 - 12t + 32$ .

0

(b) Because  $B$  is triangular, its eigenvalues are the diagonal elements 1, 2, 3; and so its characteristic polynomial is  $\Delta(t) = (t - 1)(t - 2)(t - 3)$ . Because  $\Delta(t)$  has distinct factors,  $m(t) = \Delta(t)$ .

(c) The characteristic polynomial of  $C$  is  $\Delta(t) = t^2 - 6t + 9 = (t - 3)^2$ . Hence the minimal polynomial of  $C$  is  $f(t) = t - 3$  or  $g(t) = (t - 3)^2$ . However,  $f(C) \neq 0$ ; that is,  $C - 3I \neq 0$ . Hence,

$$m(t) = g(t) = \Delta(t) = (t - 3)^2.$$

**9.29.** Suppose  $S = \{u_1, u_2, \dots, u_n\}$  is a basis of  $V$ , and suppose  $F$  and  $G$  are linear operators on  $V$  such that

$[F]$  has 0's on and below the diagonal, and  $[G]$  has  $a \neq 0$  on the superdiagonal and 0's elsewhere. That is,

$$[F] = \begin{bmatrix} 0 & a_{21} & a_{31} & \cdots & a_{n1} \\ 0 & 0 & a_{32} & \cdots & a_{n2} \\ \cdots & \cdots & \cdots & \cdots & \cdots \\ 0 & 0 & 0 & \cdots & a_{n,n-1} \\ 0 & 0 & 0 & \cdots & 0 \end{bmatrix}, \quad [G] = \begin{bmatrix} 0 & a & 0 & \cdots & 0 \\ 0 & 0 & a & \cdots & 0 \\ \cdots & \cdots & \cdots & \cdots & \cdots \\ 0 & 0 & 0 & \cdots & a \\ 0 & 0 & 0 & \cdots & 0 \end{bmatrix}$$

Show that (a)  $F^n = 0$ , (b)  $G^{n-1} \neq 0$ , but  $G^n = 0$ . (These conditions also hold for  $[F]$  and  $[G]$ .)

(a) We have  $F(u_1) = 0$  and, for  $r > 1$ ,  $F(u_r)$  is a linear combination of vectors preceding  $u_r$  in  $S$ . That is,

$$F(u_r) = a_{r1}u_1 + a_{r2}u_2 + \cdots + a_{r,r-1}u_{r-1}$$

Hence,  $F^2(u_r) = F(F(u_r))$  is a linear combination of vectors preceding  $u_{r-1}$ , and so on. Hence,  $F^r(u_r) = 0$  for each  $r$ . Thus, for each  $r$ ,  $F^n(u_r) = F^{n-r}(0) = 0$ , and so  $F^n = 0$ , as claimed.

(b) We have  $G(u_1) = 0$  and, for each  $k > 1$ ,  $G(u_k) = au_{k-1}$ . Hence,  $G^r(u_k) = a^r u_{k-r}$  for  $r < k$ . Because  $a \neq 0$ ,  $a^{n-1} \neq 0$ . Therefore,  $G^{n-1}(u_n) = a^{n-1} u_1 \neq 0$ , and so  $G^{n-1} \neq 0$ . On the other hand, by (a),  $G^n = 0$ .

**9.30.** Let  $B$  be the matrix in [Example 9.12\(a\)](#) that has 1's on the diagonal,  $a$ 's on the superdiagonal, where  $a \neq 0$ , and 0's elsewhere. Show that  $f(t) = (t - \lambda)^n$  is both the characteristic polynomial  $\Delta(t)$  and the minimum polynomial  $m(t)$  of  $A$ .

Because  $A$  is triangular with  $\lambda$ 's on the diagonal,  $\Delta(t) = f(t) = (t - \lambda)^n$  is its characteristic polynomial.

Thus,  $m(t)$  is a power of  $t - \lambda$ . By [Problem 9.29](#),  $(A - \lambda I)^{r-1} \neq 0$ . Hence,  $m(t) = \Delta(t) = (t - \lambda)^n$ .

**9.31.** Find the characteristic polynomial  $\Delta(t)$  and minimal polynomial  $m(t)$  of each matrix:

$$(a) \quad M = \begin{bmatrix} 4 & 1 & 0 & 0 & 0 \\ 0 & 4 & 1 & 0 & 0 \\ 0 & 0 & 4 & 0 & 0 \\ 0 & 0 & 0 & 4 & 1 \\ 0 & 0 & 0 & 0 & 4 \end{bmatrix}, \quad (b) \quad M' = \begin{bmatrix} 2 & 7 & 0 & 0 \\ 0 & 2 & 0 & 0 \\ 0 & 0 & 1 & 1 \\ 0 & 0 & -2 & 4 \end{bmatrix}$$

(a)  $M$  is block diagonal with diagonal blocks

$$A = \begin{bmatrix} 4 & 1 & 0 \\ 0 & 4 & 1 \\ 0 & 0 & 4 \end{bmatrix} \quad \text{and} \quad B = \begin{bmatrix} 4 & 1 \\ 0 & 4 \end{bmatrix}$$

The characteristic and minimal polynomial of  $A$  is  $f(t) = (t - 4)^3$  and the characteristic and minimal polynomial of  $B$  is  $g(t) = (t - 4)^2$ . Then

$$\Delta(t) = f(t)g(t) = (t - 4)^5 \quad \text{but} \quad m(t) = \text{LCM}[f(t), g(t)] = (t - 4)^3$$

(where LCM means least common multiple). We emphasize that the exponent in  $m(t)$  is the size of the largest block.

(b) Here  $M'$  is block diagonal with diagonal blocks

$$A' = \begin{bmatrix} 2 & 7 \\ 0 & 2 \end{bmatrix} \quad \text{and} \quad B' = \begin{bmatrix} 1 & 1 \\ -2 & 4 \end{bmatrix}$$

The characteristic and minimal polynomial of  $A'$  is  $f(t) = (t - 2)^2$ . The characteristic polynomial of  $B'$  is  $g(t) = t^2 - 5t + 6 = (t - 2)(t - 3)$ , which has distinct factors. Hence,  $g(t)$  is also the minimal polynomial of  $B$ . Accordingly,

$$\Delta(t) = f(t)g(t) = (t - 2)^3(t - 3) \quad \text{but} \quad m(t) = \text{LCM}[f(t), g(t)] = (t - 2)^2(t - 3)$$

**9.32.** Find a matrix  $A$  whose minimal polynomial is  $f(t) = t^3 - 8t^2 + 5t + 7$ .

Simply let  $A = \begin{bmatrix} 0 & 0 & -7 \\ 1 & 0 & -5 \\ 0 & 1 & 8 \end{bmatrix}$ , the companion matrix of  $f(t)$  [defined in [Example 9.12\(b\)](#)].

**9.33.** Prove [Theorem 9.15](#): The minimal polynomial  $m(t)$  of a matrix (linear operator)  $A$  divides every polynomial that has  $A$  as a zero. In particular (by the Cayley-Hamilton theorem),  $m(t)$  divides the characteristic polynomial  $\Delta(t)$  of  $A$ .

Suppose  $f(t)$  is a polynomial for which  $f(A) = 0$ . By the division algorithm, there exist polynomials  $q(t)$  and  $r(t)$  for which  $f(t) = m(t)q(t) + r(t)$  and  $r(t) = 0$  or  $\deg r(t) < \deg m(t)$ . Substituting  $t = A$  in this equation, and using that  $f(A) = 0$  and  $m(A) = 0$ , we obtain  $r(A) = 0$ . If  $r(t) \neq 0$ , then  $r(t)$  is a polynomial of degree less than  $m(t)$  that has  $A$  as a zero. This contradicts the definition of the minimal polynomial. Thus,  $r(t) = 0$ , and so  $f(t) = m(t)q(t)$ ; that is,  $m(t)$  divides  $f(t)$ .

**9.34.** Let  $m(t)$  be the minimal polynomial of an  $n$ -square matrix  $A$ . Prove that the characteristic polynomial  $\Delta(t)$  of  $A$  divides  $[m(t)]^n$ .

Suppose  $m(t) = t^r + c_1 t^{r-1} + \cdots + c_{r-1} t + c_r$ . Define matrices  $B_j$  as follows:

|                                                        |    |                                           |
|--------------------------------------------------------|----|-------------------------------------------|
| $B_0 = I$                                              | so | $I = B_0$                                 |
| $B_1 = A + c_1 I$                                      | so | $c_1 I = B_1 - A = B_1 - AB_0$            |
| $B_2 = A^2 + c_1 A + c_2 I$                            | so | $c_2 I = B_2 - A(A + c_1 I) = B_2 - AB_1$ |
|                                                        |    |                                           |
| $B_{r-1} = A^{r-1} + c_1 A^{r-2} + \cdots + c_{r-1} I$ | so | $c_{r-1} I = B_{r-1} - AB_{r-2}$          |

Then

$$-AB_{r-1} = c_r I - (A^r + c_1 A^{r-1} + \cdots + c_{r-1} A + c_r I) = c_r I - m(A) = c_r I$$

$$B(t) = t^{r-1} B_0 + t^{r-2} B_1 + \cdots + t B_{r-2} + B_{r-1}$$

Set

Then

$$\begin{aligned} (tI - A)B(t) &= (t^r B_0 + t^{r-1} B_1 + \cdots + t B_{r-1}) - (t^{r-1} A B_0 + t^{r-2} A B_1 + \cdots + A B_{r-1}) \\ &= t^r B_0 + t^{r-1} (B_1 - A B_0) + t^{r-2} (B_2 - A B_1) + \cdots + t (B_{r-1} - A B_{r-2}) - A B_{r-1} \\ &= t^r I + c_1 t^{r-1} I + c_2 t^{r-2} I + \cdots + c_{r-1} t I + c_r I = m(t) I \end{aligned}$$

Taking the determinant of both sides gives  $|tI - A| \det B(t) = |m(t)I| = [m(t)]^n$ . Because  $|B(t)|$  is a polynomial,  $|tI - A|$  divides  $[m(t)]^n$ ; that is, the characteristic polynomial of  $A$  divides  $[m(t)]^n$ .

**9.35.** Prove [Theorem 9.16](#): The characteristic polynomial  $\Delta(t)$  and the minimal polynomial  $m(t)$  of  $A$  have the same irreducible factors.

Suppose  $f(t)$  is an irreducible polynomial. If  $f(t)$  divides  $m(t)$ , then  $f(t)$  also divides  $\Delta(t)$  [because  $m(t)$  divides  $\Delta(t)$ ]. On the other hand, if  $f(t)$  divides  $\Delta(t)$ , then by [Problem 9.34](#),  $f(t)$  also divides  $[m(t)]^n$ . But  $f(t)$  is irreducible; hence,  $f(t)$  also divides  $m(t)$ . Thus,  $m(t)$  and  $\Delta(t)$  have the same irreducible factors.

**9.36.** Prove [Theorem 9.19](#): The minimal polynomial  $m(t)$  of a block diagonal matrix  $M$  with diagonal blocks  $A_i$  is equal to the least common multiple (LCM) of the minimal polynomials of the diagonal blocks  $A_i$ .

We prove the theorem for the case  $r = 2$ . The general theorem follows easily by induction. Suppose

$$M = \begin{bmatrix} A & 0 \\ 0 & B \end{bmatrix}, \text{ where } A \text{ and } B \text{ are square matrices. We}$$

need to show that the minimal polynomial  $m(t)$  of  $M$  is

the LCM of the minimal polynomials  $g(t)$  and  $h(t)$  of  $A$  and  $B$ , respectively.

Because  $m(t)$  is the minimal polynomial of  $M$ ,  $m(M) = \begin{bmatrix} m(A) & 0 \\ 0 & m(B) \end{bmatrix} = 0$ , and  $m(A) = 0$  and  $m(B) = 0$ . Because  $g(t)$  is the minimal polynomial of  $A$ ,  $g(t)$  divides  $m(t)$ . Similarly,  $h(t)$  divides  $m(t)$ . Thus  $m(t)$  is a multiple of  $g(t)$  and  $h(t)$ .

Now let  $f(t)$  be another multiple of  $g(t)$  and  $h(t)$ . Then  $f(M) = \begin{bmatrix} f(A) & 0 \\ 0 & f(B) \end{bmatrix} = \begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix} = 0$ . But  $m(t)$  is the minimal polynomial of  $M$ ; hence,  $m(t)$  divides  $f(t)$ . Thus,  $m(t)$  is the LCM of  $g(t)$  and  $h(t)$ .

**9.37.** Suppose  $m(t) = t^r + a_{r-1}t^{r-1} + \cdots + a_1t + a_0$  is the minimal polynomial of an  $n$ -square matrix  $A$ . Prove the following:

(a)  $A$  is nonsingular if and only if the constant term  $a_0 \neq 0$ .

(b) If  $A$  is nonsingular, then  $A^{-1}$  is a polynomial in  $A$  of degree  $r - 1 < n$ .

(a) The following are equivalent: (i)  $A$  is nonsingular, (ii) 0 is not a root of  $m(t)$ , (iii)  $a_0 \neq 0$ . Thus, the statement is true.

(b) Because  $A$  is nonsingular,  $a_0 \neq 0$  by (a). We have

$$\begin{aligned} m(A) &= A^r + a_{r-1}A^{r-1} + \cdots + a_1A + a_0I = 0 \\ &\quad - \frac{1}{a_0}(A^{r-1} + a_{r-1}A^{r-2} + \cdots + a_1I)A = I \end{aligned}$$

Thus,

Accordingly,

$$A^{-1} = -\frac{1}{a_0}(A^{r-1} + a_{r-1}A^{r-2} + \cdots + a_1I)$$

## SUPPLEMENTARY PROBLEMS

### Polynomials of Matrices

- 9.38.** Let  $A = \begin{bmatrix} 2 & -3 \\ 5 & 1 \end{bmatrix}$  and  $B = \begin{bmatrix} 1 & 2 \\ 0 & 3 \end{bmatrix}$ . Find  $f(A)$ ,  $g(A)$ ,  $f(B)$ ,  $g(B)$ , where  $f(t) = 2t^2 - 5t + 6$  and  $g(t) = t^3 - 2t^2 + t + 3$ .
- 9.39.** Let  $A = \begin{bmatrix} 1 & 2 \\ 0 & 1 \end{bmatrix}$ . Find  $A^2$ ,  $A^3$ ,  $A^n$ , where  $n > 3$ , and  $A^{-1}$ .
- 9.40.** Let  $B = \begin{bmatrix} 8 & 12 & 0 \\ 0 & 8 & 12 \\ 0 & 0 & 8 \end{bmatrix}$ . Find a real matrix  $A$  such that  $B = A^3$ .
- 9.41.** For each matrix, find a polynomial having the following matrix as a root:
- (a)  $A = \begin{bmatrix} 2 & 5 \\ 1 & -3 \end{bmatrix}$ , (b)  $B = \begin{bmatrix} 2 & -3 \\ 7 & -4 \end{bmatrix}$ , (c)  $C = \begin{bmatrix} 1 & 1 & 2 \\ 1 & 2 & 3 \\ 2 & 1 & 4 \end{bmatrix}$
- 9.42.** Let  $A$  be any square matrix and let  $f(t)$  be any polynomial. Prove (a)  $(P^{-1}AP)^n = P^{-1}A^nP$ .  
 (b)  $f(P^{-1}AP) = P^{-1}f(A)P$ . (c)  $f(A^T) = [f(A)]^T$ . (d) If  $A$  is symmetric, then  $f(A)$  is symmetric.
- 9.43.** Let  $M = \text{diag}[A_1, \dots, A_r]$  be a block diagonal matrix, and let  $f(t)$  be any polynomial. Show that  $f(M)$  is block

diagonal and  $f(M) = \text{diag}[f(A_1), \dots, f(A_r)]$ :

- 9.44.** Let  $M$  be a block triangular matrix with diagonal blocks  $A_1, \dots, A_r$  and let  $f(t)$  be any polynomial. Show that  $f(M)$  is also a block triangular matrix, with diagonal blocks  $f(A_1), \dots, f(A_r)$ .

## Eigenvalues and Eigenvectors

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- 9.45.** For each of the following matrices, find all eigenvalues and corresponding linearly independent eigenvectors:

(a)  $A = \begin{bmatrix} 2 & -3 \\ 2 & -5 \end{bmatrix}$ , (b)  $B = \begin{bmatrix} 2 & 4 \\ -1 & 6 \end{bmatrix}$ , (c)  $C = \begin{bmatrix} 1 & -4 \\ 3 & -7 \end{bmatrix}$

When possible, find the nonsingular matrix  $P$  that diagonalizes the matrix.

**9.46.** Let  $A = \begin{bmatrix} 2 & -1 \\ -2 & 3 \end{bmatrix}$ .

(a) Find eigenvalues and corresponding eigenvectors.

(b) Find a nonsingular matrix  $P$  such that  $D = P^{-1}AP$  is diagonal.

(c) Find  $A^8$  and  $f(A)$  where  $f(t) = t^4 - 5t^3 + 7t^2 - 2t + 5$ .

(d) Find a matrix  $B$  such that  $B^2 = A$ .

**9.47.** Repeat [Problem 9.46](#) for  $A = \begin{bmatrix} 5 & 6 \\ -2 & -2 \end{bmatrix}$ .

- 9.48.** For each of the following matrices, find all eigenvalues and a maximum set  $S$  of linearly independent eigenvectors:

$$(a) \quad A = \begin{bmatrix} 1 & -3 & 3 \\ 3 & -5 & 3 \\ 6 & -6 & 4 \end{bmatrix}, (b) \quad B = \begin{bmatrix} 3 & -1 & 1 \\ 7 & -5 & 1 \\ 6 & -6 & 2 \end{bmatrix}, (c) \quad C = \begin{bmatrix} 1 & 2 & 2 \\ 1 & 2 & -1 \\ -1 & 1 & 4 \end{bmatrix}$$

Which matrices can be diagonalized, and why?

**9.49.** Find: (a)  $2 \times 2$  matrix  $A$  with eigenvalues  $\lambda_1 = 1$  and  $\lambda_2 = -2$  and corresponding eigenvectors  $v_1 = (1, 2)$  and  $v_2 = (3, 7)$ .

(b)  $3 \times 3$  matrix  $A$  with eigenvalues  $\lambda_1 = 1$ ,  $\lambda_2 = 2$ ,  $\lambda_3 = 3$  and corresponding eigenvectors  $v_1 = (1, 0, 1)$ ,  $v_2 = (1, 1, 2)$ ,  $v_3 = (1, 2, 4)$ . [Hint: See [Problem 2.19](#).]

**9.50.** Let  $A = \begin{bmatrix} a & b \\ c & d \end{bmatrix}$  be a real matrix. Find necessary and sufficient conditions on  $a, b, c, d$  so that  $A$  is diagonalizable—that is, so that  $A$  has two (real) linearly independent eigenvectors.

**9.51.** Show that matrices  $A$  and  $A^T$  have the same eigenvalues. Give an example of a  $2 \times 2$  matrix  $A$  where  $A$  and  $A^T$  have different eigenvectors.

**9.52.** Suppose  $v$  is an eigenvector of linear operators  $F$  and  $G$ . Show that  $v$  is also an eigenvector of the linear operator  $kF + KG$ , where  $k$  and  $K$  are scalars.

**9.53.** Suppose  $v$  is an eigenvector of a linear operator  $T$  belonging to the eigenvalue  $\lambda$ . Prove

(a) For  $n > 0$ ,  $v$  is an eigenvector of  $T^n$  belonging to  $\lambda^n$ .

(b)  $f(\lambda)$  is an eigenvalue of  $f(T)$  for any polynomial  $f(t)$ .

- 9.54.** Suppose  $\lambda \neq 0$  is an eigenvalue of the composition  $F \circ G$  of linear operators  $F$  and  $G$ . Show that  $\lambda$  is also an eigenvalue of the composition  $G \circ F$ . [*Hint:* Show that  $G(v)$  is an eigenvector of  $G \circ F$ .]
- 9.55.** Let  $E: V \rightarrow V$  be a projection mapping; that is,  $E^2 = E$ . Show that  $E$  is diagonalizable and, in fact, can be represented by the diagonal matrix  $M = \begin{bmatrix} I_r & 0 \\ 0 & 0 \end{bmatrix}$ , where  $r$  is the rank of  $E$ .

## Diagonalizing Real Symmetric Matrices and Quadratic Forms

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- 9.56.** For each of the following symmetric matrices  $A$ , find an orthogonal matrix  $P$  and a diagonal matrix  $D$  such that  $D = P^{-1}AP$ :
- (a)  $A = \begin{bmatrix} 5 & 4 \\ 4 & -1 \end{bmatrix}$ , (b)  $A = \begin{bmatrix} 4 & -1 \\ -1 & 4 \end{bmatrix}$ , (c)  $A = \begin{bmatrix} 7 & 3 \\ 3 & -1 \end{bmatrix}$
- 9.57.** For each of the following symmetric matrices  $B$ , find its eigenvalues, a maximal orthogonal set  $S$  of eigenvectors, and an orthogonal matrix  $P$  such that  $D = P^{-1}BP$  is diagonal:
- (a)  $B = \begin{bmatrix} 0 & 1 & 1 \\ 1 & 0 & 1 \\ 1 & 1 & 0 \end{bmatrix}$ , (b)  $B = \begin{bmatrix} 2 & 2 & 4 \\ 2 & 5 & 8 \\ 4 & 8 & 17 \end{bmatrix}$
- 9.58.** Using variables  $s$  and  $t$ , find an orthogonal substitution that diagonalizes each of the following quadratic forms:
- (a)  $q(x, y) = 4x^2 + 8xy - 11y^2$ , (b)  $q(x, y) = 2x^2 - 6xy + 10y^2$
- 9.59.** For each of the following quadratic forms  $q(x, y, z)$ , find an orthogonal substitution expressing  $x, y, z$  in

terms of variables  $r, s, t$ , and find  $q(r, s, t)$ :

$$(a) \quad q(x, y, z) = 5x^2 + 3y^2 + 12xz, \quad (b) \quad q(x, y, z) = 3x^2 - 4xy + 6y^2 + 2xz - 4yz + 3z^2$$

**9.60.** Find a real  $2 \times 2$  symmetric matrix  $A$  with eigenvalues:

(a)  $\lambda = 1$  and  $\lambda = 4$  and eigenvector  $u = (1, 1)$  belonging to  $\lambda = 1$ ;

(b)  $\lambda = 2$  and  $\lambda = 3$  and eigenvector  $u = (1, 2)$  belonging to  $\lambda = 2$ .

In each case, find a matrix  $B$  for which  $B^2 = A$ .

## Characteristic and Minimal Polynomials

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**9.61.** Find the characteristic and minimal polynomials of each of the following matrices:

$$(a) \quad A = \begin{bmatrix} 3 & 1 & -1 \\ 2 & 4 & -2 \\ -1 & -1 & 3 \end{bmatrix}, \quad (b) \quad B = \begin{bmatrix} 3 & 2 & -1 \\ 3 & 8 & -3 \\ 3 & 6 & -1 \end{bmatrix}$$

**9.62.** Find the characteristic and minimal polynomials of each of the following matrices:

$$(a) \quad A = \begin{bmatrix} 2 & 5 & 0 & 0 & 0 \\ 0 & 2 & 0 & 0 & 0 \\ 0 & 0 & 4 & 2 & 0 \\ 0 & 0 & 3 & 5 & 0 \\ 0 & 0 & 0 & 0 & 7 \end{bmatrix}, \quad (b) \quad B = \begin{bmatrix} 4 & -1 & 0 & 0 & 0 \\ 1 & 2 & 0 & 0 & 0 \\ 0 & 0 & 3 & 1 & 0 \\ 0 & 0 & 0 & 3 & 1 \\ 0 & 0 & 0 & 0 & 3 \end{bmatrix}, \quad (c) \quad C = \begin{bmatrix} 3 & 2 & 0 & 0 & 0 \\ 1 & 4 & 0 & 0 & 0 \\ 0 & 0 & 3 & 1 & 0 \\ 0 & 0 & 1 & 3 & 0 \\ 0 & 0 & 0 & 0 & 4 \end{bmatrix}$$

**9.63.** Let  $A = \begin{bmatrix} 1 & 1 & 0 \\ 0 & 2 & 0 \\ 0 & 0 & 1 \end{bmatrix}$  and  $B = \begin{bmatrix} 2 & 0 & 0 \\ 0 & 2 & 2 \\ 0 & 0 & 1 \end{bmatrix}$ . Show that  $A$  and  $B$  have different characteristic polynomials (and so are not similar) but have the same minimal polynomial. Thus, nonsimilar matrices may have the same minimal polynomial.

- 9.64.** Let  $A$  be an  $n$ -square matrix for which  $A^k = 0$  for some  $k > n$ . Show that  $A^n = 0$ .
- 9.65.** Show that a matrix  $A$  and its transpose  $A^T$  have the same minimal polynomial.
- 9.66.** Suppose  $f(t)$  is an irreducible monic polynomial for which  $f(A) = 0$  for a matrix  $A$ . Show that  $f(t)$  is the minimal polynomial of  $A$ .
- 9.67.** Show that  $A$  is a scalar matrix  $kI$  if and only if the minimal polynomial of  $A$  is  $m(t) = t - k$ .
- 9.68.** Find a matrix  $A$  whose minimal polynomial is (a)  $t^3 - 5t^2 + 6t + 8$ , (b)  $t^4 - 5t^3 - 2t + 7t + 4$ .
- 9.69.** Let  $f(t)$  and  $g(t)$  be monic polynomials (leading coefficient one) of minimal degree for which  $A$  is a root. Show  $f(t) = g(t)$ : [Thus, the minimal polynomial of  $A$  is unique.]

## ANSWERS TO SUPPLEMENTARY PROBLEMS

*Notation:*  $M = [R_1; R_2; \dots]$  denotes a matrix  $M$  with rows  $R_1, R_2, \dots$

- 9.38.**  $f(A) = [26, -3; 5, -27]$ ,  $g(A) = [40, 39; -65, -27]$ ,  
 $f(B) = [3, 6; 0, 9]$ ,  $g(B) = [3, 12; 0, 15]$
- 9.39.**  $A^2 = [1, 4; 0, 1]$ ,  $A^3 = [1, 6; 0, 1]$ ,  $A^n = [1, -2n; 0, 1]$ ,  $A^{-1} = [1, -2; 0, 1]$
- 9.40.** Let  $A = [2, a, b; 0, 2, c; 0, 0, 2]$ . Set  $B = A^3$  and then  $a = 1$ ,  $b = -\frac{1}{2}$ ,  $c = 1$

- 9.41.** Find  $\Delta(t)$ : (a)  $t^2 + t - 11$ , (b)  $t^2 + 2t + 13$ , (c)  $t^3 - 7t^2 + 6t - 1$
- 9.45.** (a)  $\lambda = 1$ ,  $u = (3, 1)$ ;  $\lambda = -4$ ,  $v = (1, 2)$ , (b)  $\lambda = 4$ ,  $u = (2, 1)$ ,  
 (c)  $\lambda = -1$ ,  $u = (2, 1)$ ;  $\lambda = 5$ ,  $v = (2, 3)$ . Only  $A$  and  $C$  can be diagonalized; use  $P = [u, v]$ .
- 9.46.** (a)  $\lambda = 1$ ,  $u = (1, 1)$ ;  $\lambda = 4$ ,  $v = (1, -2)$ ,  
 (b)  $P = [u, v]$ ,  
 (c)  $f(A) = [3, 1; 2, 1]$ ,  $A^8 = [21\ 846, -21\ 845; -43\ 690, 43\ 691]$ ,  
 (d)  $B = \begin{bmatrix} \frac{4}{3} & -\frac{1}{3} \\ -\frac{2}{3} & \frac{5}{3} \end{bmatrix}$
- 9.47.** (a)  $\lambda = 1$ ,  $u = (3, -2)$ ;  $\lambda = 2$ ,  $v = (2, -1)$ , (b)  $P = [u, v]$ ,  
 (c)  $f(A) = [2, -6; 2, 9]$ ,  $A^8 = [1021, 1530; -510, -764]$ ,  
 (d)  $B = [-3 + 4\sqrt{2}, -6 + 6\sqrt{2}; 2 - 2\sqrt{2}, 4 - 3\sqrt{2}]$
- 9.48.** (a)  $\lambda = 2$ ,  $u = (1, 1, 0)$ ,  $v = (1, 0, -1)$ ;  $\lambda = 4$ ,  $w = (1, 1, 2)$ ,  
 (b)  $\lambda = 2$ ,  $u = (1, 1, 0)$ ;  $\lambda = 4$ ,  $v = (0, 1, 1)$ ,  
 (c)  $\lambda = 3$ ,  $u = (1, 1, 0)$ ,  $v = (1, 0, 1)$ ;  $\lambda = 1$ ,  $w = (2, -1, 1)$ . Only  $A$  and  $C$  can be diagonalized; use  $P = [u, v, w]$ .
- 9.49.** (a)  $[19, -9; 42, -20]$ , (b)  $[1, 1, 0; -2, 0, 2; -4, -2, 5]$
- 9.50.** We need  $[\text{tr}(A)^2 - 4[\det(A)]] \geq 0$  or  $(a - d)^2 + 4bc \geq 0$ .
- 9.51.**  $A = [1, 1; 0, 1]$

**9.56.** (a)  $P = [2, -1; 1, 2]/\sqrt{5}, \quad D = [7, 0; 0, 3],$   
 (b)  $P = [1, 1; 1, -1]/\sqrt{2}, \quad D = [3, 0; 0, 5],$   
 (c)  $P = [3, -1; 1, 3]/\sqrt{10}, \quad D = [8, 0; 0, 2]$

**9.57.** (a)  $\lambda = 1, u = (1, -1, 0), v = (1, 1, -2); \lambda = 2, w = (1, 1, 1),$

(b)  $\lambda = 1; u = (2, 1, -1), v = (2, -3, 1); \lambda = 22; w = (1, 2, 4);$

Normalize  $u, v, w$ , obtaining  $\hat{u}, \hat{v}, \hat{w}$ , and set  $P = [\hat{u}, \hat{v}, \hat{w}]$ . (Remark:  $u$  and  $v$  are not unique.)

**9.58.**

(a)  $x = (4s + t)/\sqrt{17}, \quad y = (-s + 4t)/\sqrt{17}, \quad q(s, t) = 5s^2 - 12t^2,$

(b)  $x = (3s - t)/\sqrt{10}, \quad y = (s + 3t)/\sqrt{10}, \quad q(s, t) = s^2 + 11t^2$

**9.59.**

(a)  $x = (3s + 2t)/\sqrt{13}, \quad y = r, \quad z = (2s - 3t)/\sqrt{13}, \quad q(r, s, t) = 3r^2 + 9s^2 - 4t^2,$

(b)

$x = 5Ks + Lt, \quad y = Jr + 2Ks - 2Lt, \quad z = 2Jr - Ks - Lt, \quad \text{where } J = 1/\sqrt{5}, \quad K = 1/\sqrt{30},$   
 $L = 1/\sqrt{6}; \quad q(r, s, t) = 2r^2 + 2s^2 + 8t^2$

**9.60.** (a)  $A = \frac{1}{2}[5, -3; -3, 5], \quad B = \frac{1}{2}[3, -1; -1, 3],$

(b)  $A = \frac{1}{5}[14, -2; -2, 11], \quad B = \frac{1}{5}[\sqrt{2} + 4\sqrt{3}, 2\sqrt{2} - 2\sqrt{3}; 2\sqrt{2} - 2\sqrt{3}, 4\sqrt{2} + \sqrt{3}]$

**9.61.** (a)  $\Delta(t) = m(t) = (t - 2)^2(t - 6),$

(b)  $\Delta(t) = (t - 2)^2(t - 6), \quad m(t) = (t - 2)(t - 6)$

**9.62.** (a)  $\Delta(t) = (t - 2)^3(t - 7)^2, \quad m(t) = (t - 2)^2(t - 7),$

(b)  $\Delta(t) = (t - 3)^5, \quad m(t) = (t - 3)^3,$

(c)  $\Delta(t) = (t - 2)^2(t - 4)^2(t - 5); \quad m(t) = (t - 2)(t - 4)(t - 5)$

**9.68.** Let  $A$  be the companion matrix [Example 9.12(b)] with last column: (a)  $[8, -6, 5]^T$ , (b)  $[4, -7, 2, 5]^T$

**9.69.** *Hint:*  $A$  is a root of  $h(t) = f(t) - g(t)$ , where  $h(t) = 0$  or the degree of  $h(t)$  is less than the degree of  $f(t)$ :

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